NEHRU INSTITUTE OF INFORMATION TECHNOLOGY AND MANAGEMENT

(*Approved by AICTE , New Delhi & Affiliated to Anna University Chennai)*

"Nehru Gardens" , T.M. Palayam P.O.

COIMBATORE-641105

MASTER OF COMPUTER APPLICATIONS

REGISTER NO :

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CLASS : I MCA

SEMESTER : II

SUBJECT CODE : MC25204

SUBJECT NAME : MOBILE APPLICATION DEVELOPMENT

NEHRU INSTITUTE OF INFORMATION TECHNOLOGY AND MANAGEMENT

BONAFIDE CERTIFICATE

It is certified that this is a Bonafide record work for the subject **MOBILE APPLICATION DEVELOPMENT** Subject Code:**MC25204** done by _____Reg no_____of I MCA during the period_____.

FACULTY SUPERVISED **HEAD OF THE DEPARTMENT**

Submitted to the Anna University Practical Examination _____ 2026 held on _____at Nehru Institute of Information Technology and Management ,Coimbatore.

INTERNAL EXAMINER **EXTERNAL EXAMINER**

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EX.NO: 01

DATE:

SIMPLE ANDROID APPLICATION USINGGUICOMPONENTS

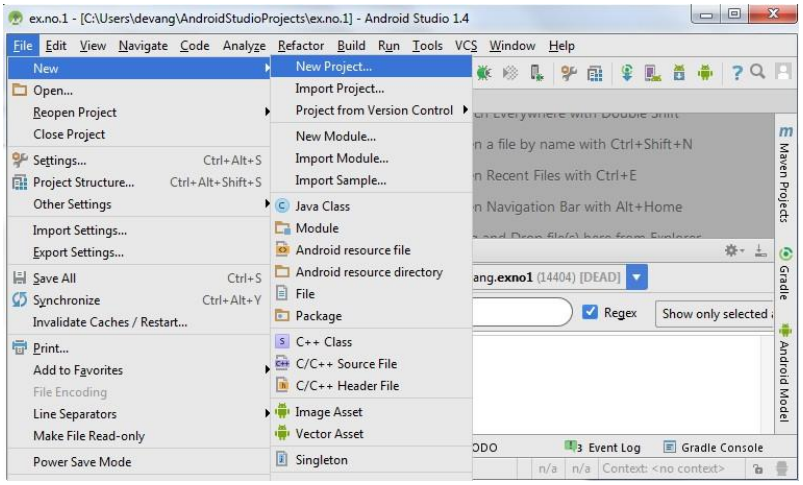
AIM:

To develop a Simple Android Application that uses GUI components , Frontend Colors.

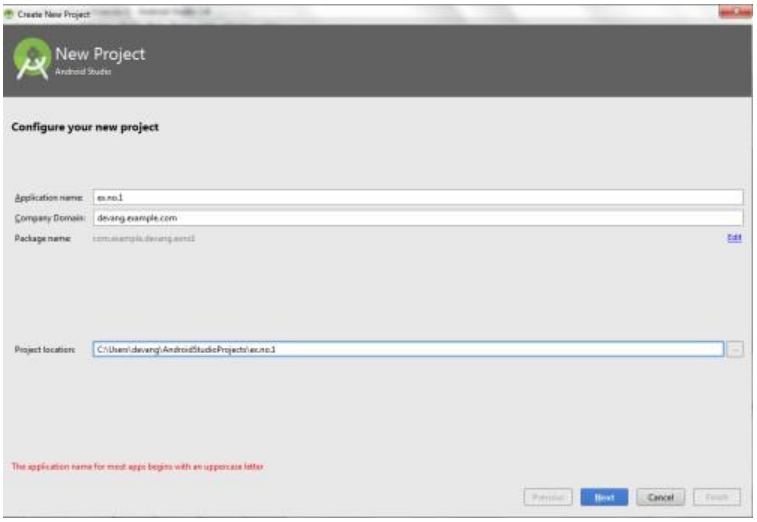
PROCEDURE:

Creating a New project:

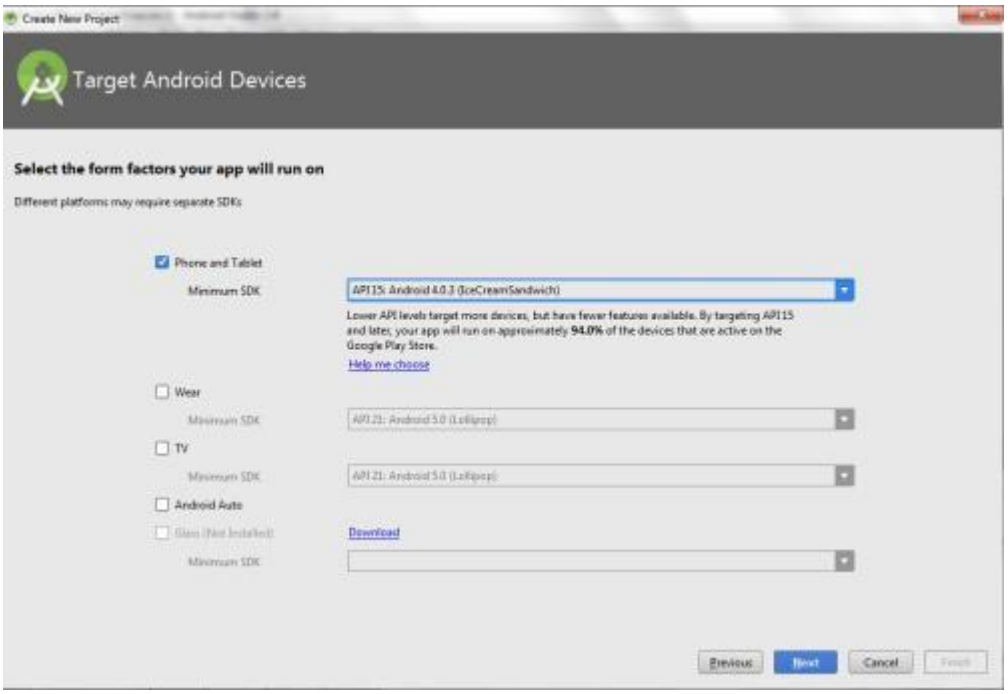
- Open Android Studio and then click on **File->New->New project.**



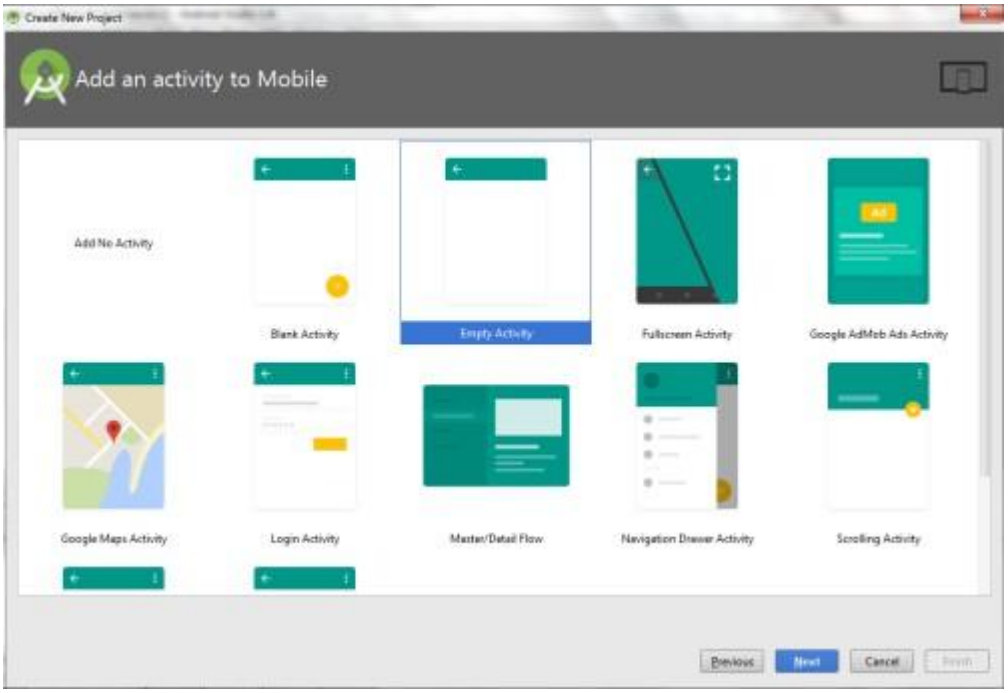
- Then type the Application name as "**ex.no.1"**and click **Next.**



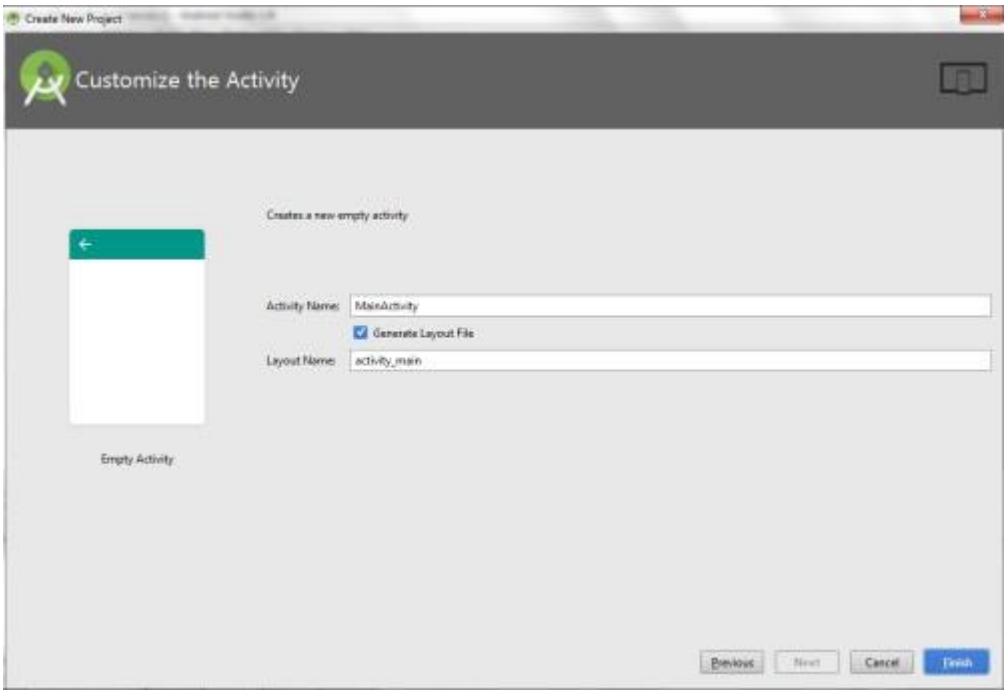
- Then select the **Minimum SDK** as shown below and click **Next**.



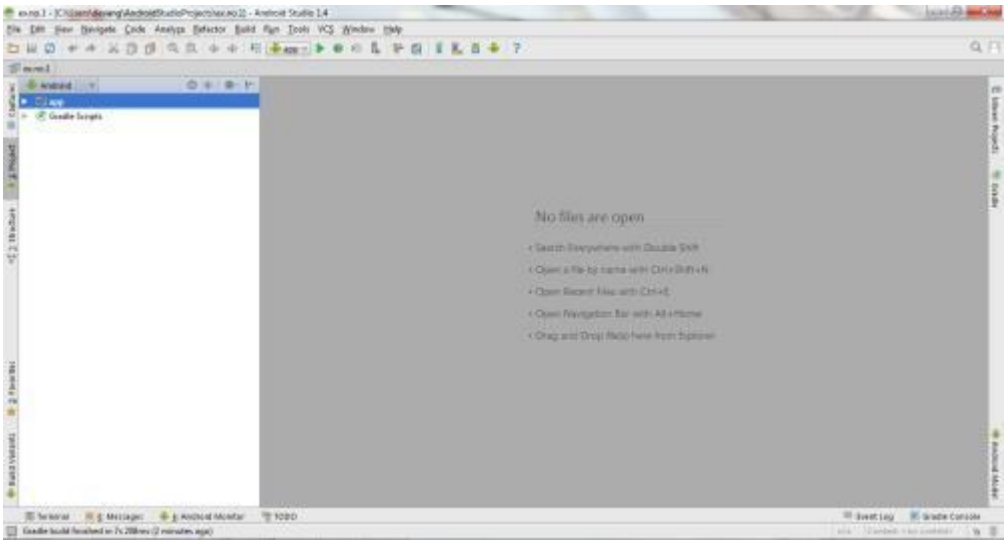
- Then select the **Empty Activity** and click **Next.**



- Finally click F**inish**.

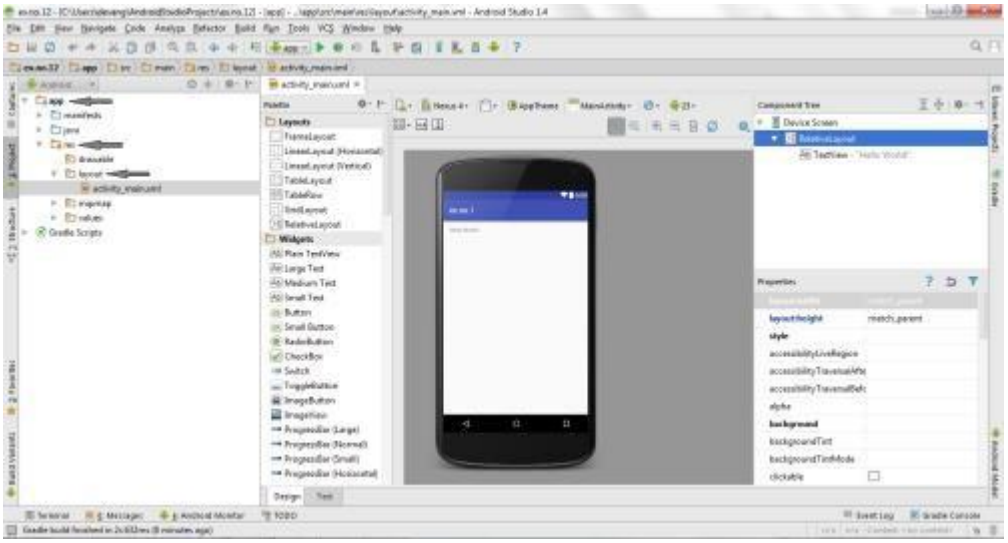


- It will take sometime to build and load the project.
- After completion it will look as given below.

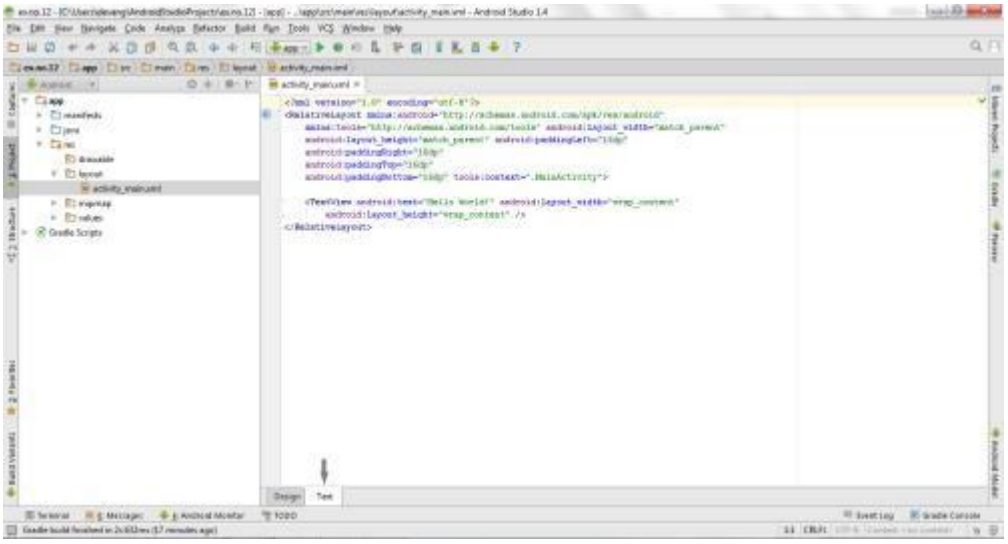


Designing layout for the Android Application:

- Clickon **app->res-> layout->activity_main.xml.**



- Now click on **Text** as shown below.



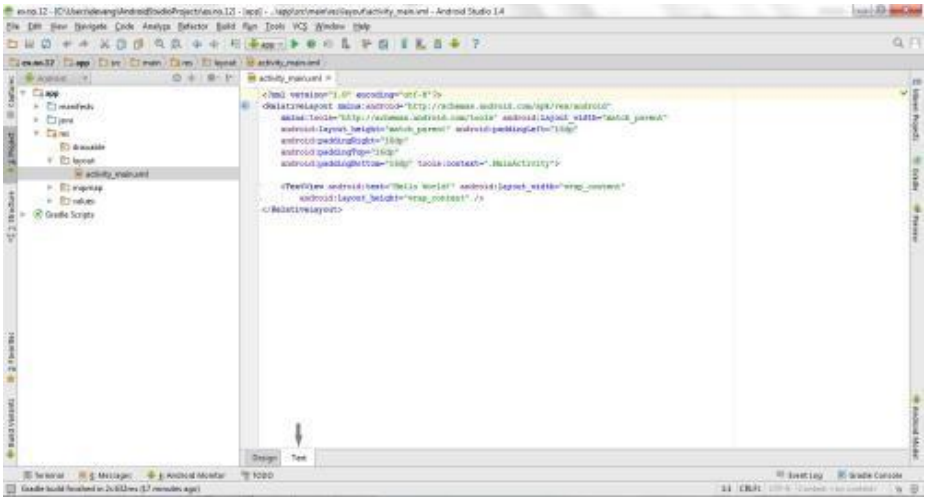
- Then delete the code which is there and type the code as given below.

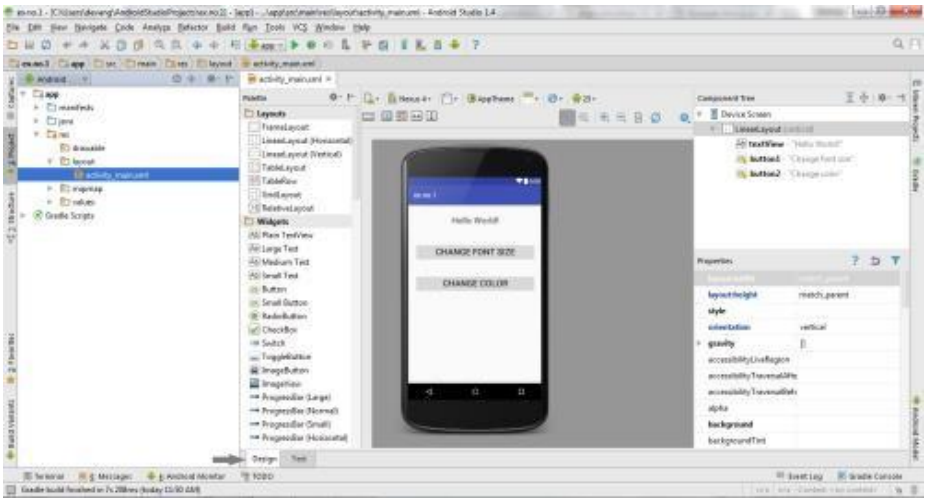
Code forActivity_main.xml:

```
<?xmlversion="1.0"encoding="utf-8"?>
<LinearLayoutxmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="30dp"
```

```
        android:gravity="center"android:text="Hel
        loWorld!"android:textSize="25sp"android:t
        extStyle="bold"/>

    <Button
        android:id="@+id/button1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:gravity="center"android:text="Cha
        nge font size" android:textSize="25sp"/>
    <Button
        android:id="@+id/button2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:gravity="center"android:text="Cha
        nge color" android:textSize="25sp"/>
</LinearLayout>
```

- Now click on Design and your application will look as given below.





- Then delete the code which is there and type the code as given below.

Code forActivity_main.java

```
packagecom.example.exno1;

importandroid.graphics.Color;
importandroid.support.v7.app.AppCompatActivity;
importandroid.os.Bundle;
importandroid.view.View;
importandroid.widget.Button;
importandroid.widget.TextView;

publicclassMainActivityextendsAppCompatActivity
{
    intch=1;
    floatfont=30;
    @Override
    protectedvoidonCreate(BundlesavedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        finalTextViewt=(TextView)findViewById(R.id.textView); Button
        b1= (Button) findViewById(R.id.button1);
        b1.setOnClickListener(newView.OnClickListener() {
            @Override
            publicvoidonClick(Viewv){
                t.setTextSize(font);
                font = font + 5;
                if(font==50)
                    font = 30;
            }
        });
        Button b2= (Button) findViewById(R.id.button2);
        b2.setOnClickListener(newView.OnClickListener(){
            @Override
            publicvoidonClick(Viewv){
                switch(ch) {
                    case1:
                        t.setTextColor(Color.RED);
                        break;
                    case2:
                        t.setTextColor(Color.GREEN);
                        break;
                    case3:
                        t.setTextColor(Color.BLUE);
                        break;
                    case4:
                        t.setTextColor(Color.CYAN);
                        break;
                    case5:
                        t.setTextColor(Color.YELLOW);
                        break;
                    case6:
                        t.setTextColor(Color.MAGENTA);
                        break;
                }
```

```
                    ch++;
                    if(ch==7)
                        ch=1;

                }
            });
        }
}
```

- So now the Coding part is also completed.
- Now run the application to see the output

SAMPLEOUTPUT:

RESULT:

Thus a Simple Android Application using GUI components is developed and executed successfully.

EX. NO: 02

DATE:

ANDROID APPLICATION FOR LAYOUT MANAGERS AND EVENT LISTENERS.

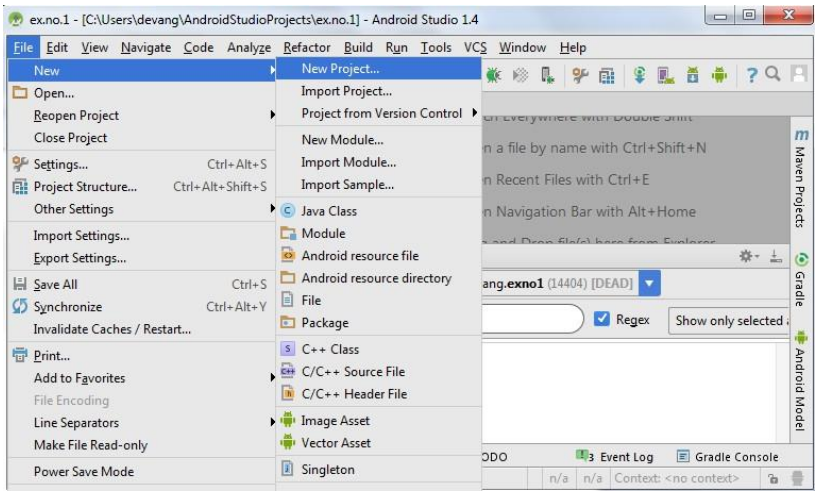
AIM:

To develop a Simple Android Application that uses Layout Managers and Event Listeners.

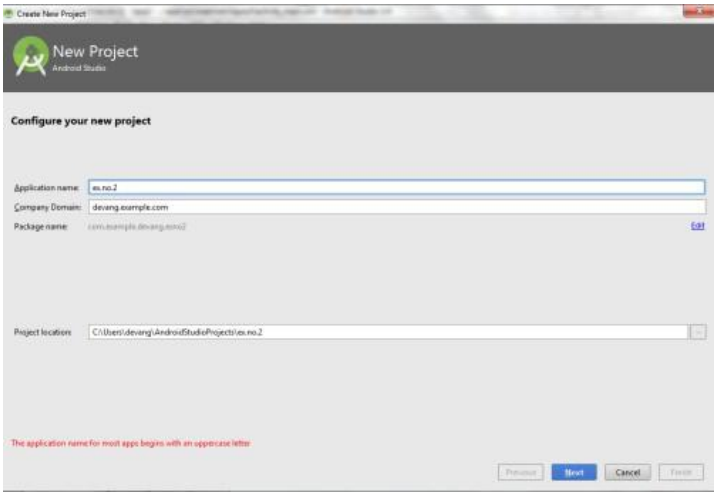
PROCEDURE:

Creating a New project:

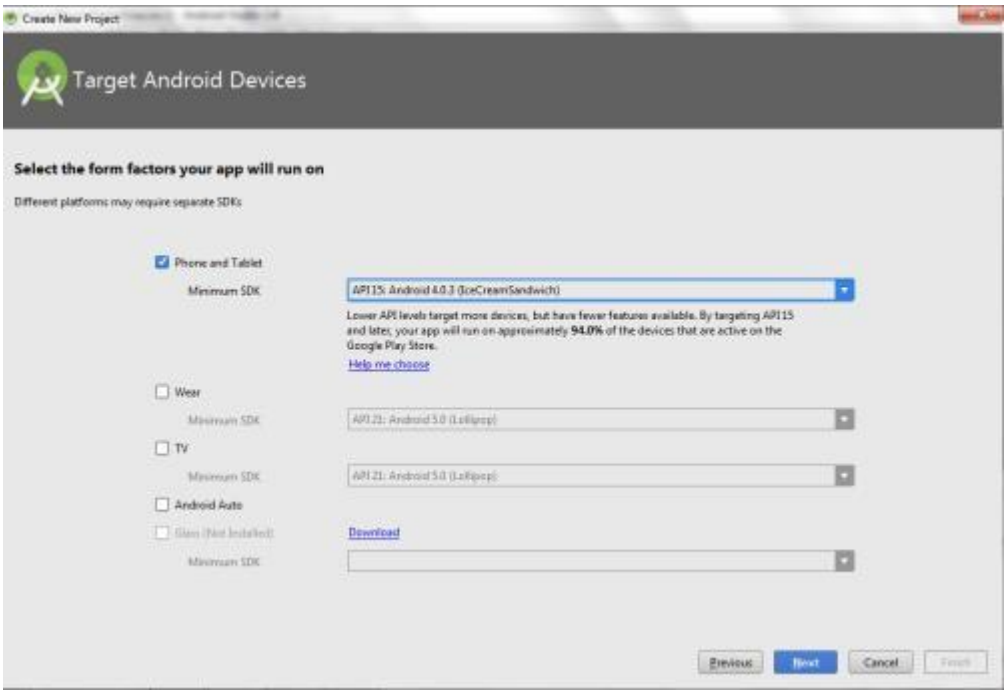
- Open Android Studio and then click on **File->New->New project.**



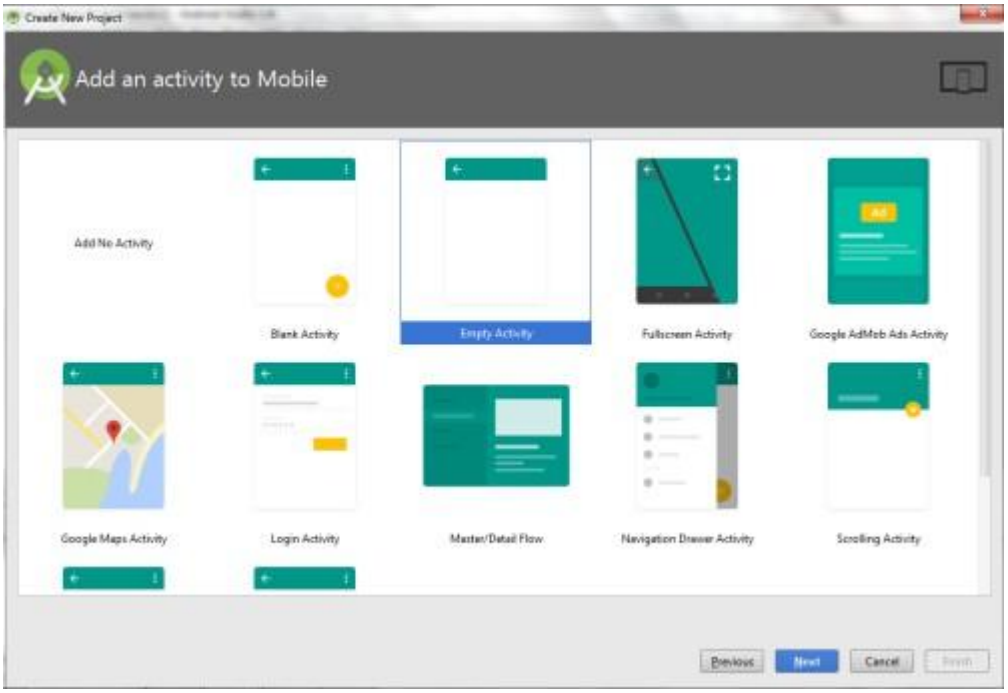
- Then type the Application name as "**ex.no.2"**and click **Next.**



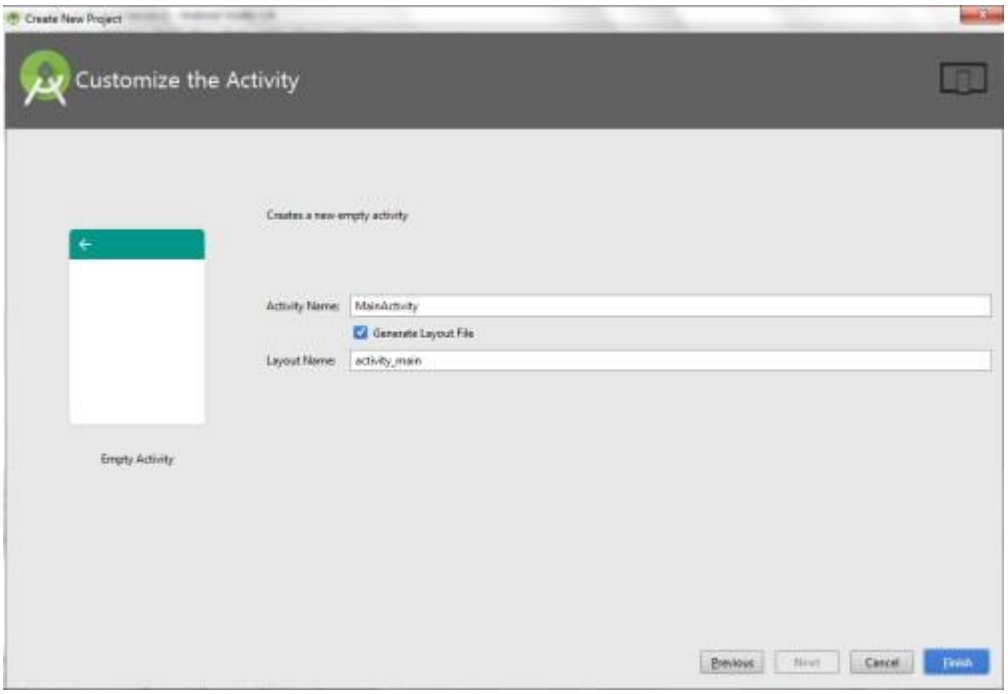
- Then select the **Minimum SDK** as shown below and click **Next**.



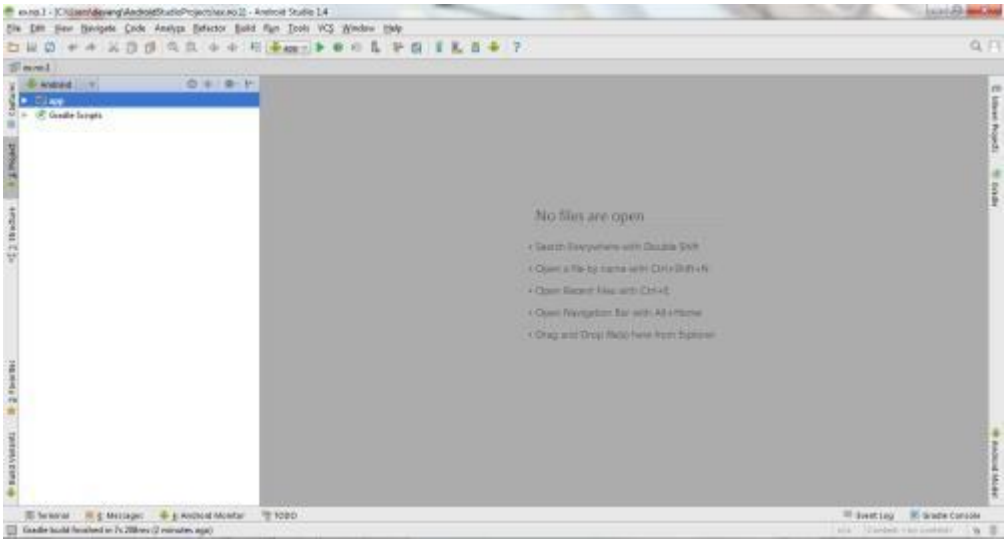
- Then select the **Empty Activity** and click **Next.**



- Finally click F**inish**.

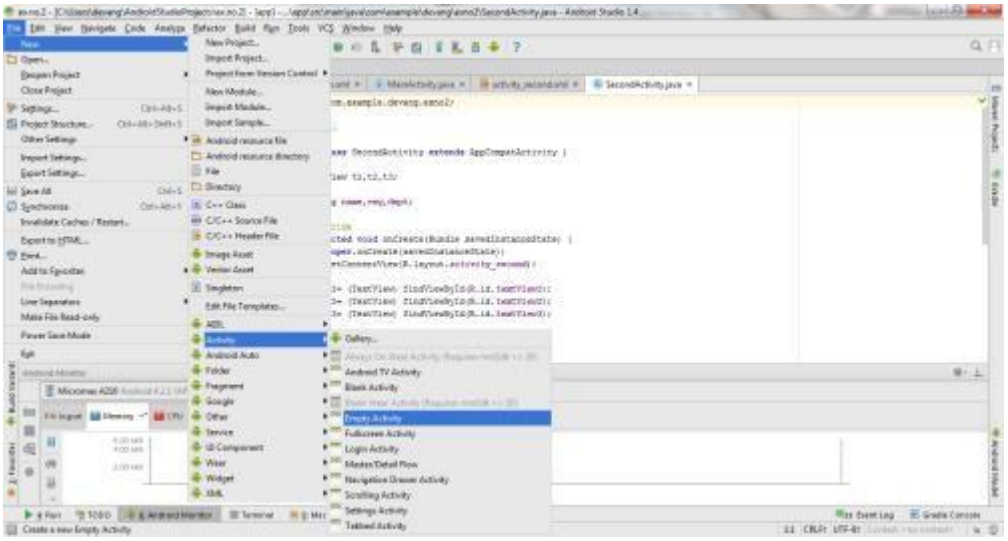


- It will take sometime to build and load the project.
- After completion it will look as given below.

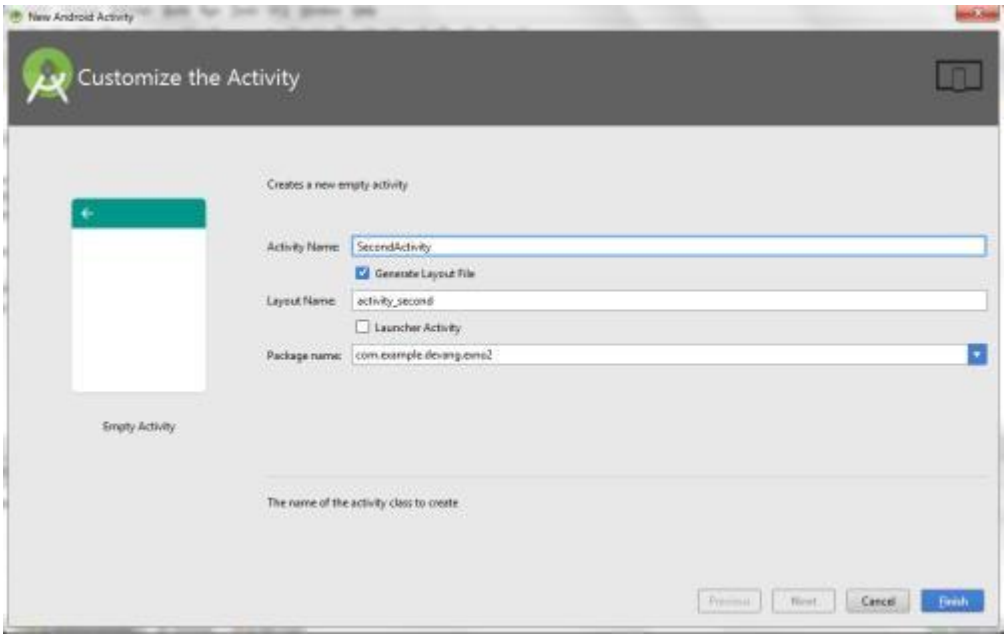


Creating Second Activity for the Android Application:

- Clickon **File->New ->Activity->Empty Activity.**



- Type the Activity Name as **Second Activity** and click **Finish** button.

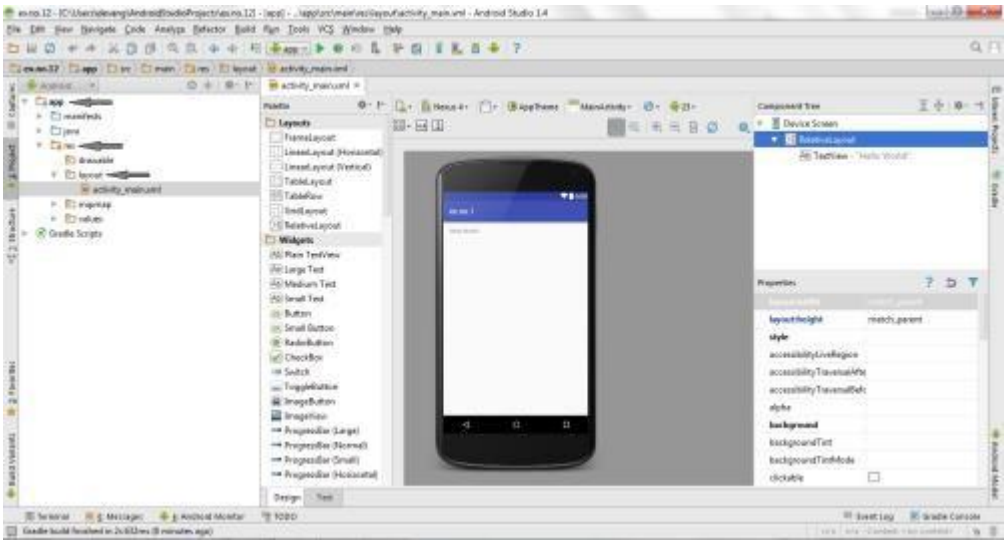


- Thus Second Activity For the application is created.

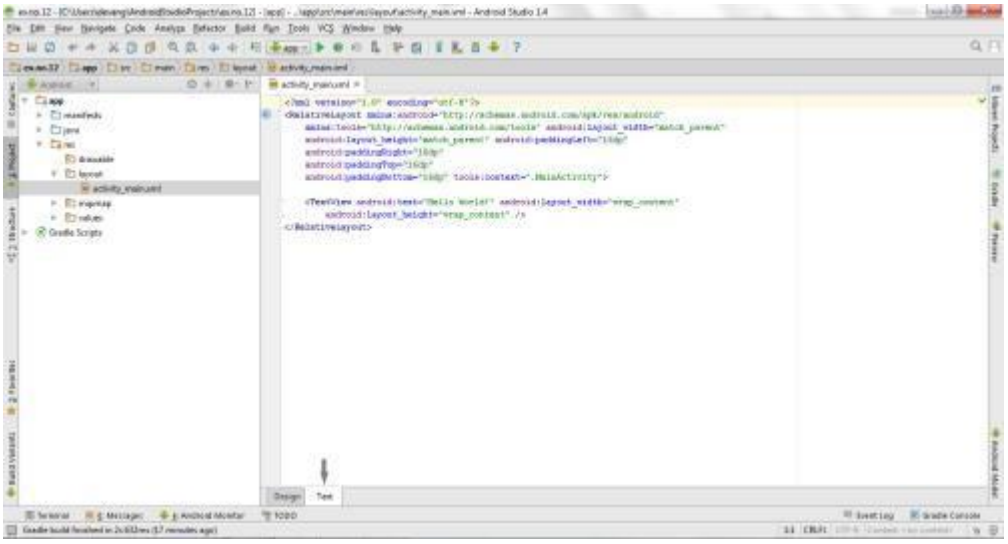
Designing layout for the Android Application:

Designing Layout for Main Activity:

- Click on **app->res-> layout->activity_main.xml.**



- Now click on **Text** as shown below.



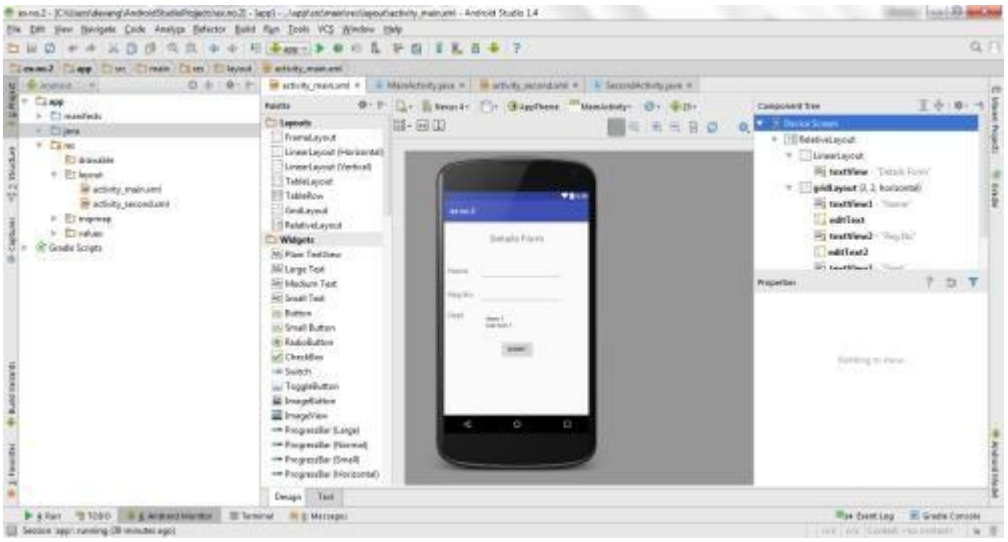
Then delete the code which is there and type the code as given below.

```
<?xmlversion="1.0"encoding="utf-8"?>
<RelativeLayoutxmlns:android="http://schemas.android.com/apk/res/android"x
    mlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="100dp">
        <TextView
            android:id="@+id/textView"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_margin="30dp"android:text
            ="Details Form"android:textSize="25sp"
            android:gravity="center"/>
```

```
</LinearLayout>

<GridLayout
    android:id="@+id/gridLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_marginTop="100dp"
    android:layout_marginBottom="200dp"
    android:columnCount="2"
    android:rowCount="3">
    <TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:layout_row="0"
        android:layout_column="0"
        android:text="Name"
        android:textSize="20sp"
        android:gravity="center"/>

    <EditText
        android:id="@+id/editText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:layout_row="0"
        android:layout_column="1"
        android:ems="10"/>

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:layout_row="1"
        android:layout_column="0"
        android:text="Reg.No"
        android:textSize="20sp"
        android:gravity="center"/>

    <EditText
        android:id="@+id/editText2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:layout_row="1"
        android:layout_column="1"
        android:inputType="number"
        android:ems="10"/>

    <TextView
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:layout_row="2"
        android:layout_column="0"
        android:text="Dept"
        android:textSize="20sp"
```

```
            android:gravity="center"/>

        <Spinner
            android:id="@+id/spinner"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_margin="10dp"
            android:layout_row="2"
            android:layout_column="1"
            android:spinnerMode="dropdown"/>

    </GridLayout>

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentBottom="true"
        android:layout_centerInParent="true"
        android:layout_marginBottom="150dp"
        android:text="Submit"/>

</RelativeLayout>
```

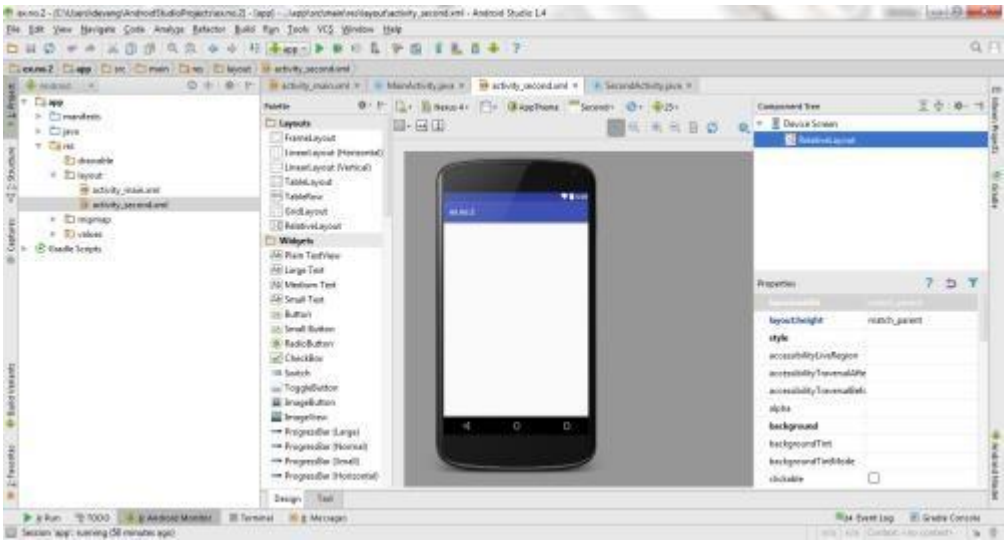
- Now click on Design and your activity will look as given below.



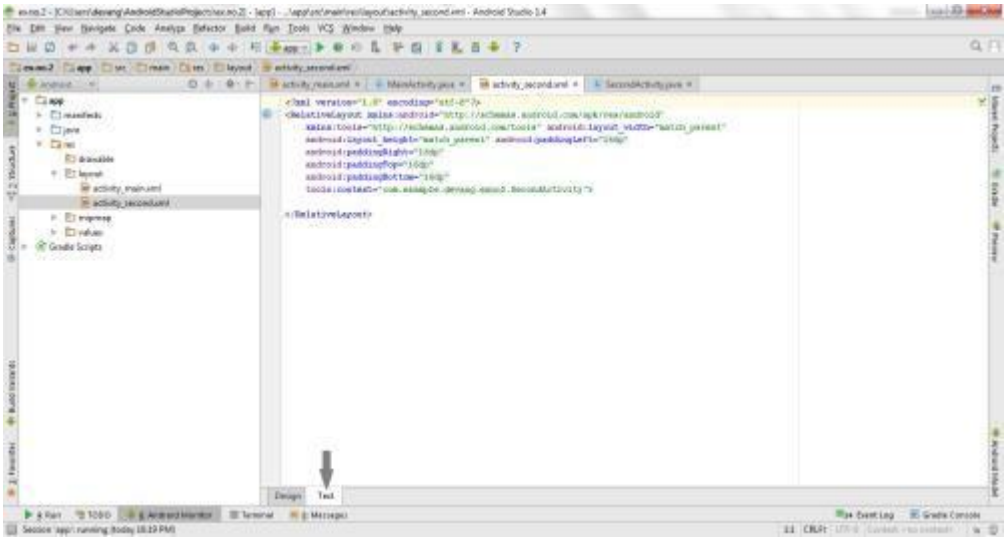
- So now the designing part of Main Activity is completed.

Designing Layout for Second Activity:

- Click on **app->res-> layout->activity_second.xml.**



- Now click on **Text** as shown below.

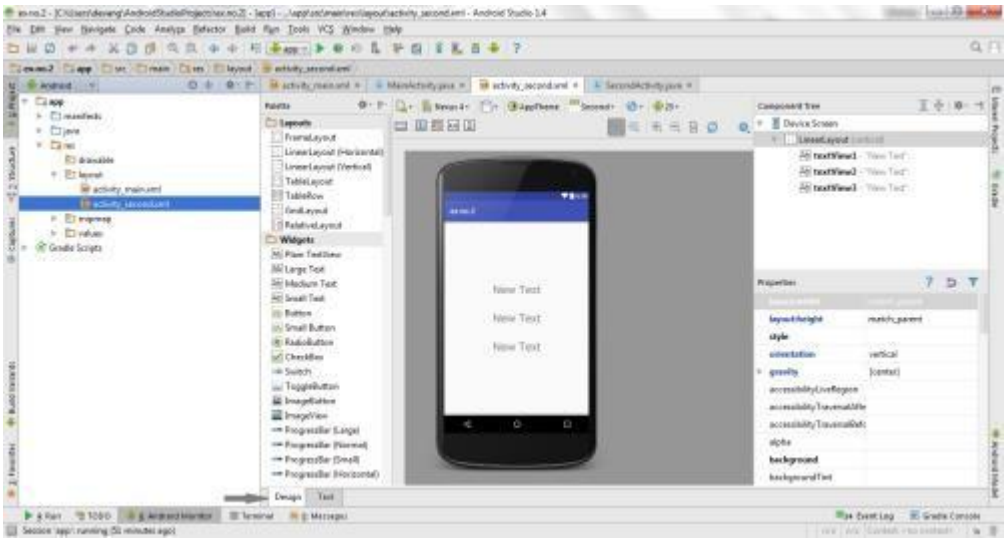


Then delete the code which is there and type the code as given below.

```
<?xmlversion="1.0"encoding="utf-8"?>
<LinearLayoutxmlns:android="http://schemas.android.com/apk/res/android"xml
    ns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context="com.example.devang.exno2.SecondActivity"
    android:orientation="vertical"
    android:gravity="center">

    <TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"android:text="Ne
        w Text"android:textSize="30sp"/>

    <TextView
        android:id="@+id/textView2"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"android:text="Ne
        w Text"android:textSize="30sp"/>

    <TextView
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"android:text="Ne
        w Text"android:textSize="30sp"/>

</LinearLayout>
```

- Now click on Design and your activity will look as given below.

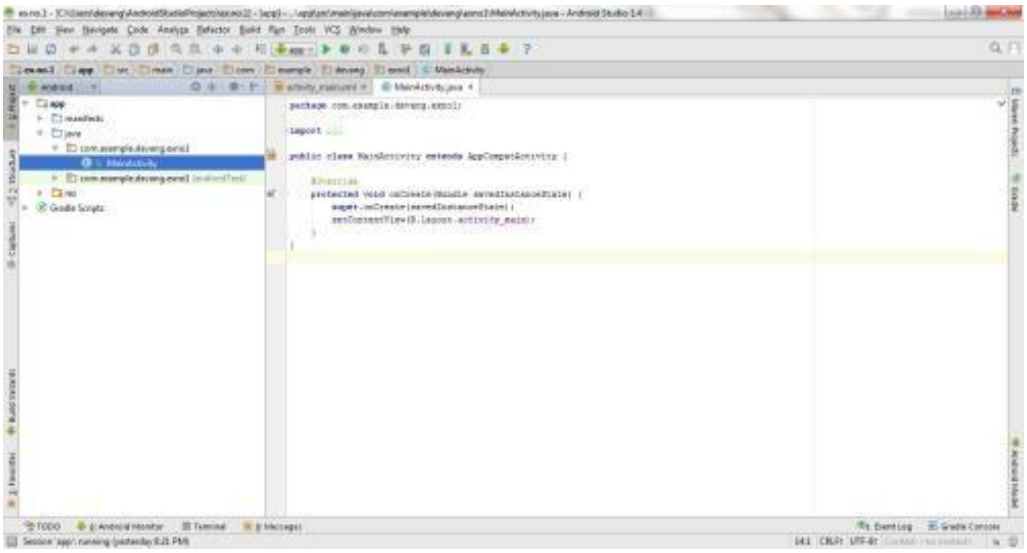


- So now the designing part of Second Activity is also completed.

Java Coding for the Android Application:

Java Coding for Main Activity:

- Click on **app->java->com.example.exno2 ->Main Activity.**



- Then delete the code which is there and type the code as given below.

Code forMainActivity.java:

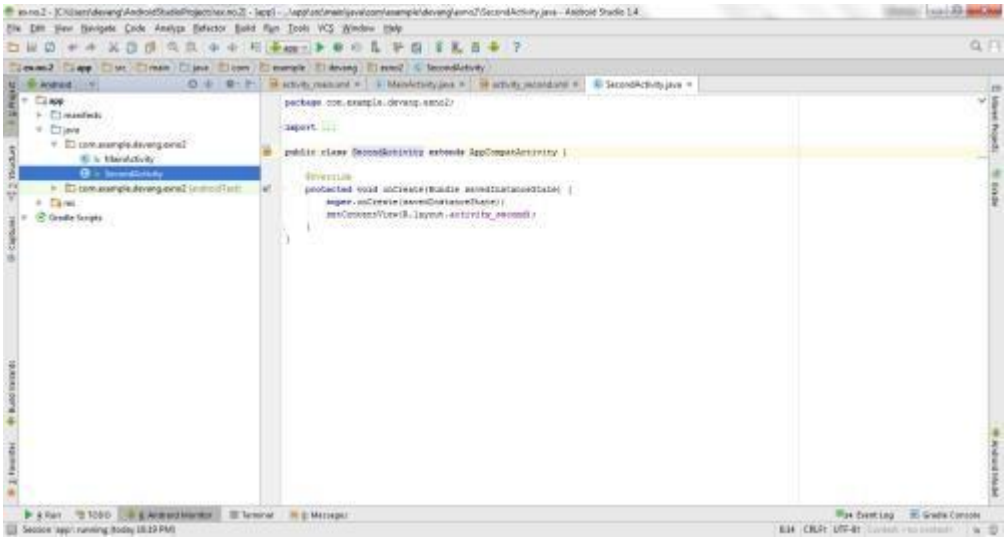
```
packagecom.example.exno2;

importandroid.content.Intent;
importandroid.support.v7.app.AppCompatActivity;
importandroid.os.Bundle;
importandroid.view.View;
importandroid.widget.ArrayAdapter;
importandroid.widget.Button;
importandroid.widget.EditText;
importandroid.widget.Spinner;

publicclassMainActivityextendsAppCompatActivity{

    //DefiningtheViews
    EditText e1,e2;
    Button bt;
    Spinners;

    //DataforpopulatinginSpinner
    String[]dept_array={"CSE","ECE","IT","Mech","Civil"};

    String name,reg,dept;

    @Override
    protectedvoidonCreate(BundlesavedInstanceState){
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        //ReferringtheViews
        e1= (EditText) findViewById(R.id.editText);
        e2=(EditText)findViewById(R.id.editText2);

        bt= (Button) findViewById(R.id.button);

        s=(Spinner)findViewById(R.id.spinner);
```

```
        //CreatingAdapterforSpinnerforadaptingthedatafromarrayto
Spinner
        ArrayAdapteradapter=new
ArrayAdapter(MainActivity.this,android.R.layout.simple_spinner_item,dept_ar
ray);
        s.setAdapter(adapter);

        //Creating Listener for Button
        bt.setOnClickListener(newView.OnClickListener(){
            @Override
            publicvoidonClick(Viewv){

                //GettingtheValuesfromViews(Edittext&Spinner)
                name=e1.getText().toString();
                reg=e2.getText().toString();
                dept=s.getSelectedItem().toString();

                //IntentForNavigatingtoSecondActivity
                Intent i = new
Intent(MainActivity.this,SecondActivity.class);

                //ForPassingtheValuestoSecondActivity
                i.putExtra("name_key", name);
                i.putExtra("reg_key",reg);
                i.putExtra("dept_key", dept);

                startActivity(i);

            }
        });
    }
}
```

So now the Coding part of Main Activity is completed.

Java Coding for Second Activity:

- Click on **app->java->com.example.exno2 ->Second Activity.**



- Then delete the code which is there and type the code as given below.

Code forSecondActivity.java:

```
packagecom.example.exno2;

importandroid.content.Intent;
importandroid.support.v7.app.AppCompatActivity;
importandroid.os.Bundle;
importandroid.widget.TextView;

publicclassSecondActivityextendsAppCompatActivity{

    TextView t1,t2,t3;

    Stringname,reg,dept;

    @Override
    protectedvoidonCreate(BundlesavedInstanceState){
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);

        t1=(TextView)findViewById(R.id.textView1);
        t2=(TextView)findViewById(R.id.textView2);
        t3=(TextView)findViewById(R.id.textView3);

        //Getting the Intent
        Intenti=getIntent();

        //GettingtheValuesfromFirstActivityusingtheIntentreceived
        name=i.getStringExtra("name_key");reg=i.getStringExtra("reg_key");
        dept=i.getStringExtra("dept_key");

        //SettingtheValuestoIntent
        t1.setText(name);
        t2.setText(reg);
        t3.setText(dept);

    }
}
```

- So now the Coding part of Second Activity is also completed.
- Now run the application to see the output.

RESULT:

Thus a Simple Android Application that uses Layout Managers and Event Listeners is developed and executed successfully.

EX. NO: 03

DATE:

SIMPLE ANDROID APPLICATION FOR NATIVE CALCULATOR

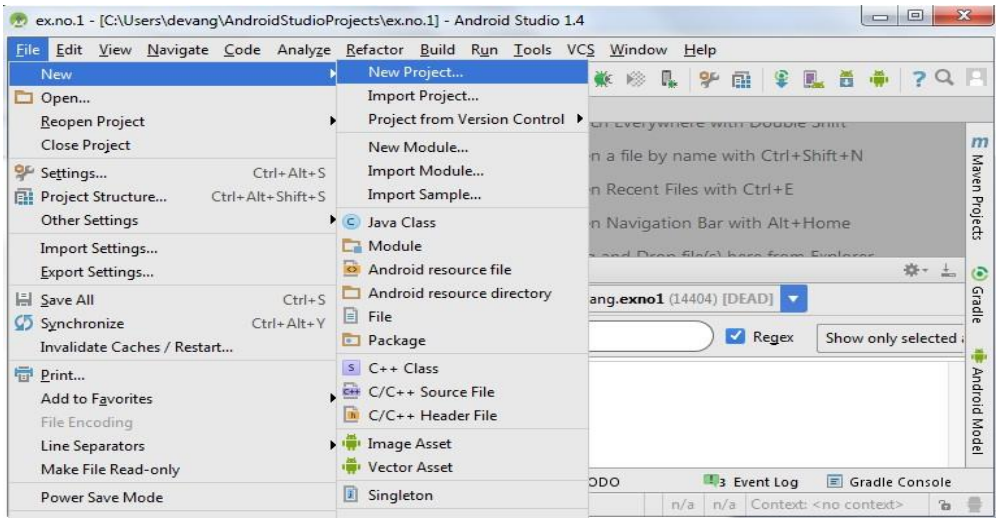
AIM:

To develop a Simple Android Application for Native Calculator.

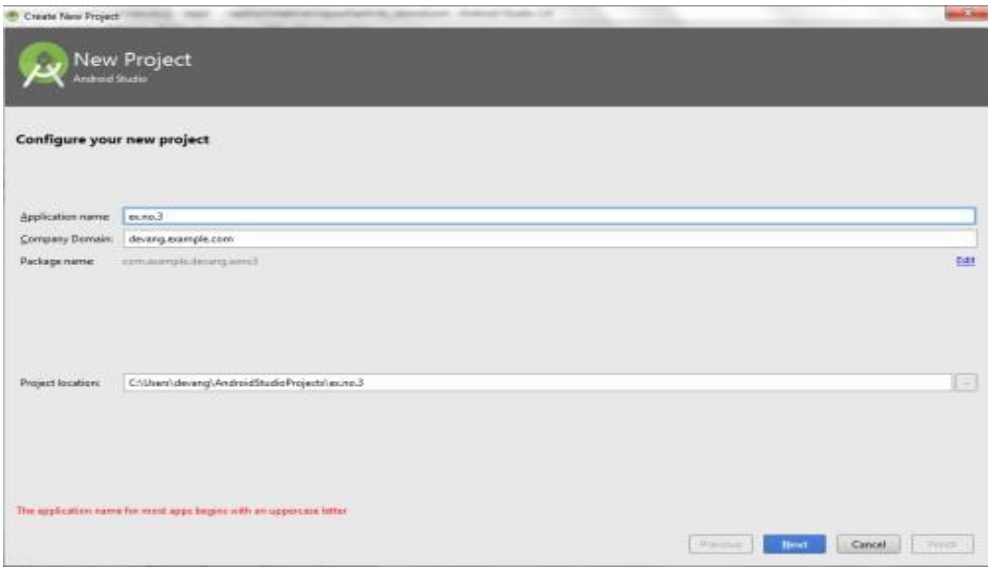
PROCEDURE:

Creating a New project:

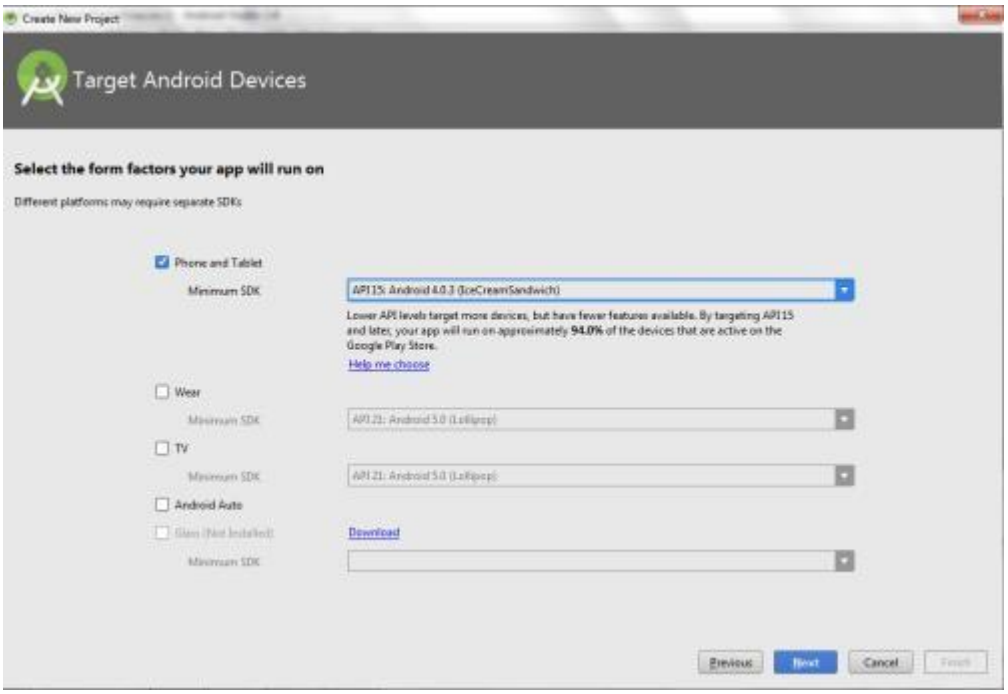
- Open Android Studio and then click on **File->New->New project.**



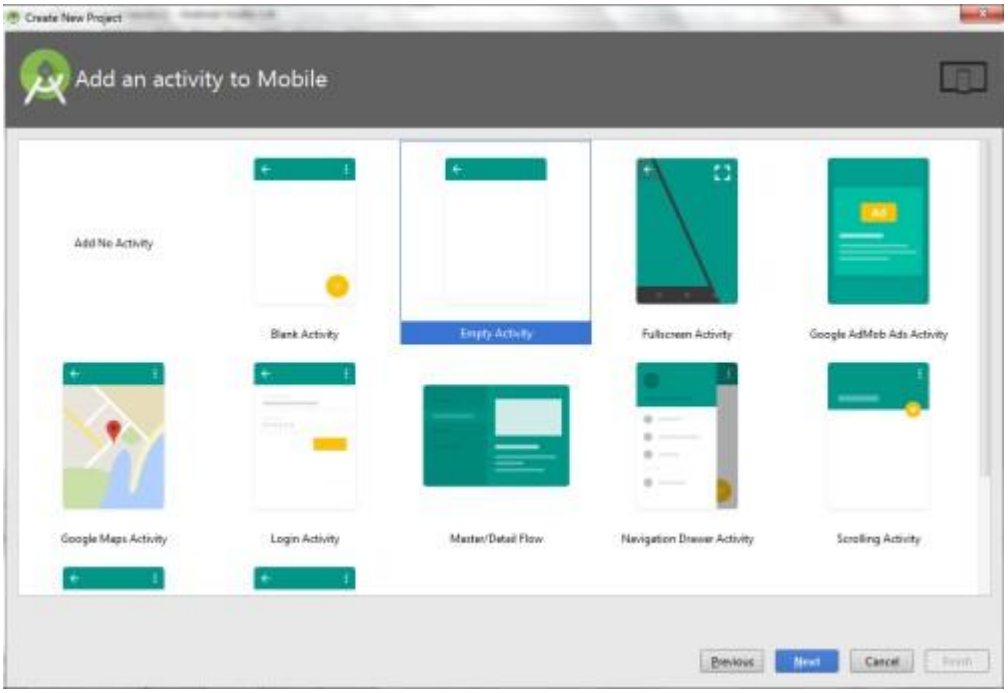
- Then type the Application name as "**ex.no.3"**and click **Next.**



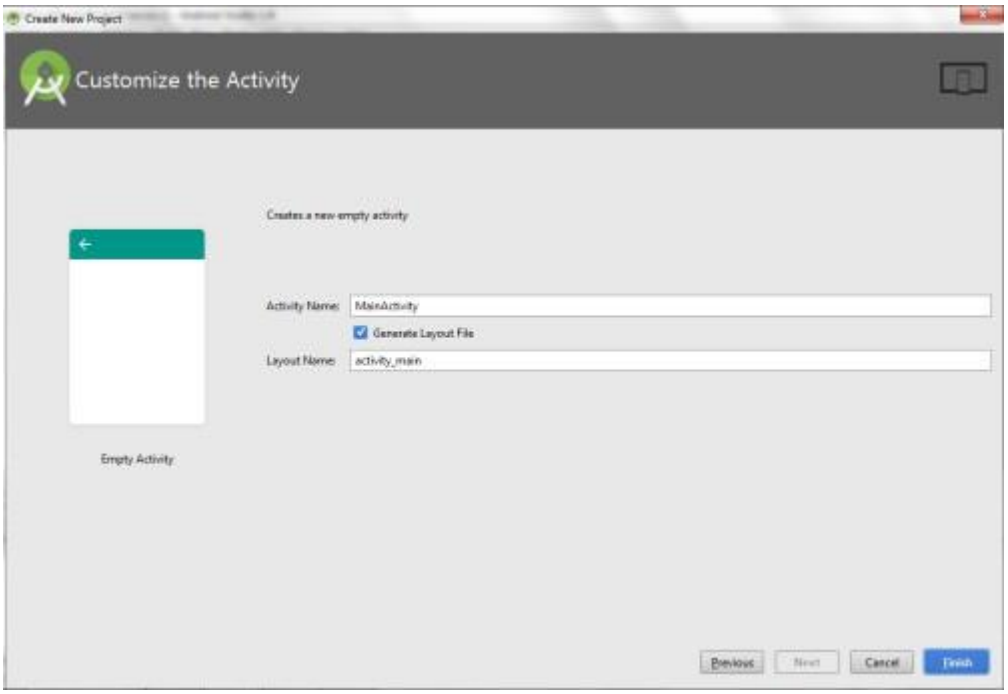
- Then select the **Minimum SDK** as shown below and click **Next**.



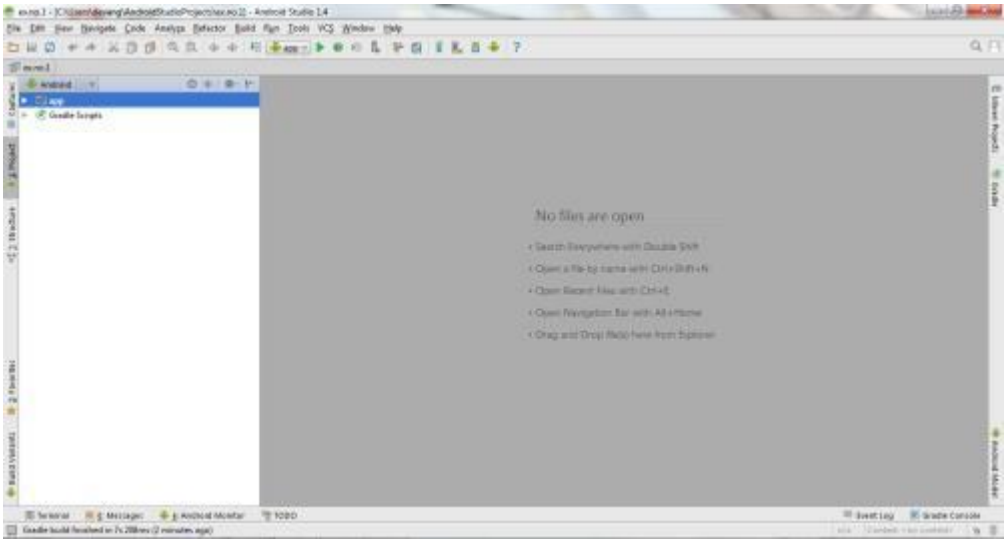
- Then select the **Empty Activity** and click **Next.**



- Finally click F**inish**.

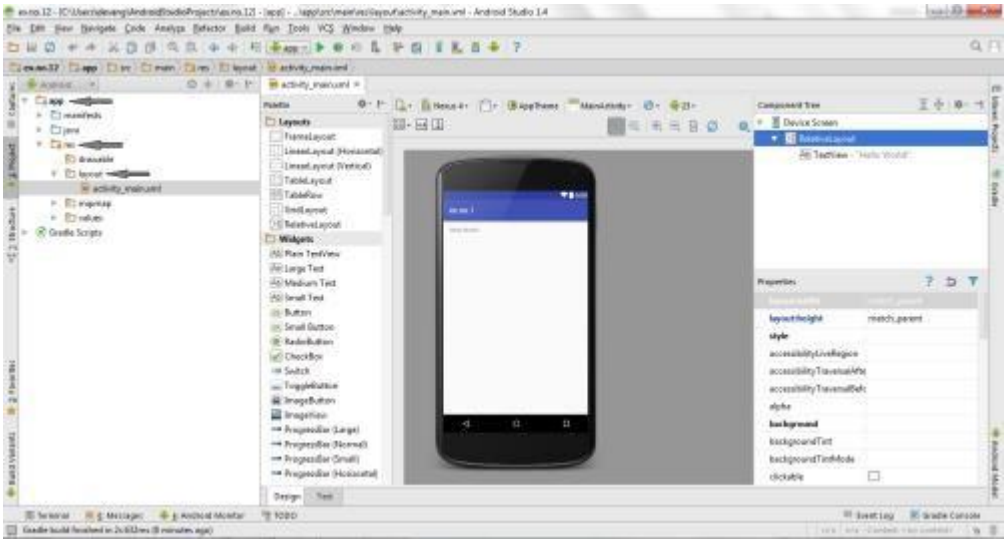


- It will take sometime to build and load the project.
- After completion it will look as given below

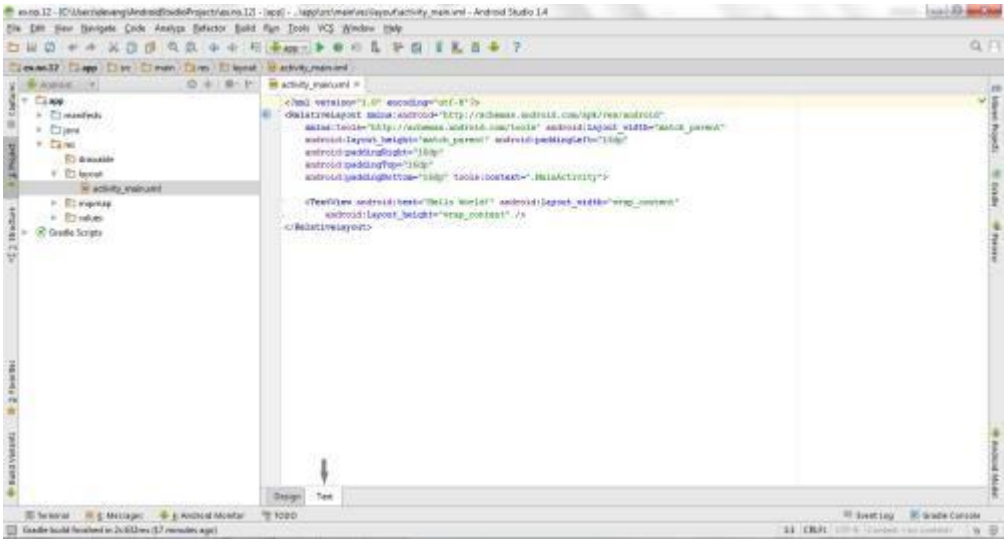


Designing layout for the Android Application:

- Clickon **app->res-> layout->activity_main.xml.**



- Now click on **Text** as shown below.



- Then delete the code which is there and type the code as given below

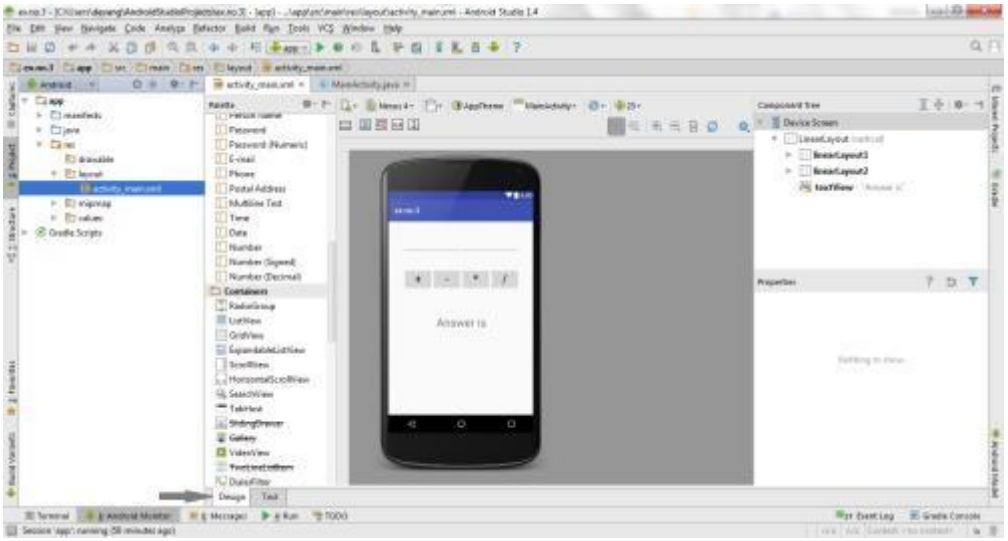
Code forActivity_main.xml:

```
<?xmlversion="1.0"encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="20dp">

    <LinearLayout
        android:id="@+id/linearLayout1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="20dp">

        <EditText
```

```
        android:id="@+id/editText1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:inputType="numberDecimal"android
        :textSize="20sp"/>

    <EditText
        android:id="@+id/editText2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:inputType="numberDecimal"android
        :textSize="20sp"/>

</LinearLayout>

<LinearLayout
    android:id="@+id/linearLayout2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="20dp">

    <Button
        android:id="@+id/Add"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="+"
        android:textSize="30sp"/>

    <Button
        android:id="@+id/Sub"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="-"
        android:textSize="30sp"/>

    <Button
        android:id="@+id/Mul"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="*"
        android:textSize="30sp"/>

    <Button
        android:id="@+id/Div"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="/"
        android:textSize="30sp"/>

</LinearLayout>

<TextView
    android:id="@+id/textView"
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="50dp"android:text=
        "Answer is"android:textSize="30sp"
        android:gravity="center"/>

</LinearLayout>
```

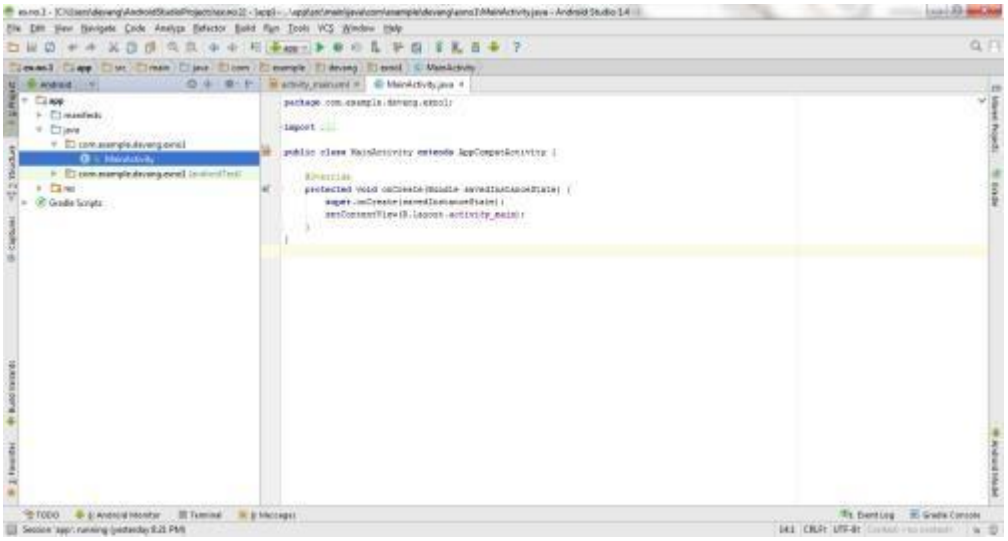
- Now click on Design and your application will look as given below.



- So now the designing part is completed.

Java Coding for the Android Application:

- Click on **app->java->com.example.exno3 ->Main Activity.**



- Then delete the code which is there and type the code as given below.

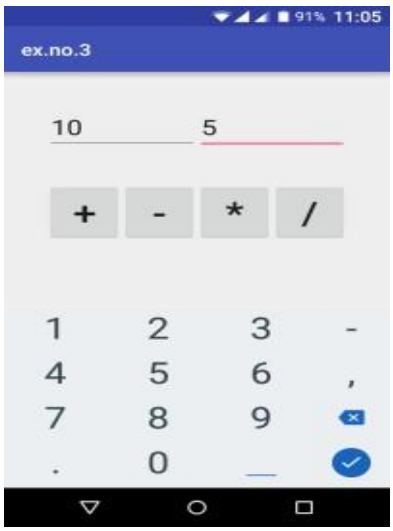
Code for MainActivity.java:

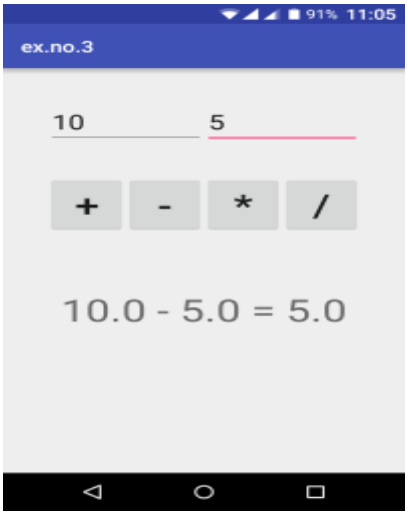
```
packagecom.example.devang.exno3;

importandroid.os.Bundle;
importandroid.support.v7.app.AppCompatActivity;
importandroid.text.TextUtils;
importandroid.view.View;
importandroid.view.View.OnClickListener;
importandroid.widget.Button;
importandroid.widget.EditText;
importandroid.widget.TextView;

publicclassMainActivityextendsAppCompatActivityimplements
OnClickListener
{
    //DefiningtheViews
    EditText Num1;
    EditText Num2;
    Button Add;
    ButtonSub;
    ButtonMul;
    ButtonDiv;
    TextViewResult;

    @Override
    publicvoidonCreate(BundlesavedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        //ReferringtheViews
        Num1=(EditText)findViewById(R.id.editText1);
        Num2=(EditText)findViewById(R.id.editText2);
        Add = (Button) findViewById(R.id.Add);
        Sub=(Button)findViewById(R.id.Sub);
        Mul=(Button)findViewById(R.id.Mul);
        Div=(Button)findViewById(R.id.Div);
        Result=(TextView)findViewById(R.id.textView);

        // set a listener
        Add.setOnClickListener(this);
        Sub.setOnClickListener(this);
        Mul.setOnClickListener(this);
        Div.setOnClickListener(this);
    }

    @Override
    publicvoidonClick(Viewv)
    {

        floatnum1 = 0;
        floatnum2 = 0;
        floatresult = 0;
        Stringoper="";

        //checkifthefieldsareempty
        if(TextUtils.isEmpty(Num1.getText().toString())||
TextUtils.isEmpty(Num2.getText().toString()))
```

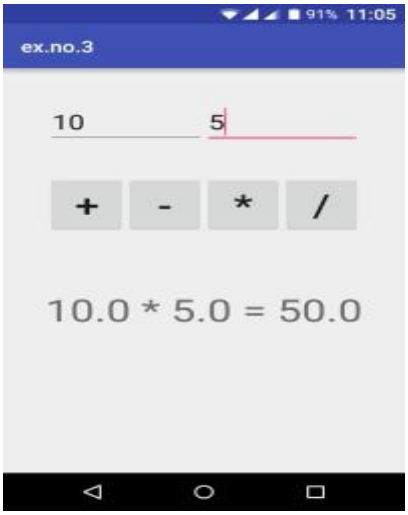
```
                return;

        // read EditText and fill variables with numbers
        num1=Float.parseFloat(Num1.getText().toString());
        num2=Float.parseFloat(Num2.getText().toString());

        //definesthebuttonthathasbeenclickedandperformsthe
corresponding operation









        //writeoperationintooper,wewilluseitlaterforoutput
        switch(v.getId())
        {
            caseR.id.Add:
                oper="+";
                result=num1+num2;
                break;
            caseR.id.Sub:
                oper="-";
                result=num1-num2;
                break;
            caseR.id.Mul:
                oper="*";
                result=num1*num2;
                break;
            caseR.id.Div:
                oper="/";
                result=num1/num2;
                break;
            default:
                break;
        }
        //formtheoutput line
        Result.setText(num1+""+oper+""+num2+"="+result);
    }
}
```

- So now the Coding part is also completed.
- Now run the application to see the output.

SAMPLEOUTPUT:







RESULT:

Thus a Simple Android Application for Native Calculator is developed and executed successfully.

EX. NO: 04

DATE:

ANDROID APPLICATION TO DRAW BASIC GRAPHICAL PRIMITIVES

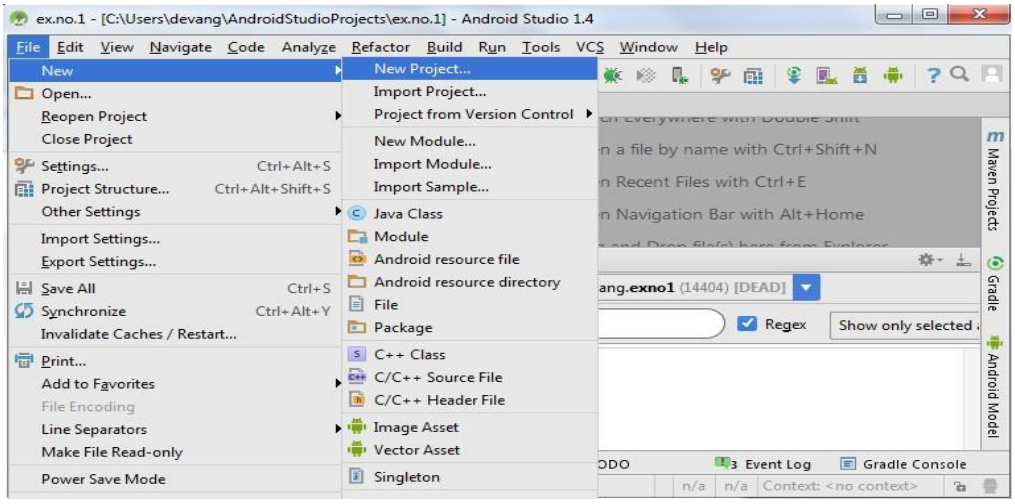
AIM:

To develop a Simple Android Application that draws basic Graphical Primitives on the screen.

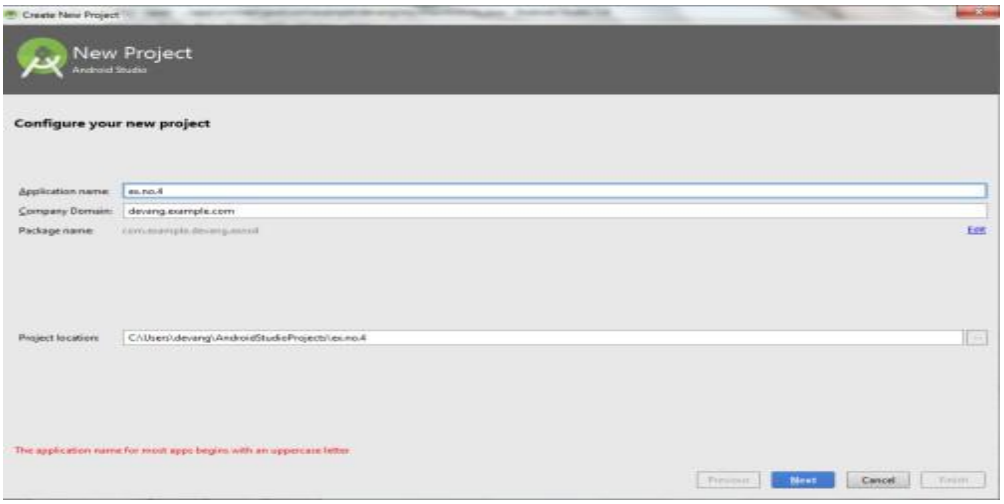
PROCEDURE:

Creating a New project:

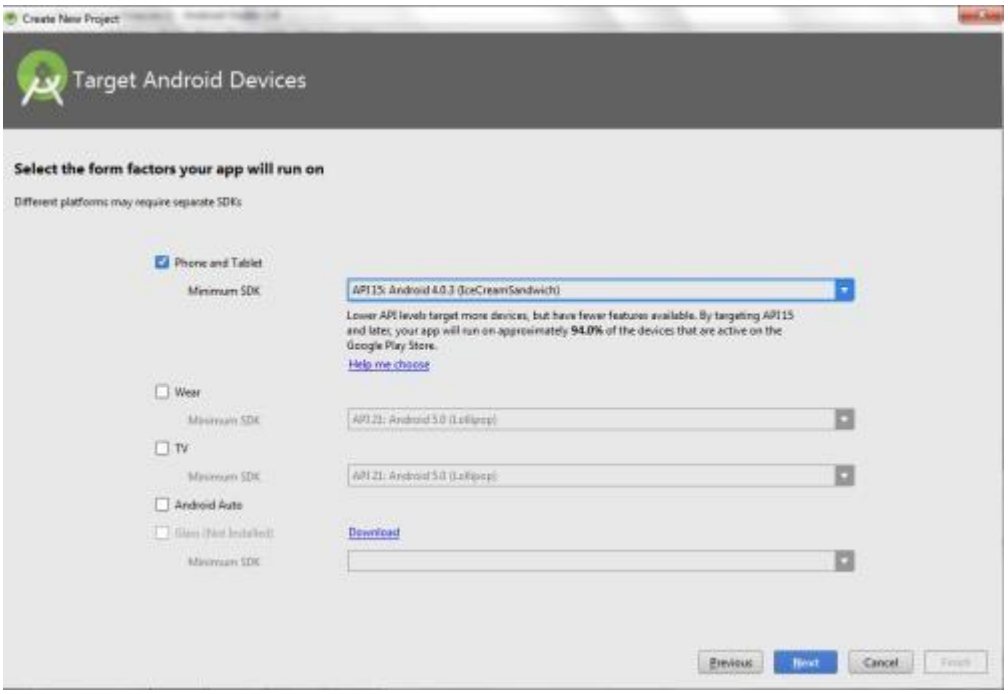
- Open Android Studio and then click on **File->New->New project.**



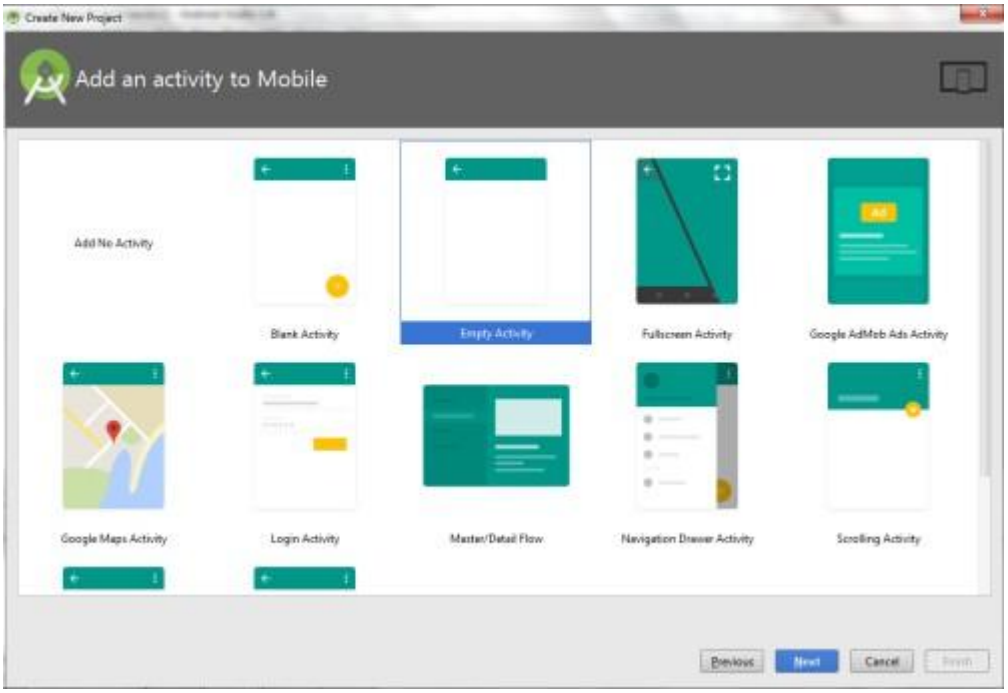
- Then type the Application name as "**ex.no.4"**and click **Next.**



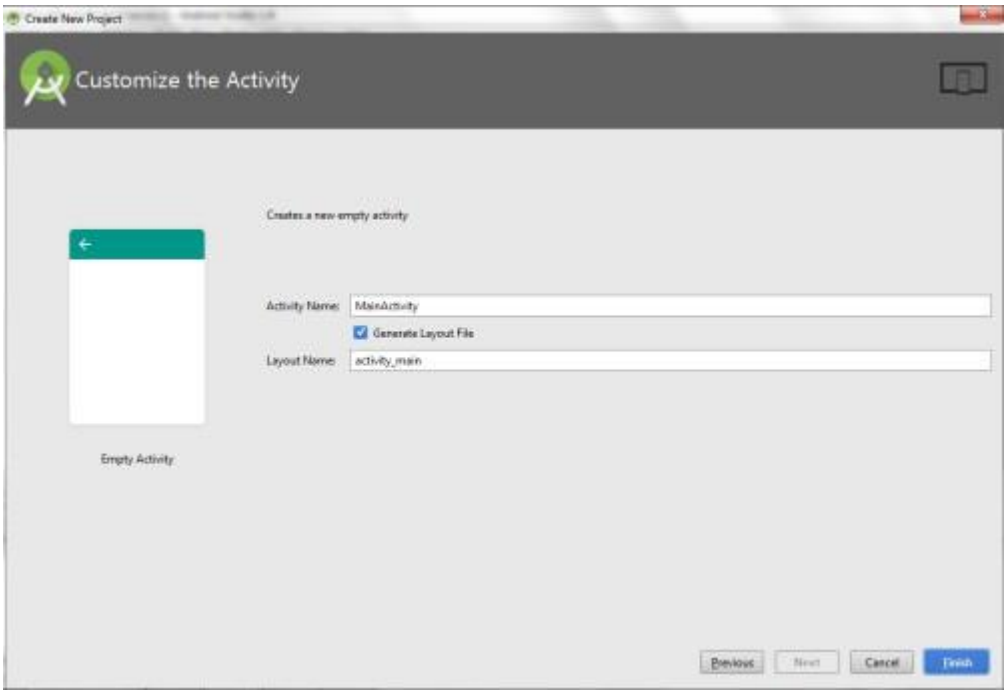
- Then select the **Minimum SDK** as shown below and click **Next**.



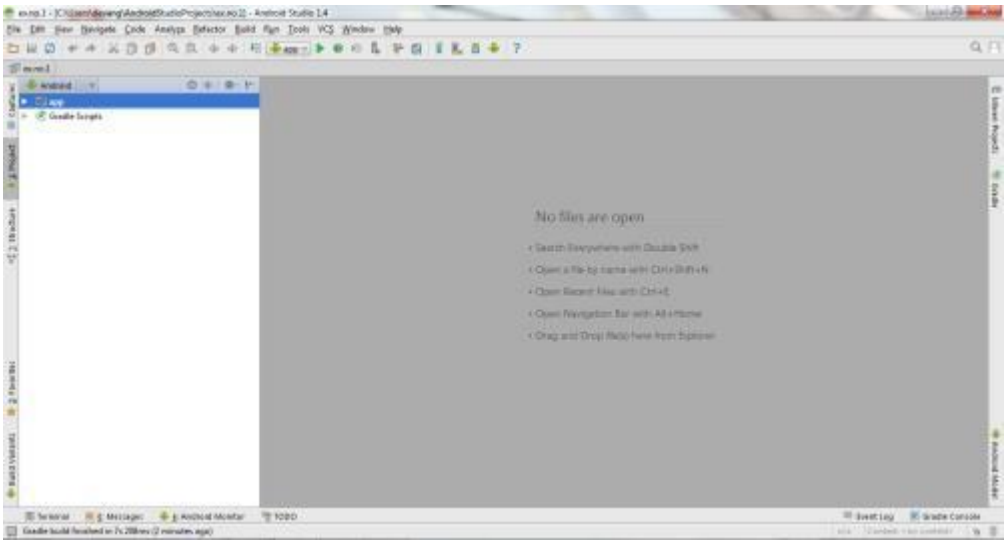
- Then select the **Empty Activity** and click **Next.**



- Finally click **Finish**.

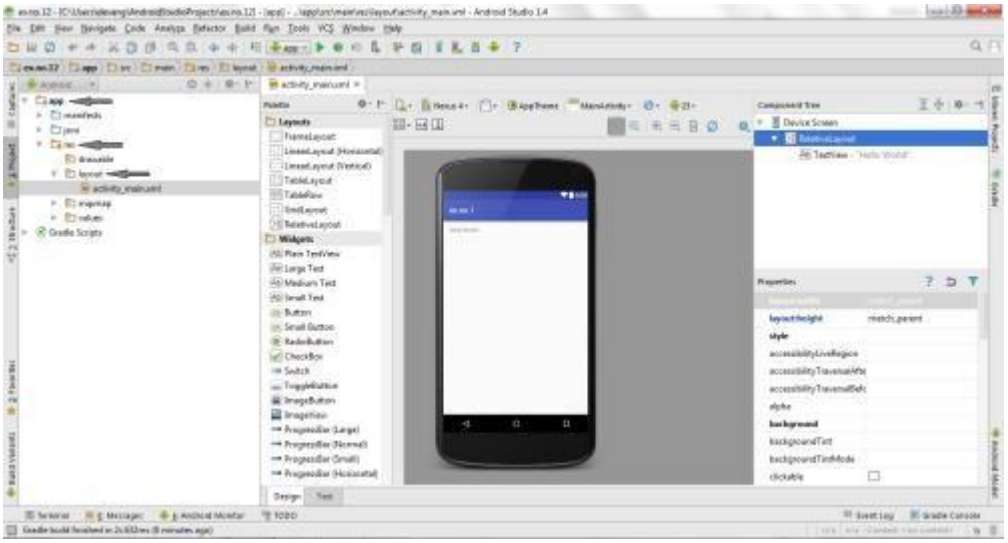


- It will take some time to build and load the project.
- After completion it will look as given below.

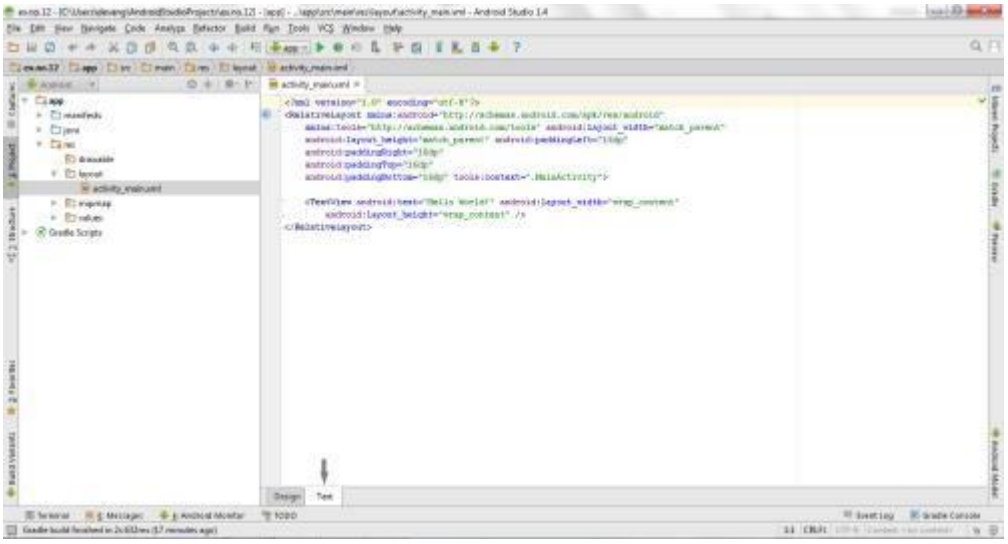


Designing layout for the Android Application:

- Click on **app->res-> layout->activity_main.xml.**



- Now click on **Text** as shown below.

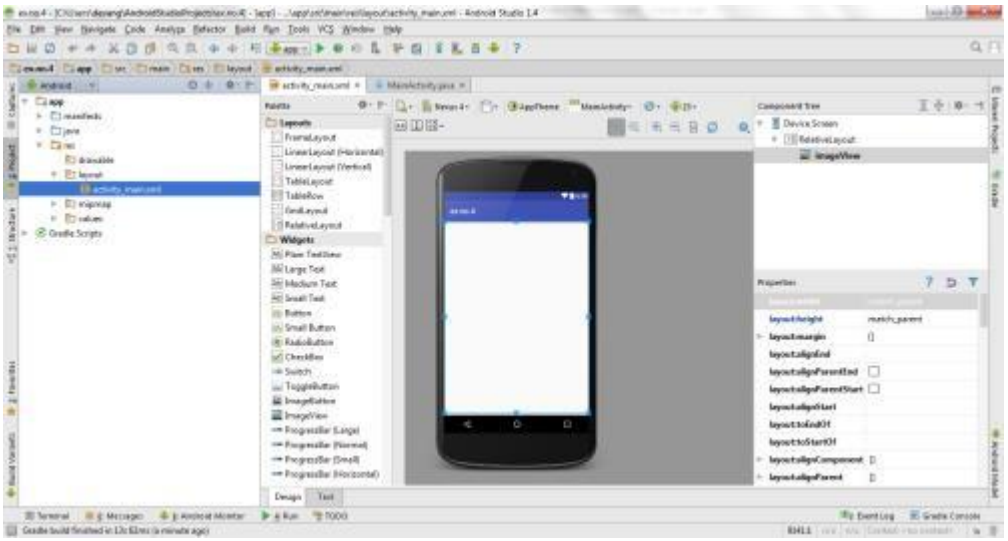


- Then delete the code which is there and type the code as given below.

Code forActivity_main.xml:

```
<?xmlversion="1.0"encoding="utf-8"?>
<RelativeLayoutxmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <ImageView
        android:layout_width="match_parent"
        android:layout_height="match_parent"android:id="@+id/imageView
        "/>
</RelativeLayout>
```

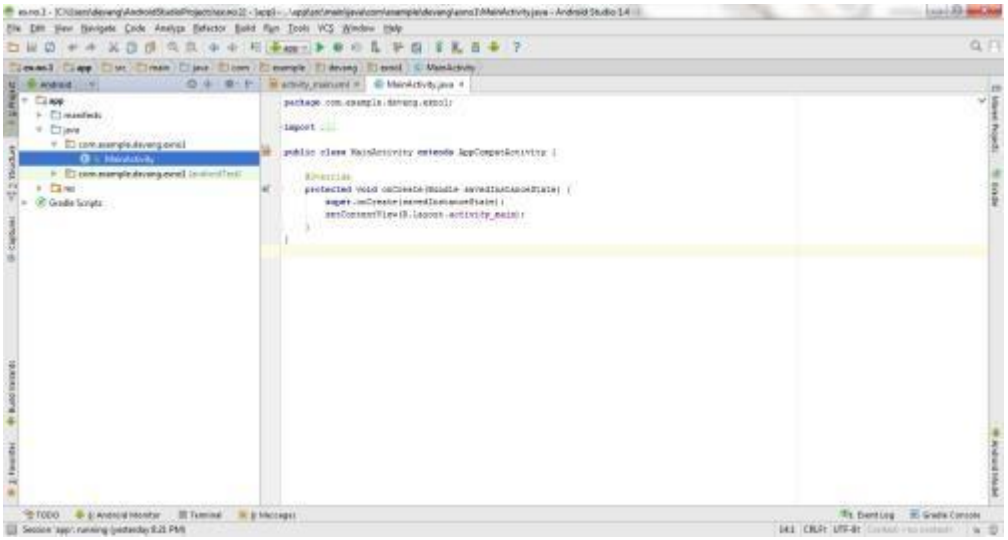
- Now click on **Design** and your application will look as given below.



- So now the designing part is completed.

Java Coding for the Android Application:

- Click on **app->java->com.example.exno4 ->Main Activity.**



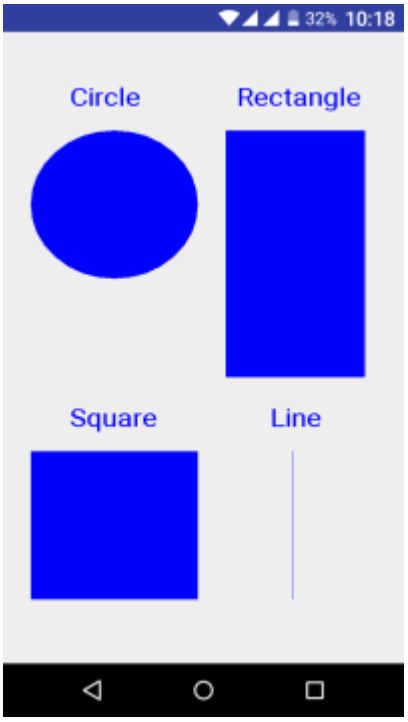
- Then delete the code which is there and type the code as given below.

Code for Main Activity.java:

```
packagecom.example.exno4;

importandroid.app.Activity;
importandroid.graphics.Bitmap;
importandroid.graphics.Canvas;
importandroid.graphics.Color;
importandroid.graphics.Paint;
importandroid.graphics.drawable.BitmapDrawable;
importandroid.os.Bundle;
importandroid.widget.ImageView;
```

```
publicclassMainActivityextendsActivity
{
    @Override
    publicvoidonCreate(BundlesavedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        //CreatingaBitmap
        Bitmapbg=Bitmap.createBitmap(720,1280,
Bitmap.Config.ARGB_8888);

        //Setting the Bitmap as background for the ImageView
        ImageViewi=(ImageView)findViewById(R.id.imageView);
        i.setBackgroundDrawable(newBitmapDrawable(bg));

        //Creating the Canvas Object
        Canvascanvas=newCanvas(bg);

        //CreatingthePaintObjectandsetitscolor&TextSize Paint paint
        = newPaint();
        paint.setColor(Color.BLUE);
        paint.setTextSize(50);

        //To draw a Rectangle
        canvas.drawText("Rectangle",420,150,paint);
        canvas.drawRect(400,200,650,700,paint);

        //To draw a Circle
        canvas.drawText("Circle",120,150,paint);
        canvas.drawCircle(200,350,150,paint);

        //To draw a Square
        canvas.drawText("Square",120,800,paint);
        canvas.drawRect(50,850,350,1150,paint);

        //TodrawaLine
        canvas.drawText("Line",480,800,paint);
        canvas.drawLine(520,850,520,1150,paint);
    }
}
```

SAMPLE OUTPUT:



RESULT:

Thus a Simple Android Application that draws basic Graphical Primitives on the screen is developed and executed successfully.

EX. NO.05

DATE:

SIMPLE ANDROID APPLICATION THAT MAKESUSE OF DATABASE.

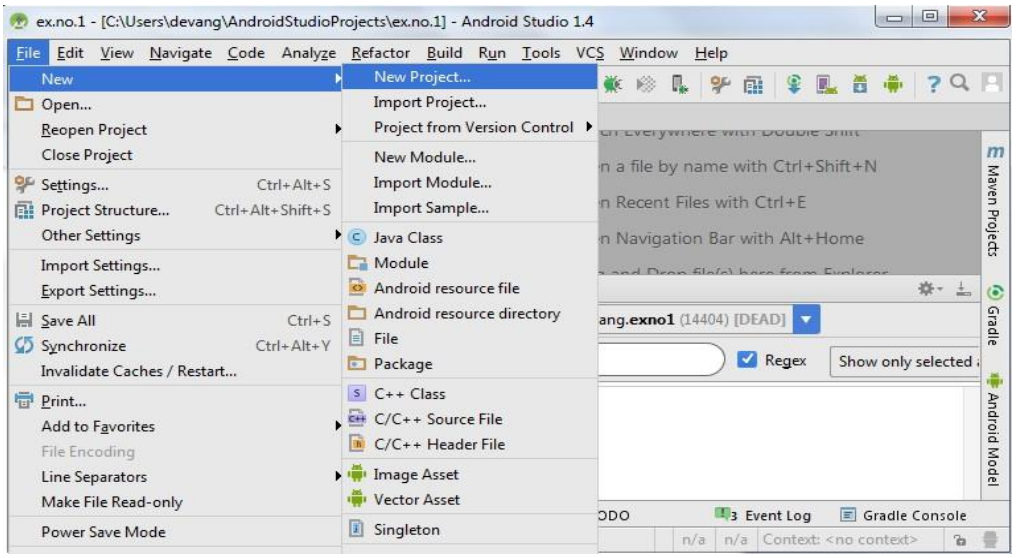
AIM:

To develop a Simple Android Application that makes use of Database.

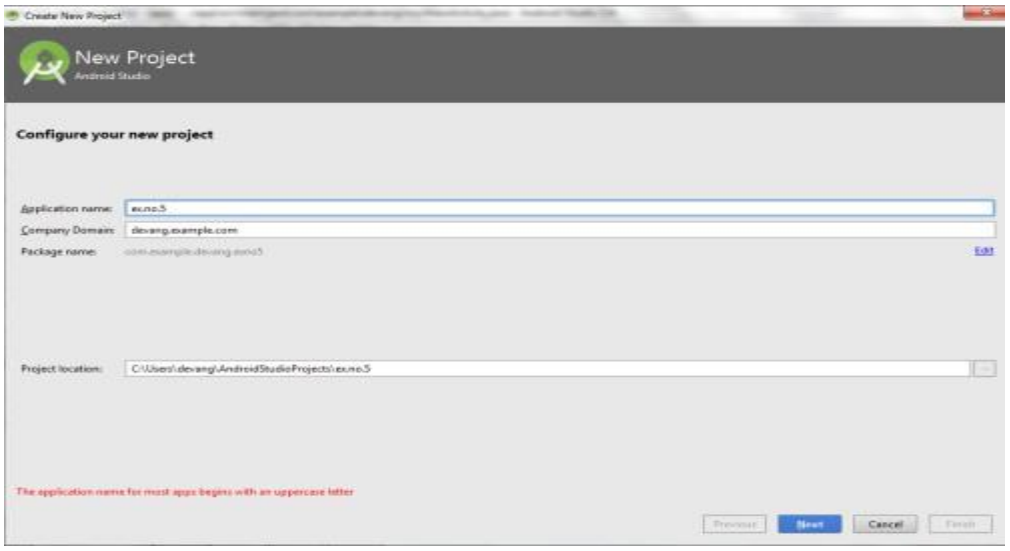
PROCEDURE:

Creating a New project:

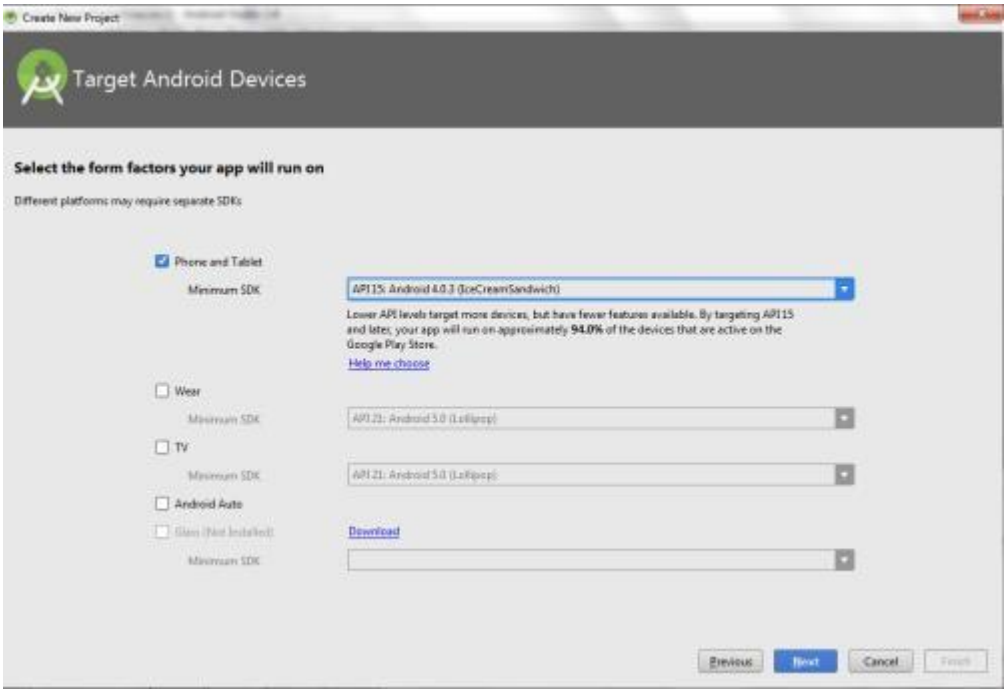
- Open Android Studio and then click on **File->New->New project.**



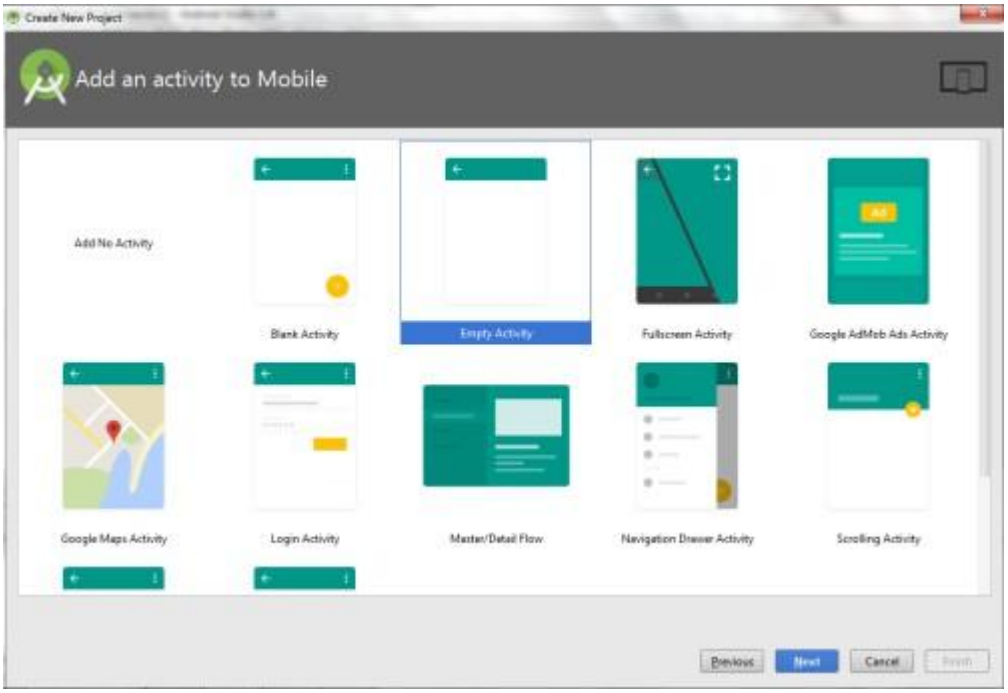
- Then type the Application name as“**ex.no.5″**and click



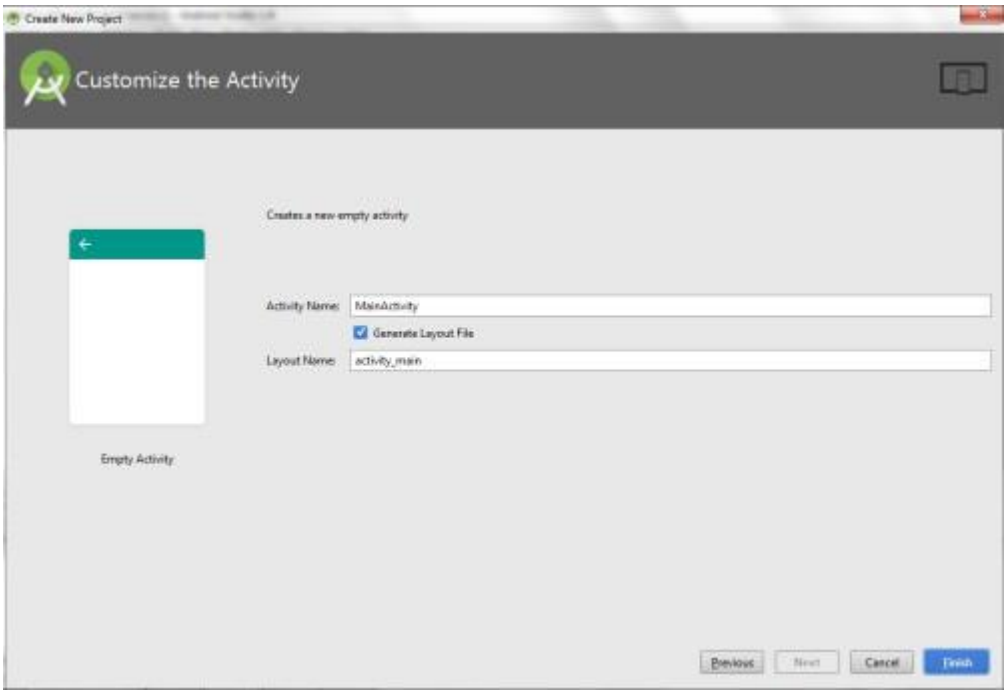
- Then select the **Minimum SDK** as shown below and click **Next**.



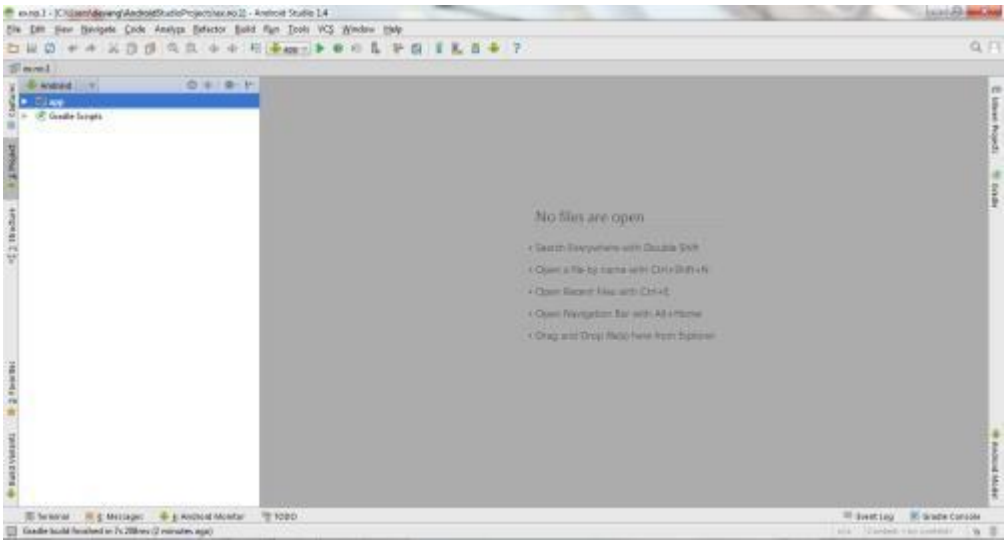
- Then select the **Empty Activity** and click **Next.**



- Finally click F**inish**.

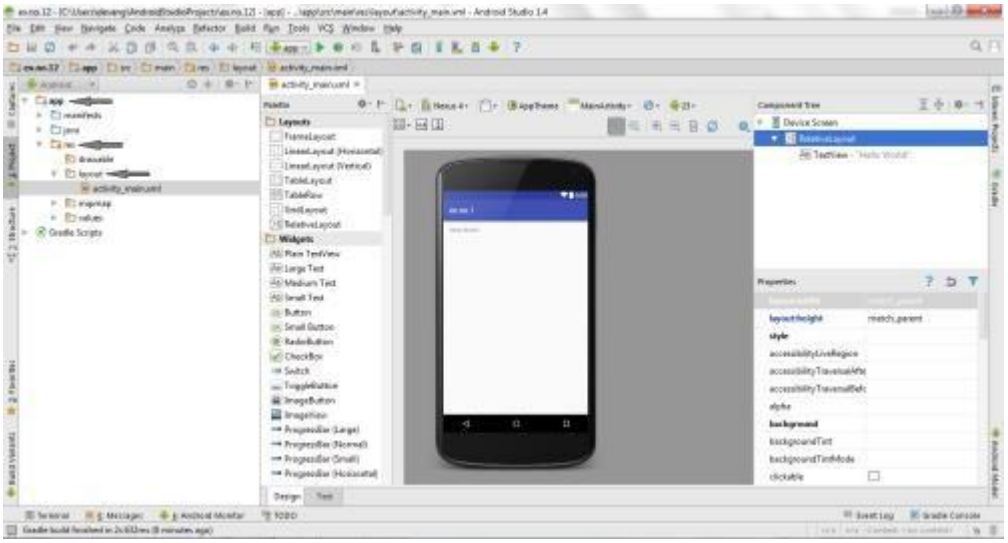


- It will take sometime to build and load the project.
- After completion it will look as given below.

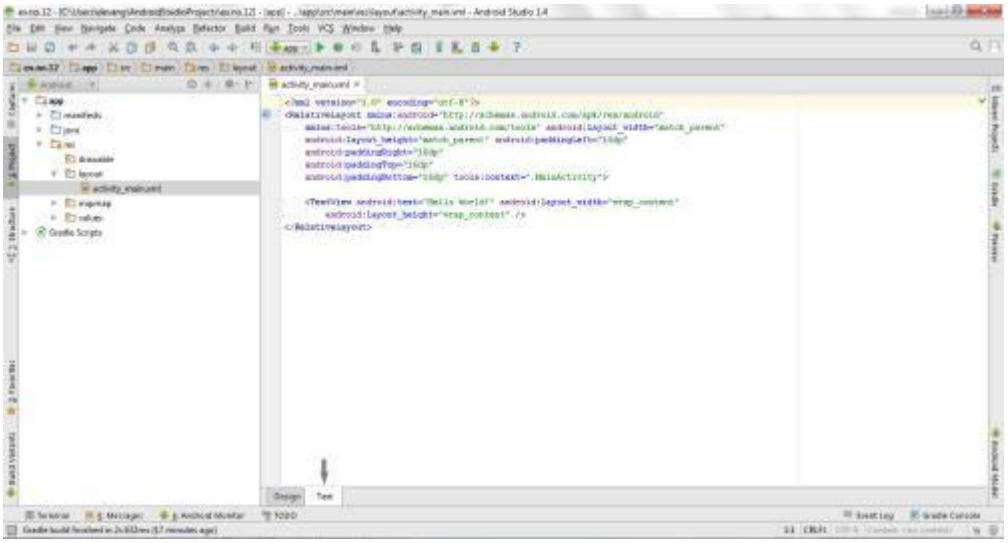


Designing layout for the Android Application:

- Click on **app->res-> layout->activity_main.xml.**



- Now click on **Text** as shown below.



- Then delete the code which is there and type the code as given below

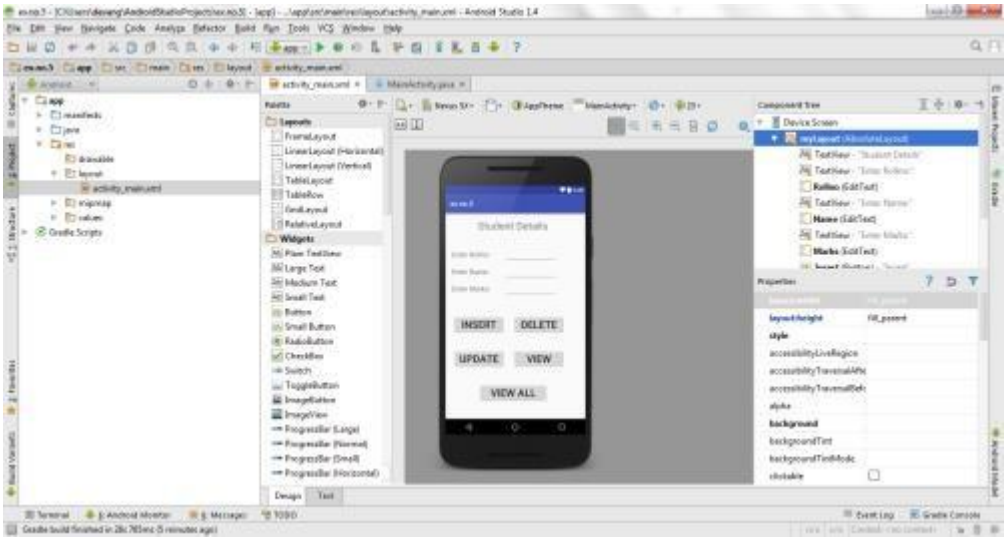
Code for Activity_main.xml:

```
<?xmlversion="1.0"encoding="utf-8"?>
<AbsoluteLayoutxmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="50dp"
        android:layout_y="20dp"android:text=
        "Student Details"
        android:textSize="30sp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="20dp"





        android:layout_y="110dp"android:text
        ="EnterRollno:"
        android:textSize="20sp"/>

    <EditText
        android:id="@+id/Rollno"
        android:layout_width="150dp"
        android:layout_height="wrap_content"
        android:layout_x="175dp"
        android:layout_y="100dp"
        android:inputType="number"android:textSize="
        20sp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="20dp"
        android:layout_y="160dp"android:text
        ="Enter Name:"
        android:textSize="20sp"/>

    <EditText
        android:id="@+id/Name"
        android:layout_width="150dp"
        android:layout_height="wrap_content"
        android:layout_x="175dp"
        android:layout_y="150dp"
        android:inputType="text"android:textSize="20
        sp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="20dp"
        android:layout_y="210dp"android:text
        ="Enter Marks:"
        android:textSize="20sp"/>
```

```
<EditText
    android:id="@+id/Marks"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="175dp"
    android:layout_y="200dp"
    android:inputType="number"android:textSize="
    20sp"/>

<Button
    android:id="@+id/Insert"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="25dp"
    android:layout_y="300dp"
    android:text="Insert"android:textSiz
    e="30dp"/>

<Button
    android:id="@+id/Delete"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="200dp"
    android:layout_y="300dp"





    android:text="Delete"
    android:textSize="30dp"/>

<Button
    android:id="@+id/Update"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="25dp"
    android:layout_y="400dp"
    android:text="Update"android:textSiz
    e="30dp"/>

<Button
    android:id="@+id/View"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="200dp"
    android:layout_y="400dp"
    android:text="View"android:textSize=
    "30dp"/>

<Button
    android:id="@+id/ViewAll"
    android:layout_width="200dp"
    android:layout_height="wrap_content"
    android:layout_x="100dp"
    android:layout_y="500dp"android:text
    ="View All"
    android:textSize="30dp"/>
```

```
</AbsoluteLayout>
```

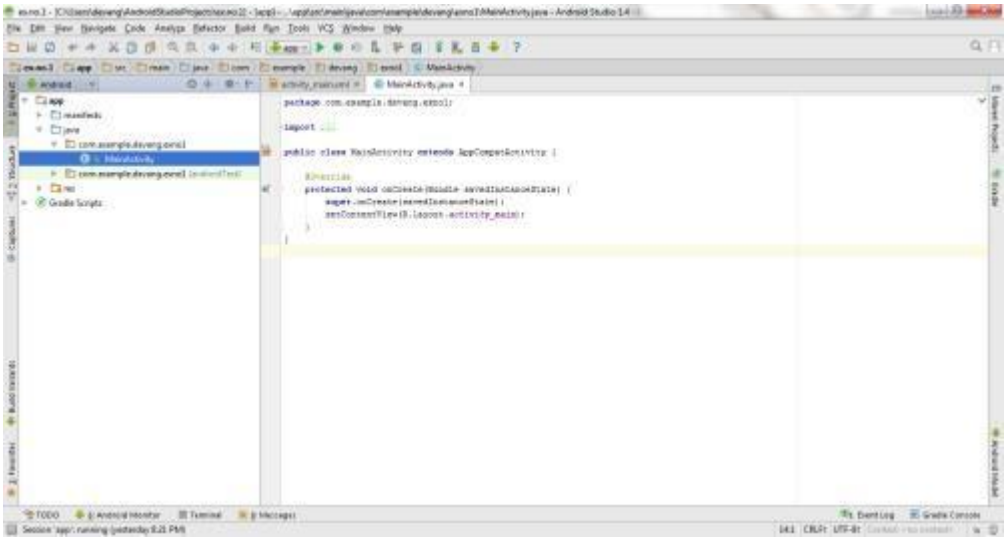
- Now click on **Design** and your application will look as given below.



- So now the designing part is completed.

Java Coding for the Android Application:

- Click on **app->java->com.example.exno5 ->Main Activity.**



- Then delete the code which is there and type the code as given below.

Code for MainActivity.java:

```
packagecom.example.exno5;

importandroid.app.Activity;
importandroid.app.AlertDialog.Builder;
importandroid.content.Context;
importandroid.database.Cursor;
importandroid.database.sqlite.SQLiteDatabase;
importandroid.os.Bundle;
importandroid.view.View;
importandroid.view.View.OnClickListener;
importandroid.widget.Button;
importandroid.widget.EditText;

publicclassMainActivityextendsActivityimplementsOnClickListener
{
    EditTextRollno,Name,Marks;
    ButtonInsert,Delete,Update,View,ViewAll;
    SQLiteDatabase db;
    /**Calledwhentheactivityisfirstcreated.*/ @Override
    publicvoidonCreate(BundlesavedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Rollno=(EditText)findViewById(R.id.Rollno);
        Name=(EditText)findViewById(R.id.Name);
        Marks=(EiditText)findViewById(R.id.Marks);
        Insert=(Button)findViewById(R.id.Insert);
        Delete=(Button)findViewById(R.id.Delete);
        Update=(Button)findViewById(R.id.Update);
        View=(Button)findViewById(R.id.View);
        ViewAll=(Button)findViewById(R.id.ViewAll);

        Insert.setOnClickListener(this);
        Delete.setOnClickListener(this);
        Update.setOnClickListener(this);





        View.setOnClickListener(this);
        ViewAll.setOnClickListener(this);

        // Creating database and table
        db=openOrCreateDatabase("StudentDB", Context.MODE_PRIVATE, null);
        db.execSQL("CREATETABLEIFNOTEXISTSstudent(rollnoVARCHAR,name
VARCHAR,marksVARCHAR);");
    }
    publicvoidonClick(Viewview)
    {
        //InsertingarecordtotheStudenttable
        if(view==Insert)
        {
            //Checkingforemptyfields
```

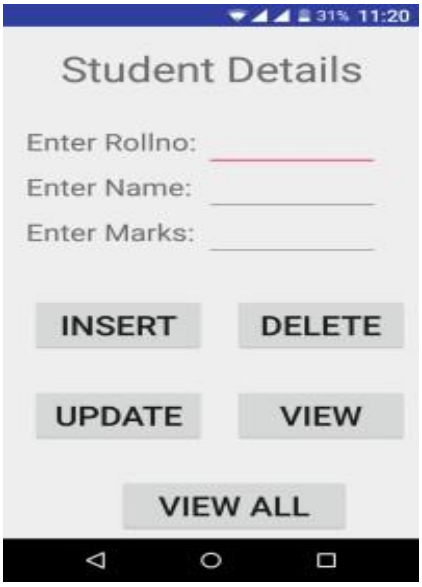
```
            if(Rollno.getText().toString().trim().length()==0||
                    Name.getText().toString().trim().length()==0||
                    Marks.getText().toString().trim().length()==0)
            {
                showMessage("Error","Pleaseenterallvalues"); return;
            }
            db.execSQL("INSERT INTO student
VALUES('"+Rollno.getText()+"','"+Name.getText()+
                    "','"+Marks.getText()+"');");
            showMessage("Success","Recordadded");
            clearText();
        }
        //DeletingarecordfromtheStudenttable
        if(view==Delete)
        {
            // Checking for empty roll number
            if(Rollno.getText().toString().trim().length()==0)
            {
                showMessage("Error","PleaseenterRollno");
                return;
            }
            Cursorc=db.rawQuery("SELECT*FROMstudentWHERE
rollno='"+Rollno.getText()+"'", null);
            if(c.moveToFirst())
            {
                db.execSQL("DELETEFROMstudentWHERE rollno='"+Rollno.getText()+"'");
                showMessage("Success","RecordDeleted");
            }
            else
            {
                showMessage("Error","InvalidRollno");
            }
            clearText();
        }
        //UpdatingarecordintheStudenttable
        if(view==Update)
        {
            // Checking for empty roll number
            if(Rollno.getText().toString().trim().length()==0)
            {
                showMessage("Error","PleaseenterRollno");
                return;
            }
            Cursorc=db.rawQuery("SELECT*FROMstudentWHERE
rollno='"+Rollno.getText()+"'", null);
            if(c.moveToFirst()){









                db.execSQL("UPDATEstudentSETname='"+Name.getText()+ "',marks='"+
Marks.getText() +
                        "'WHERErollno='"+Rollno.getText()+"'");
                showMessage("Success", "Record Modified");
            }
```

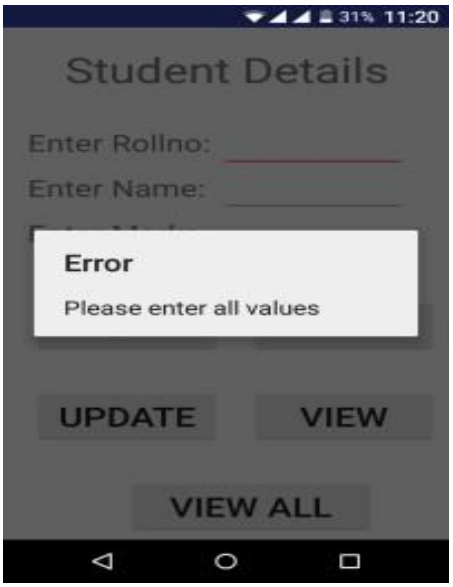
```
            else{
                showMessage("Error","InvalidRollno");
            }
            clearText();
        }
        //DisplayarecordfromtheStudenttable
        if(view==View)
        {
            // Checking for empty roll number
            if(Rollno.getText().toString().trim().length()==0)
            {
                showMessage("Error","PleaseenterRollno");
                return;
            }
            Cursorc=db.rawQuery("SELECT*FROMstudentWHERE
rollno='"+Rollno.getText()+"'", null);
            if(c.moveToFirst())
            {
                Name.setText(c.getString(1));
                Marks.setText(c.getString(2));
            }
            else
            {
                showMessage("Error","InvalidRollno");
                clearText();
            }
        }
        //Displayingalltherecords if(view==ViewAll)
        {
            Cursorc=db.rawQuery("SELECT*FROMstudent",null);
            if(c.getCount()==0)
            {
                showMessage("Error","Norecordsfound");
                return;
            }
            StringBufferbuffer=newStringBuffer();
            while(c.moveToNext())
            {
                buffer.append("Rollno: "+c.getString(0)+"\n");
                buffer.append("Name: "+c.getString(1)+"\n");
                buffer.append("Marks:"+c.getString(2)+"\n\n");
            }
            showMessage("StudentDetails",buffer.toString());
        }
    }
    publicvoidshowMessage(Stringtitle,Stringmessage)
    {
        Builderbuilder=newBuilder(this);
        builder.setCancelable(true);
        builder.setTitle(title);
        builder.setMessage(message);
        builder.show();
    }
    publicvoidclearText()
    {



        Rollno.setText("");
```

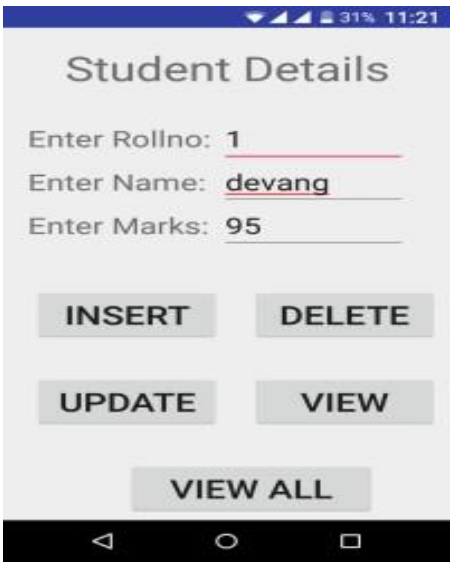
```
        Name.setText("");
        Marks.setText("");
        Rollno.requestFocus();
    }
}
```

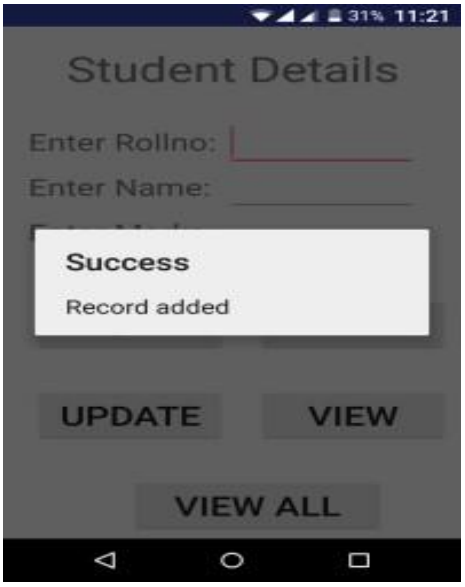
- So now the Coding part is also completed.
- Now run the application to see the output.

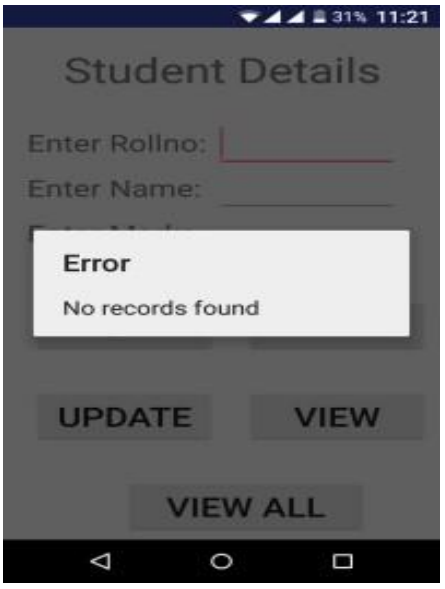
SAMPLEOUTPUT:



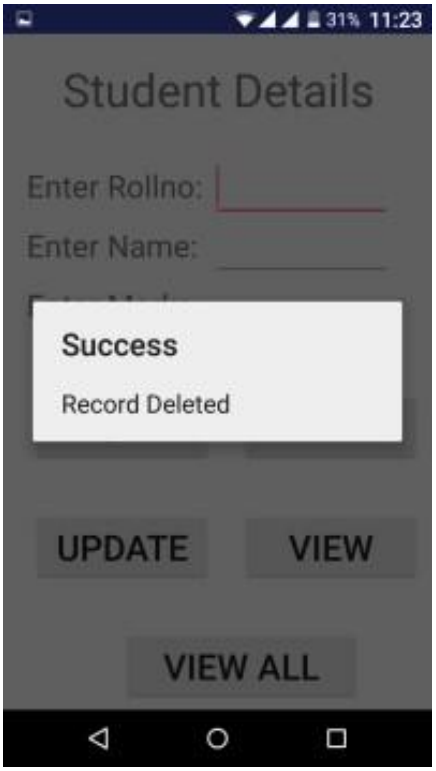


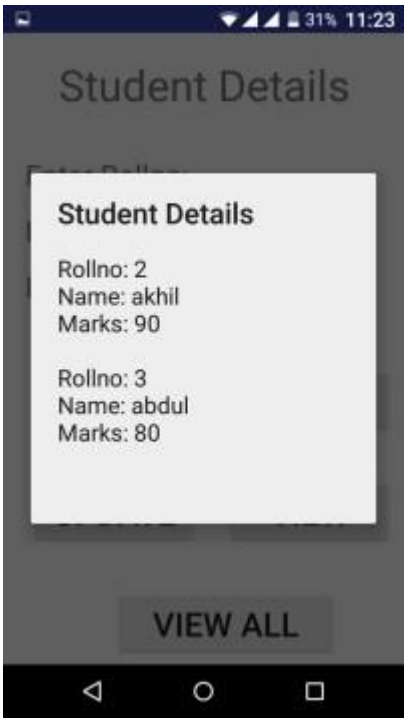


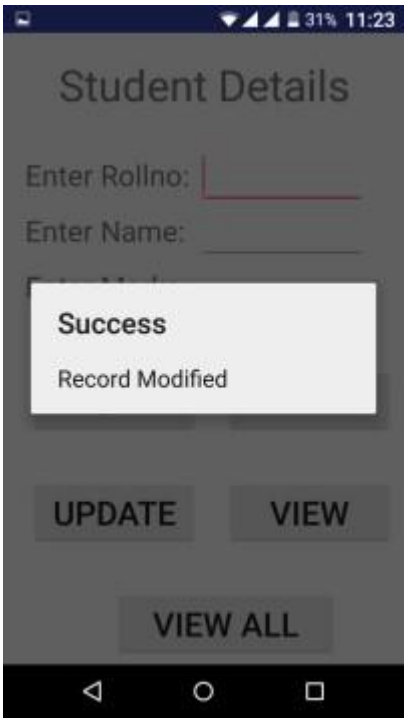

31% 11:21
Student Details
Enter Rollno:
Enter Name:
Success
Record added
UPDATE
VIEW
VIEW ALL

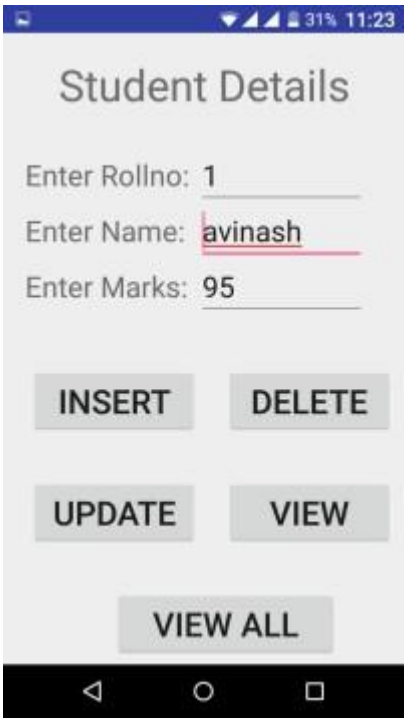

31% 11:21
Student Details
Enter Rollno:
Enter Name:
Error
No records found
UPDATE
VIEW
VIEW ALL

31% 11:22
Student Details
Student Details
Rollno: 1
Name: devang
Marks: 95
Rollno: 2
Name: akhil
Marks: 90
Rollno: 3
Name: abdul
Marks: 80
VIEW ALL


31% 11:23
Student Details
Enter Rollno:
Enter Name:
Success
Record Deleted
UPDATE
VIEW
VIEW ALL







Result:

Thus a Simple Android Application that makes use of Database is developed and executed successfully.

EX.NO:06

DATE:

ANDROID APPLICATION THAT MAKESUSE OF RSSFEED

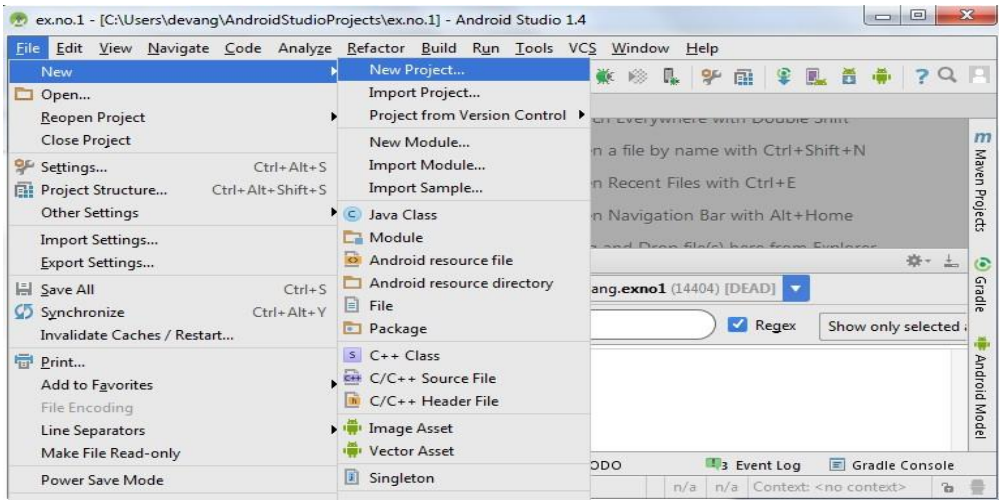
AIM:

To develop a Android Application that makes use of RSS Feed.

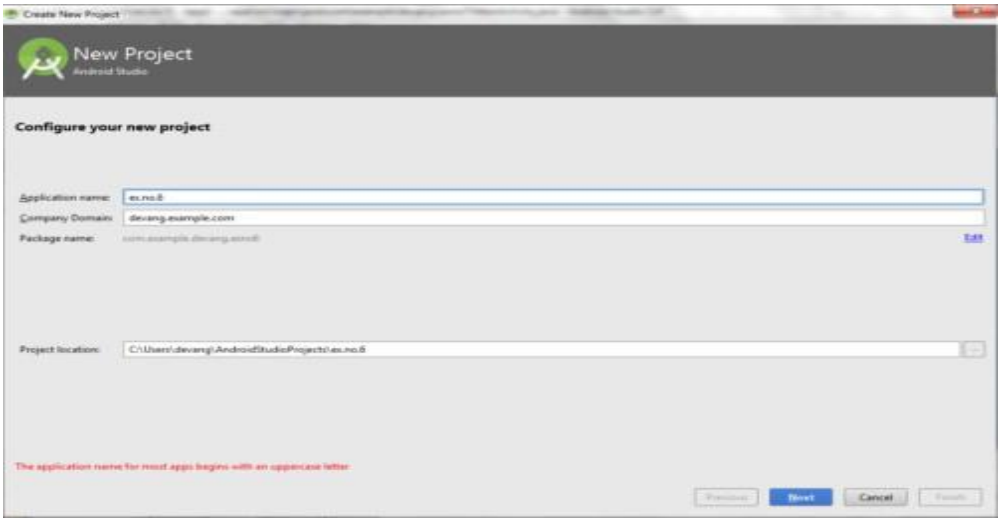
PROCEDURE:

Creating a New project:

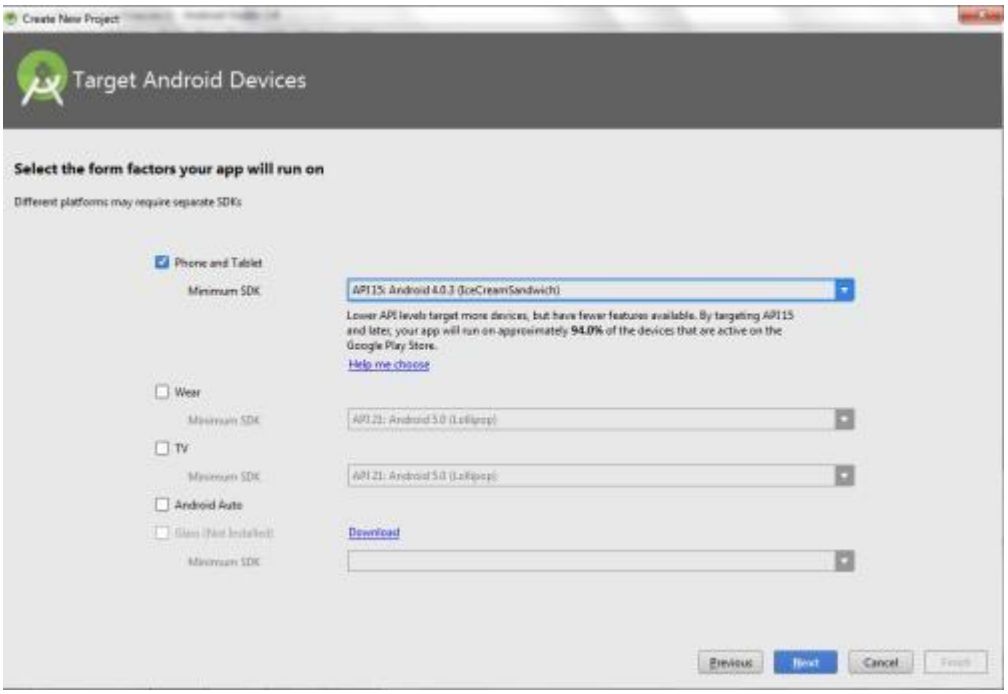
- Open Android Studio and then click on **File->New->New project.**



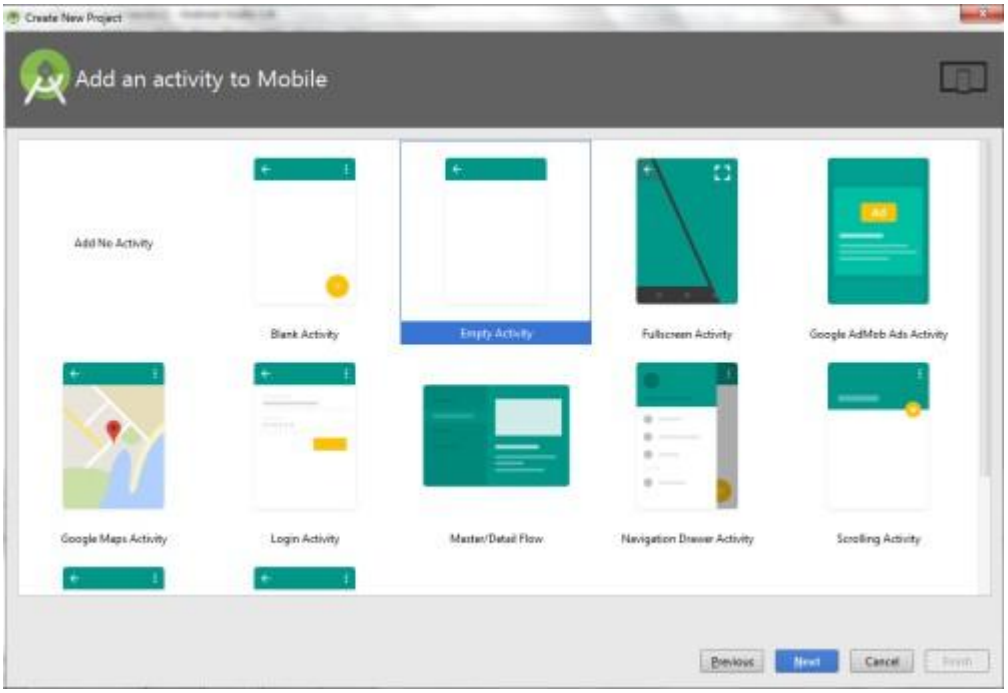
- Then type the Application name as "**ex.no.6"**and click **Next.**



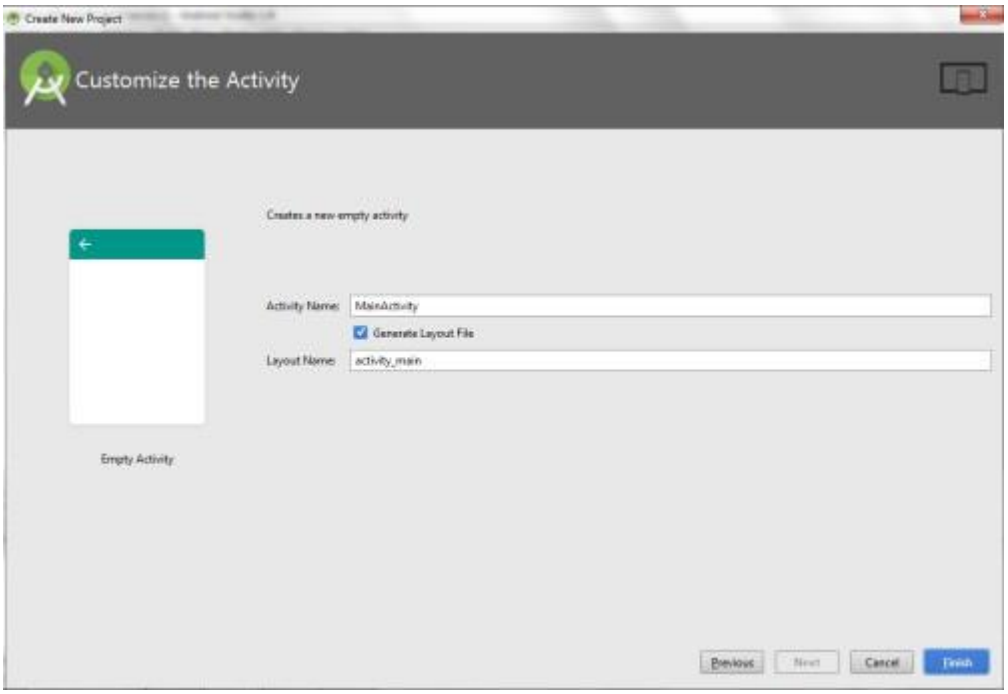
- Then select the **Minimum SDK** as shown below and click **Next**.



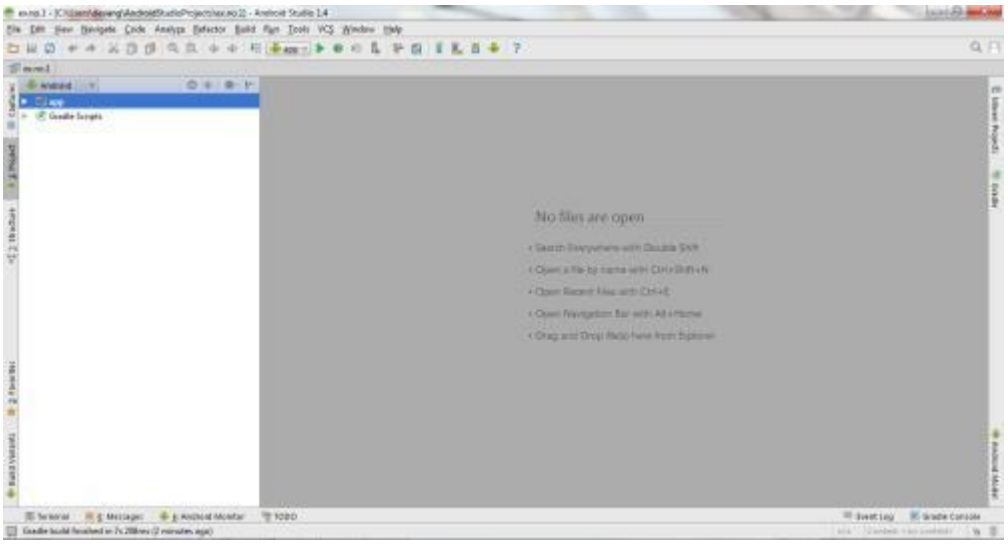
- Then select the **Empty Activity** and click **Next.**



- Finally click F**inish**.

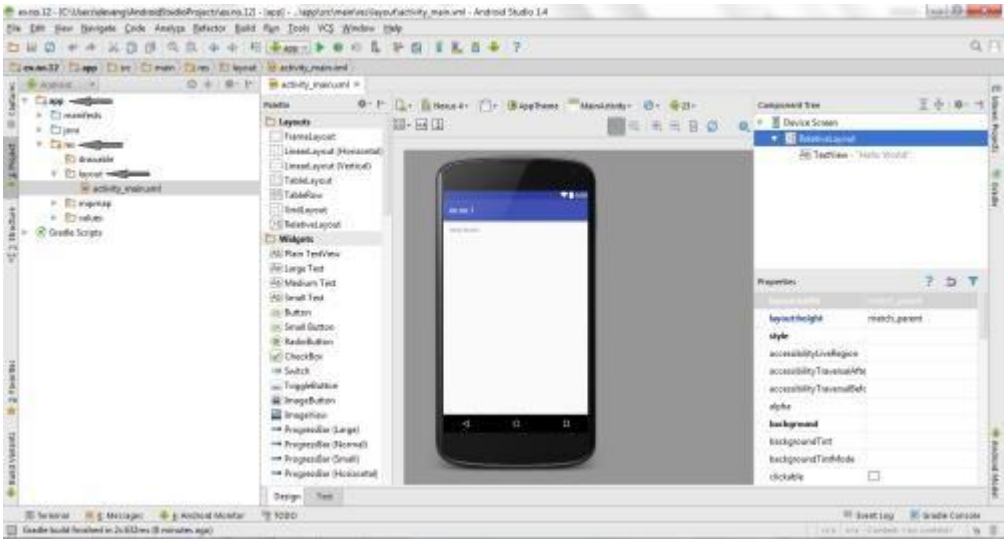


- It will take sometime to build and load the project.
- After completion it will look as given below.

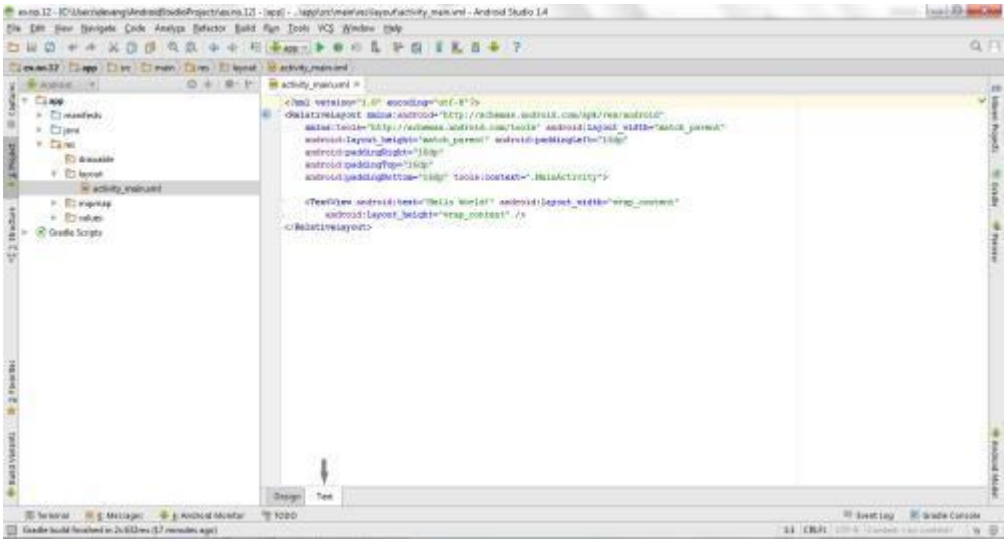


Designing layout for the Android Application:

- Click on **app->res-> layout->activity_main.xml**



- Now click on **Text** as shown below.

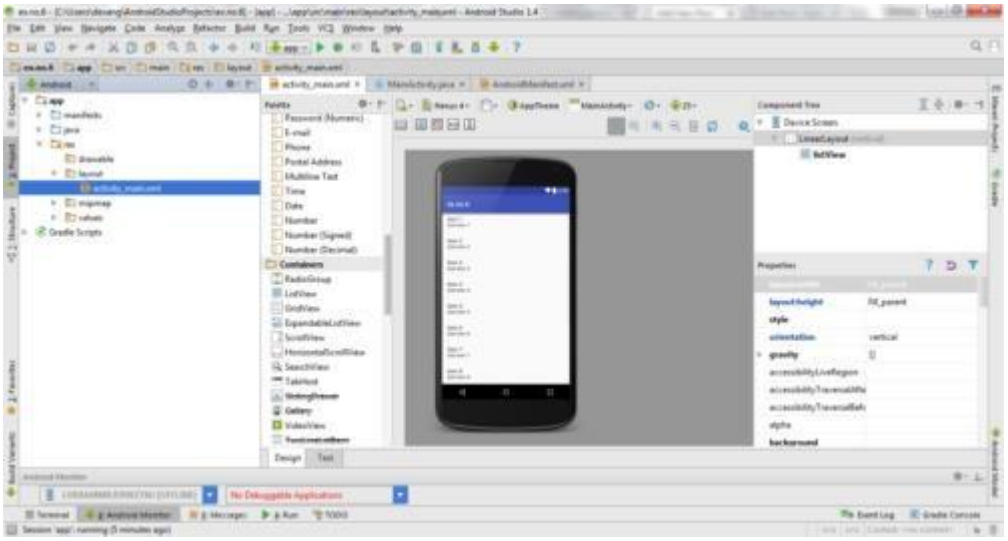


- Then delete the code which is there and type the code as given below.

Code forActivity_main.xml:

```
<?xmlversion="1.0"encoding="utf-8"?>
<LinearLayoutxmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent"
    android:orientation="vertical">
     <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
 </LinearLayout>
```

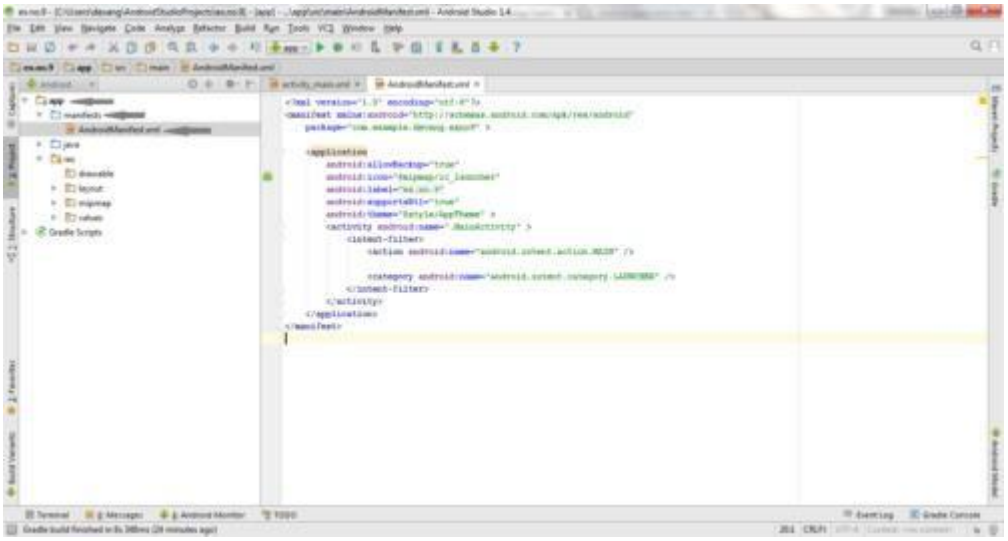
- Now click on **Design** and your application will look as given below.



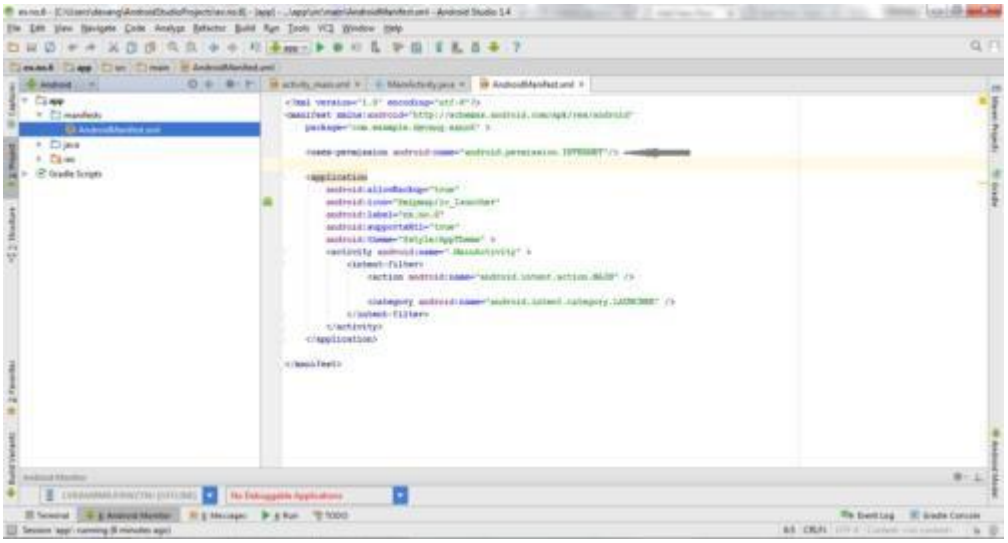
- So now the designing part is completed.

Adding permissions in Manifest for the Android Application:

- Click on **app ->manifests->AndroidManifest.xml**



- Now include the **INTERNET** permissions in the Android Manifest.xmlfile as shown below



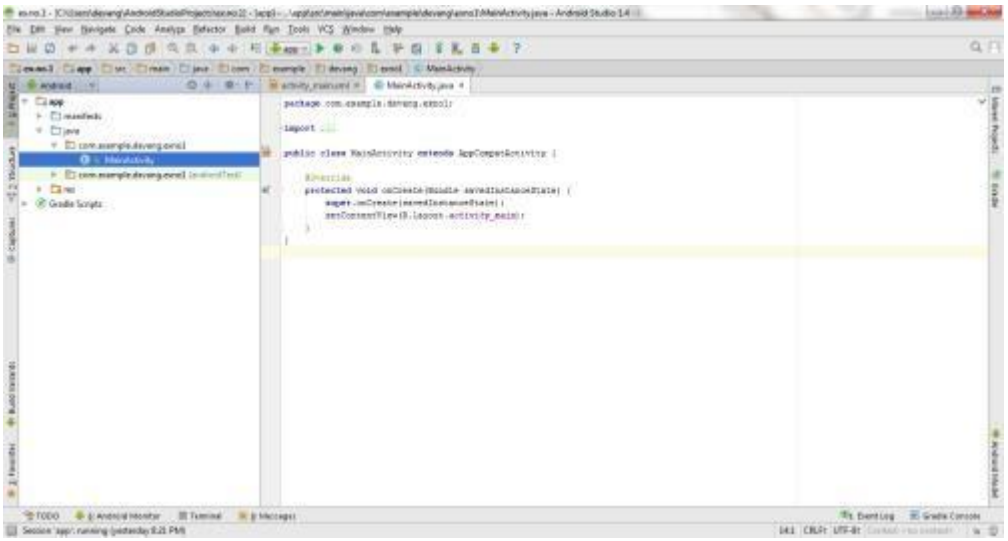
Code for Android Manifest.xml:

```
<?xmlversion="1.0"encoding="utf-8"?>
<manifestxmlns:android="http://schemas.android.com/apk/res/android"package="com.exa
    mple.exno6">

    <uses-permissionandroid:name="android.permission.INTERNET"/>

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:supportsRtl="true"android:theme="@st
        yle/AppTheme">
        <activityandroid:name=".MainActivity">
            <intent-filter>
                <actionandroid:name="android.intent.action.MAIN"/>

                <categoryandroid:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>
    </application>

</manifest>
```

- So now the Permissions are added in the Manifest.

Java Coding for the Android Application:

- Click on **app->java->com.example.exno6 ->MainActivity.**



- Then delete the code which is there and type the code as given below.

Code forMainActivity.java:

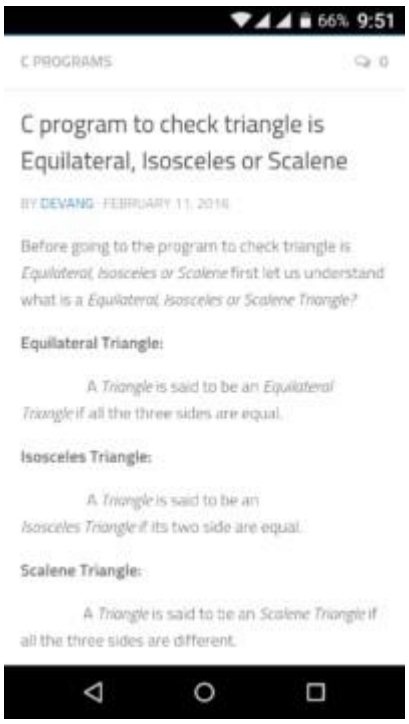
```
packagecom.example.exno6;

importandroid.app.ListActivity;
importandroid.content.Intent;
importandroid.net.Uri;
importandroid.os.AsyncTask;
importandroid.os.Bundle;
importandroid.view.View;
importandroid.widget.ArrayAdapter;
importandroid.widget.ListView;
importorg.xmlpull.v1.XmlPullParser;
importorg.xmlpull.v1.XmlPullParserException;
importorg.xmlpull.v1.XmlPullParserFactory;
importjava.io.IOException;
importjava.io.InputStream;
importjava.net.MalformedURLException;
importjava.net.URL;
importjava.util.ArrayList;
importjava.util.List;

publicclassMainActivityextendsListActivity
{
    Listheadlines;
    Listlinks;

    @Override
    protectedvoidonCreate(BundlesavedInstanceState)
    {
        super.onCreate(savedInstanceState);
        newMyAsyncTask().execute();
    }

    classMyAsyncTaskextendsAsyncTask<Object,Void,ArrayAdapter>
    {
```

```
        @Override
        protectedArrayAdapterdoInBackground(Object[]params)
        {





            headlines=newArrayList(); links
            = newArrayList();
            try
            {
                URLurl=newURL("https://codingconnect.net/feed");
                XmlPullParserFactory factory =
XmlPullParserFactory.newInstance();
                factory.setNamespaceAware(false);
                XmlPullParserxpp=factory.newPullParser();

                //WewillgettheXMLfromaninputstream
                xpp.setInput(getInputStream(url),"UTF_8");
                booleaninsideItem = false;

                //Returnsthetypeofcurrentevent:START_TAG,END_TAG,
etc..
                inteventType=xpp.getEventType();
                while(eventType!=XmlPullParser.END_DOCUMENT)
                {
                    if(eventType==XmlPullParser.START_TAG)
                    {
                        if(xpp.getName().equalsIgnoreCase("item"))
                        {
                             insideItem=true;
                        }
                        elseif(xpp.getName().equalsIgnoreCase("title"))
                        {
                            if(insideItem)
                                headlines.add(xpp.nextText());//extract
theheadline
                        }
                        elseif(xpp.getName().equalsIgnoreCase("link"))
                        {
                            if(insideItem)
                                links.add(xpp.nextText());//extractthe
linkofarticle
                        }
                    }
                    elseif(eventType==XmlPullParser.END_TAG&&xpp.getName().equalsI
gnoreCase("item"))
                    {
                        insideItem=false;
                    }
                    eventType=xpp.next();//movetonextelement
                }

            }
            catch(MalformedURLException e)
            {
                e.printStackTrace();
            }
```

```
            catch(XmlPullParserExceptione)
            {
                e.printStackTrace();
            }
            catch(IOExceptione)
            {
                e.printStackTrace();
            }
            returnnull;
        }
        protectedvoidonPostExecute(ArrayAdapteradapter)
        {





            adapter=newArrayAdapter(MainActivity.this,
android.R.layout.simple_list_item_1, headlines);
            setListAdapter(adapter);
        }
    }

    @Override
    protectedvoidonListItemClick(ListViewl,Viewv,intposition,longid)
    {
        Uriuri=Uri.parse((links.get(position)).toString());
        Intent intent = newIntent(Intent.ACTION_VIEW, uri);
        startActivity(intent);
    }

    publicInputStreamgetInputStream(URLurl)
    {
        try
        {
            returnurl.openConnection().getInputStream();
        }
        catch(IOExceptione)
        {
            returnnull;
        }
    }
}
```

- So now the Coding part is also completed.
- Now run the application to see the output.

SAMPLEOUTPUT:



RESULT:

Thus Android Application that makes use of RSS Feed is developed and executed successfully.

EX. NO: 07

DATE:

ANDROID APPLICATION THAT IMPLEMENTS MULTITHREADING

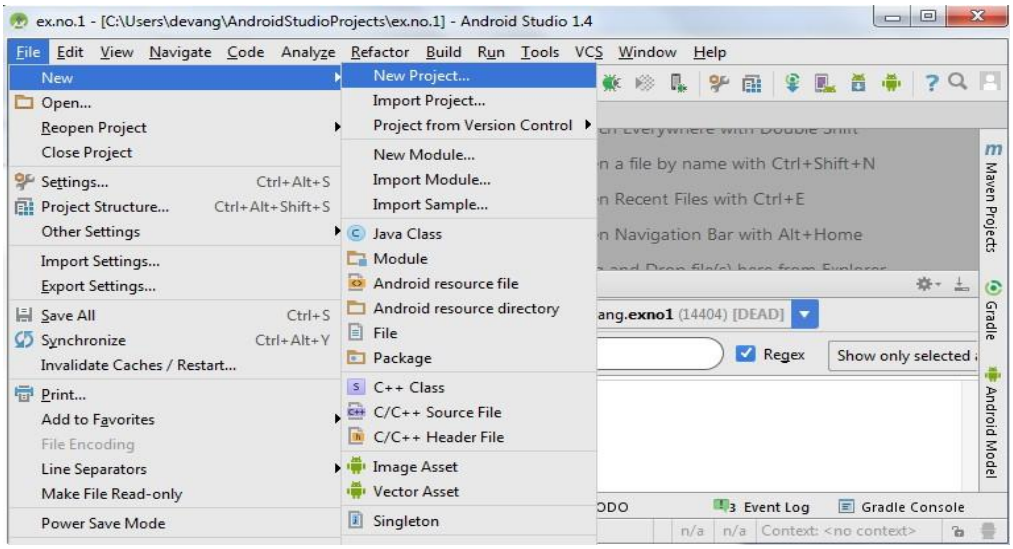
AIM:

To develop a Android Application that implements Multi threading.

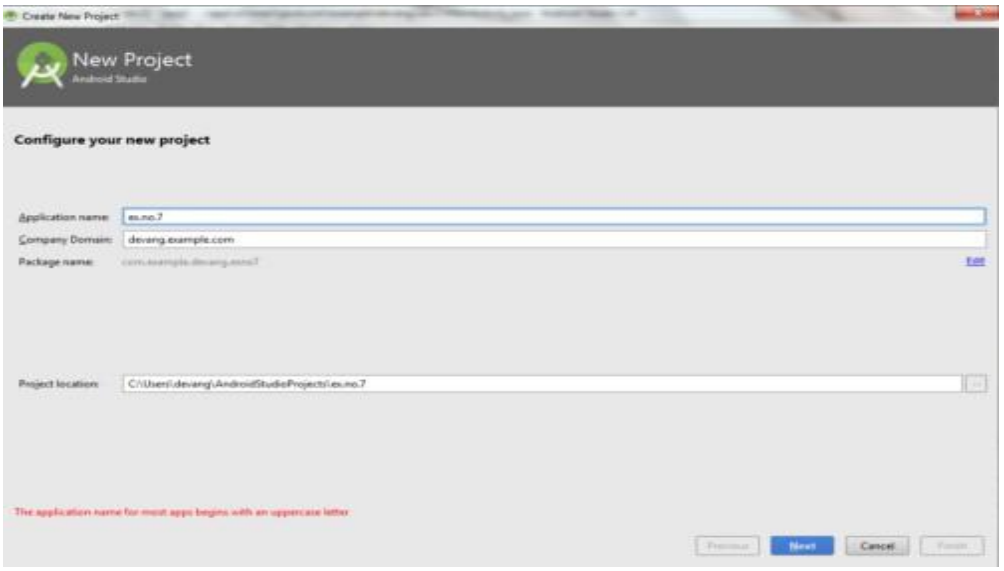
PROCEDURE:

Creating a New project:

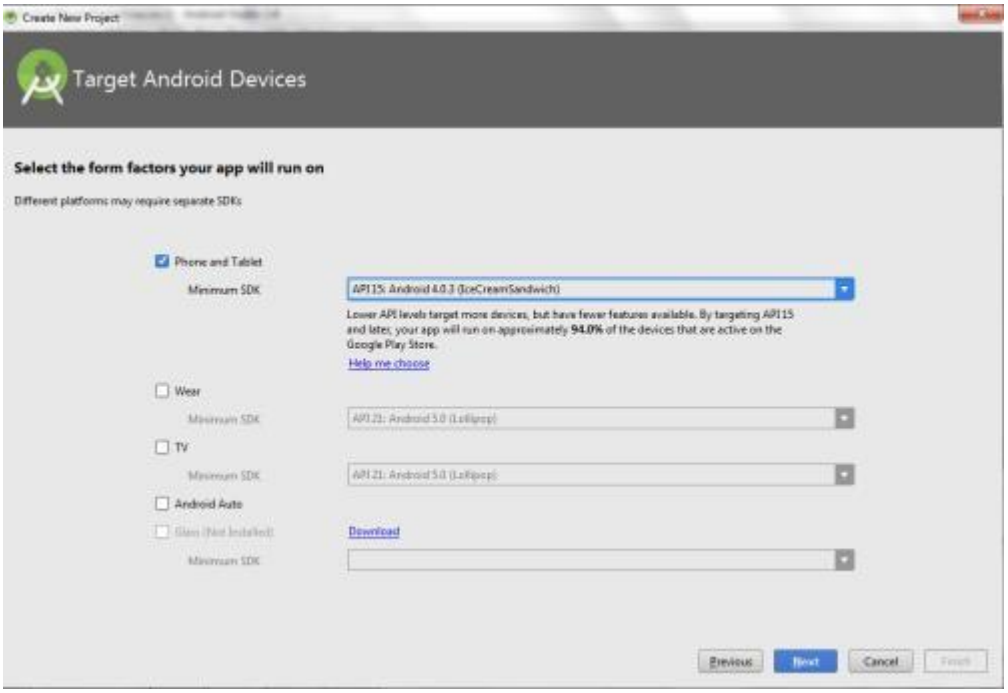
- Open Android Studio and then click on **File->New->New project.**



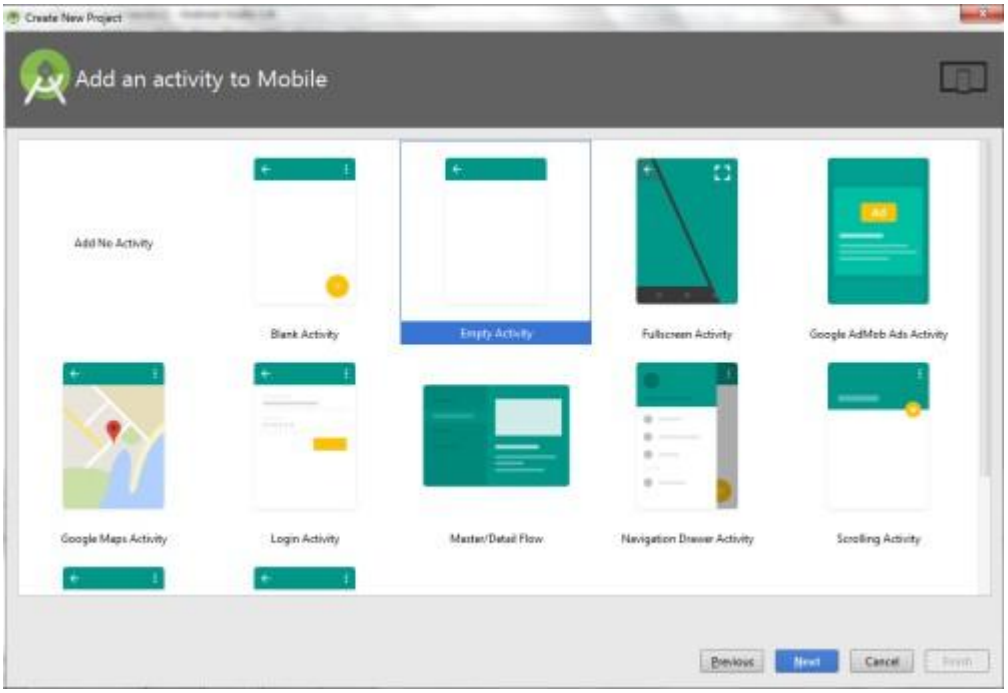
- Then type the Application name as "**ex.no.7"**and click**Next.**



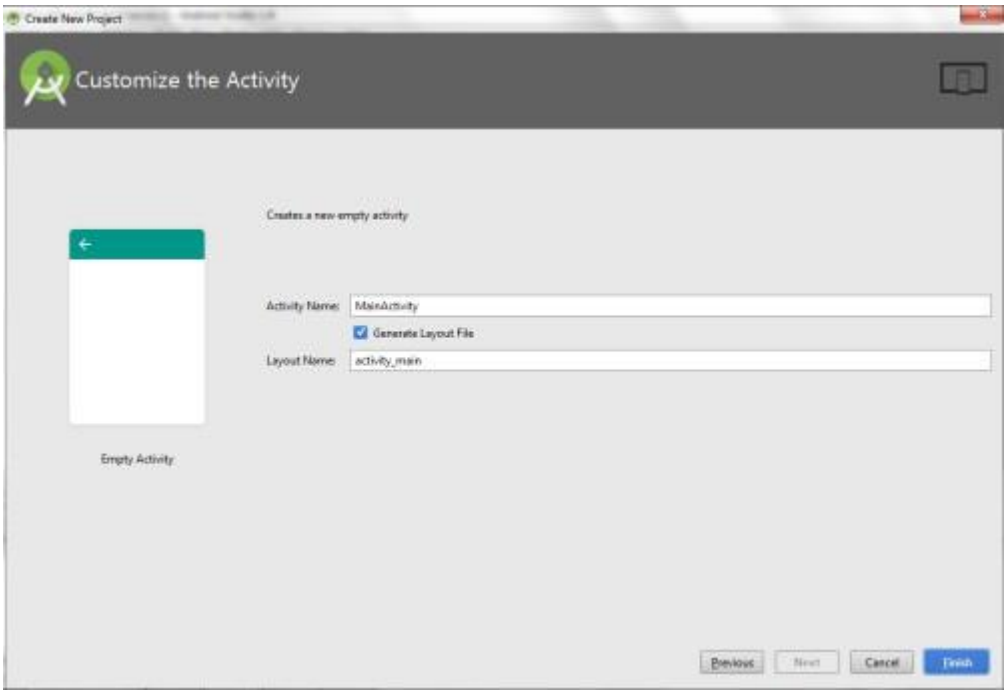
- Then select the **Minimum SDK** as shown below and click **Next**.



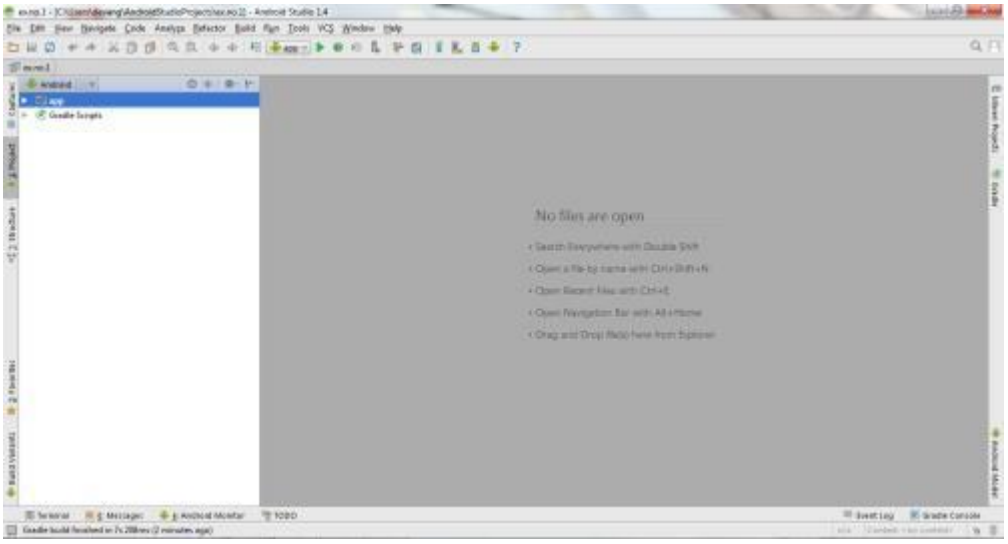
- Then select the **Empty Activity** and click **Next.**



- Finally click **Finish**.

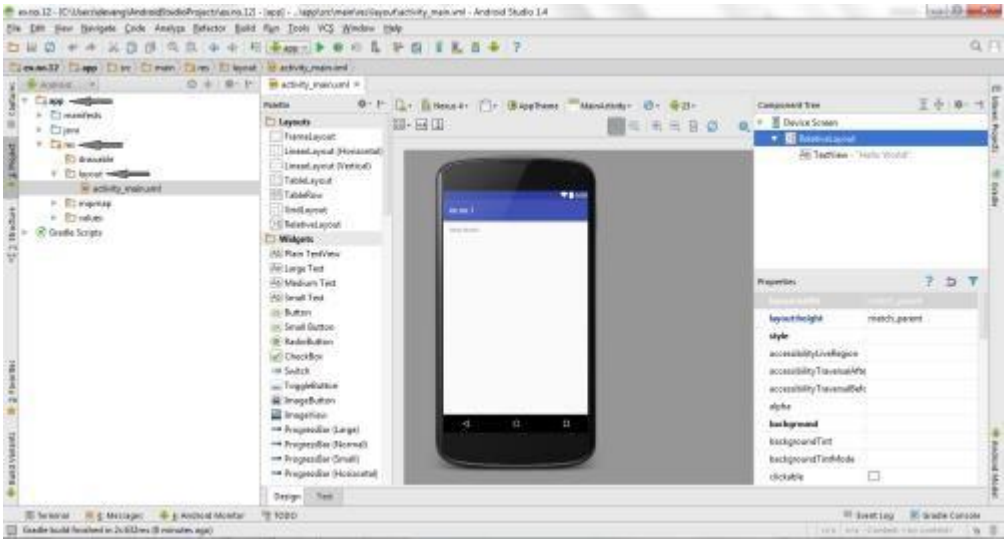


- It will take sometime to build and load the project.
- After completion it will look as given below.

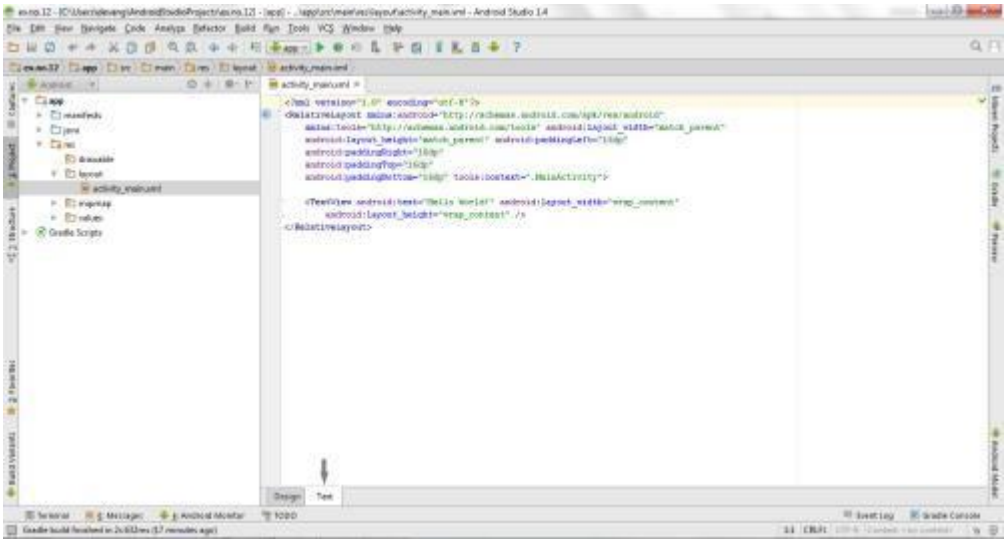


Designing layout for the Android Application:

- Clickon **app->res-> layout->activity_main.xml**



- Now click on **Text** as shown below.



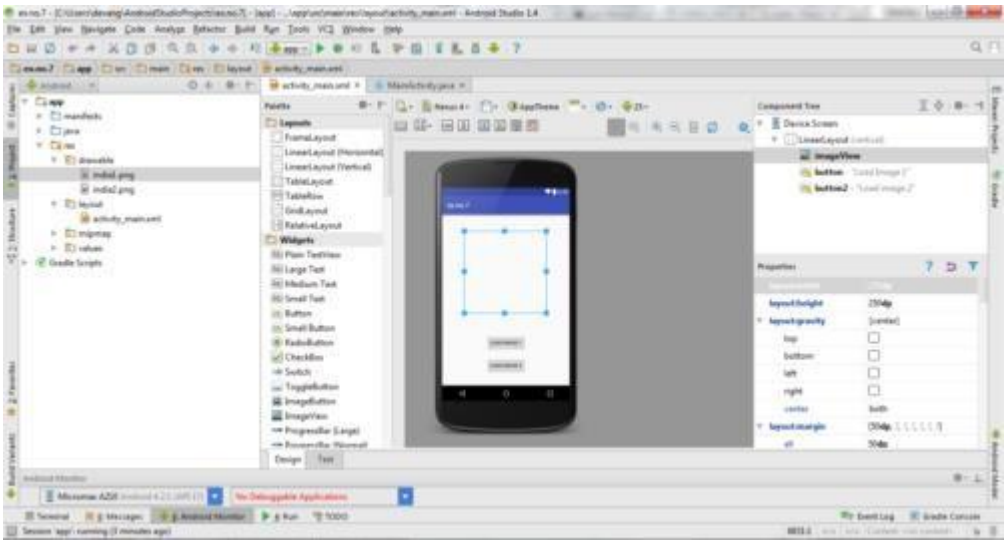
- Then delete the code which is there and type the code as given below.

Code forActivity_main.xml:

```
<?xmlversion="1.0"encoding="utf-8"?>
<LinearLayoutxmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="250dp"
        android:layout_height="250dp"
        android:layout_margin="50dp"
        android:layout_gravity="center"/>

    <Button
        android:id="@+id/button"
```

```
        android:layout_width="wrap_content"




        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:layout_gravity="center"android:text=
        "Load Image 1"/>

    <Button
        android:id="@+id/button2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:layout_gravity="center"android:text=
        "Load image 2"/>

</LinearLayout>
```

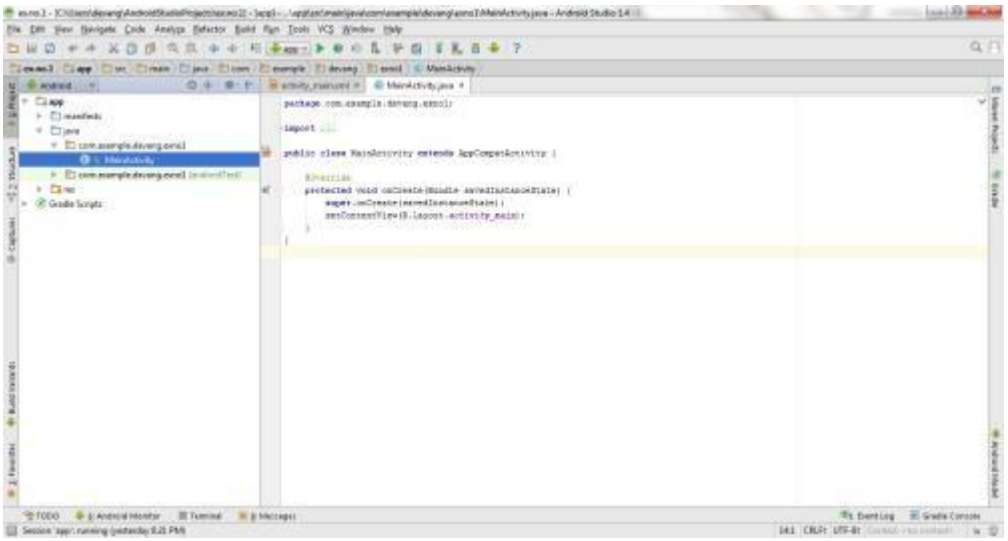
- Now click on **Design** and your application will look as given below.



- So now the designing part is completed.

Java Coding for the Android Application:

- Click on **app->java->com.example.exno7 ->Main Activity.**



- Then delete the code which is there and type the code as given below.

Code forMainActivity.java:

```
packagecom.example.exno7;

importandroid.os.Bundle;
importandroid.support.v7.app.AppCompatActivity;
importandroid.view.View;
importandroid.widget.Button;
importandroid.widget.ImageView;
publicclassMainActivityextendsAppCompatActivity
{
    ImageView img;
    Buttonbt1,bt2;
    @Override
    protectedvoidonCreate(BundlesavedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        bt1 = (Button)findViewById(R.id.button);
        bt2=(Button)findViewById(R.id.button2);
        img= (ImageView)findViewById(R.id.imageView);

        bt1.setOnClickListener(newView.OnClickListener()
        {
            @Override
            publicvoidonClick(Viewv)
            {
                newThread(newRunnable()
                {
                    @Override
                    publicvoidrun()
                    {
                        img.post(newRunnable()
                        {
                            @Override
```

```
                        publicvoidrun()
                        {
                            img.setImageResource(R.drawable.india1);
                        }
                    });
                }
            }).start();
        }
    });

    bt2.setOnClickListener(newView.OnClickListener()
    {
        @Override
        publicvoidonClick(Viewv)
        {
            newThread(newRunnable()
            {
                @Override
                publicvoidrun()
                {
                    img.post(newRunnable()
                    {
                        @Override
                        publicvoidrun()
                        {




                            img.setImageResource(R.drawable.india2);
                        }
                    });
                }
            }).start();
        }
    });
  }
}
```

- So now the Coding part is also completed.
- Now run the application to seethe output.

Note: Before Running the Application, Copy the Images given below and Pasteit in *"app ->res-> drawable "*by pressing" right click mouse button on *drawable "* and selecting the "*Paste*" option.

SAMPLEOUTPUT:\

RESULT:

Thus Android Application that implements Multithreading is developed and executed successfully.

EX NO.08

DATE:

ANDROID PROGRAM FOR GPS LOCATION TRACKING

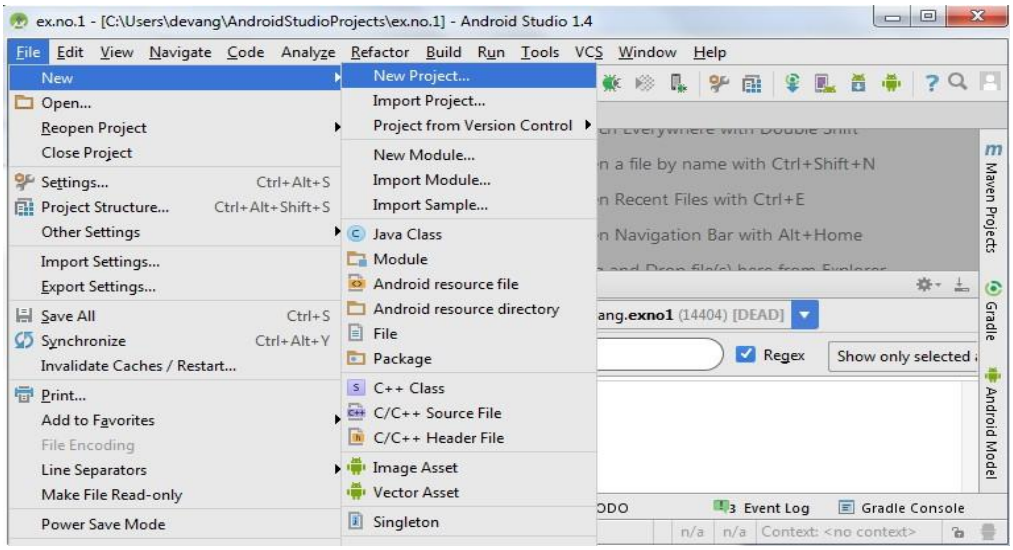
AIM:

To access current location information through location providers, we need to set permissions with android_manifest file

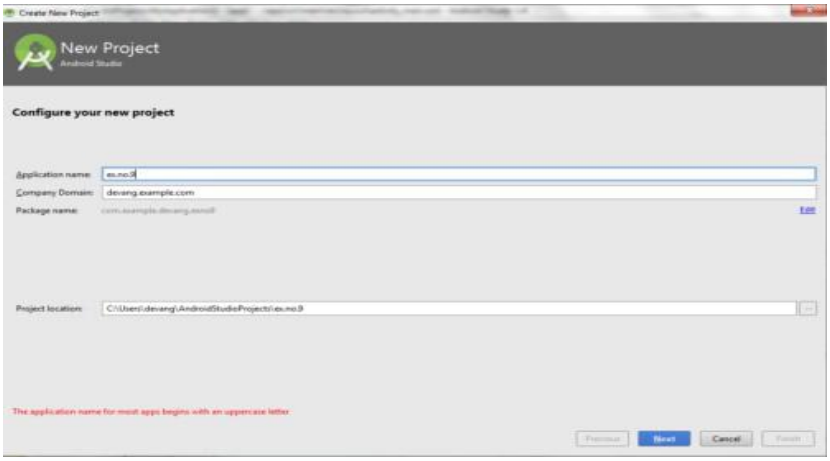
PROCEDURE:

Creating a New project:

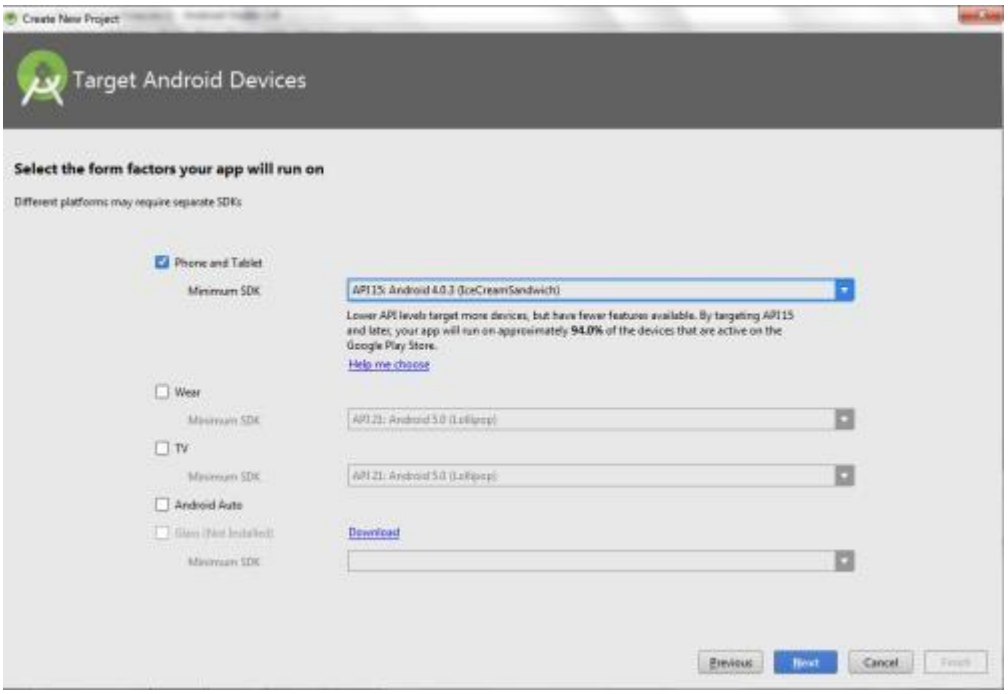
- Open Android Studio and then click on **File->New->New project.**



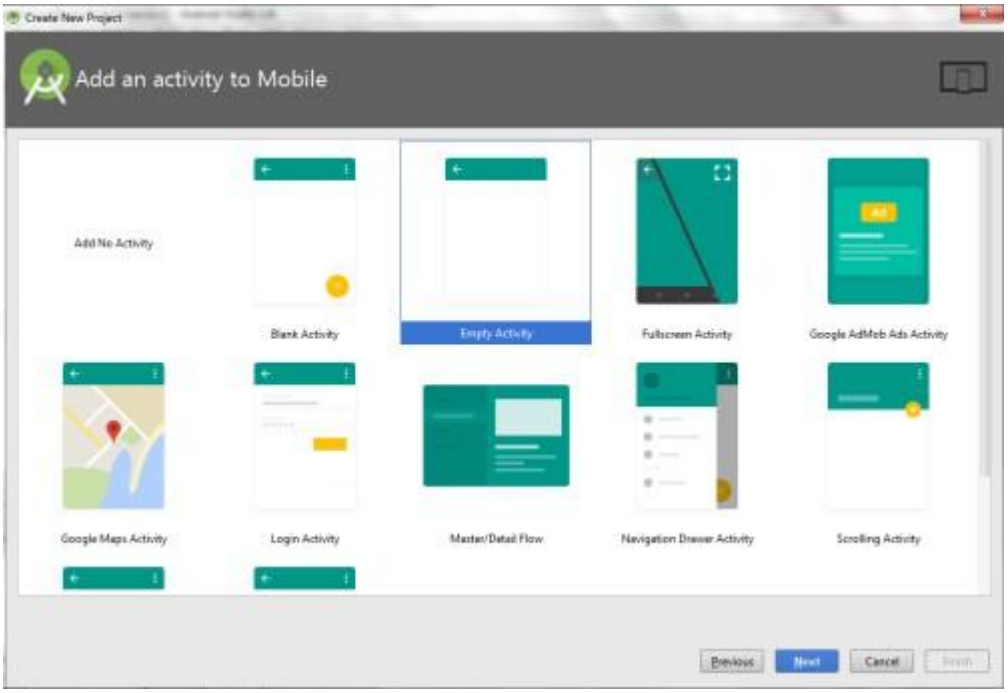
- Then type the Application name as "**ex.no.9"**and click **Next.**



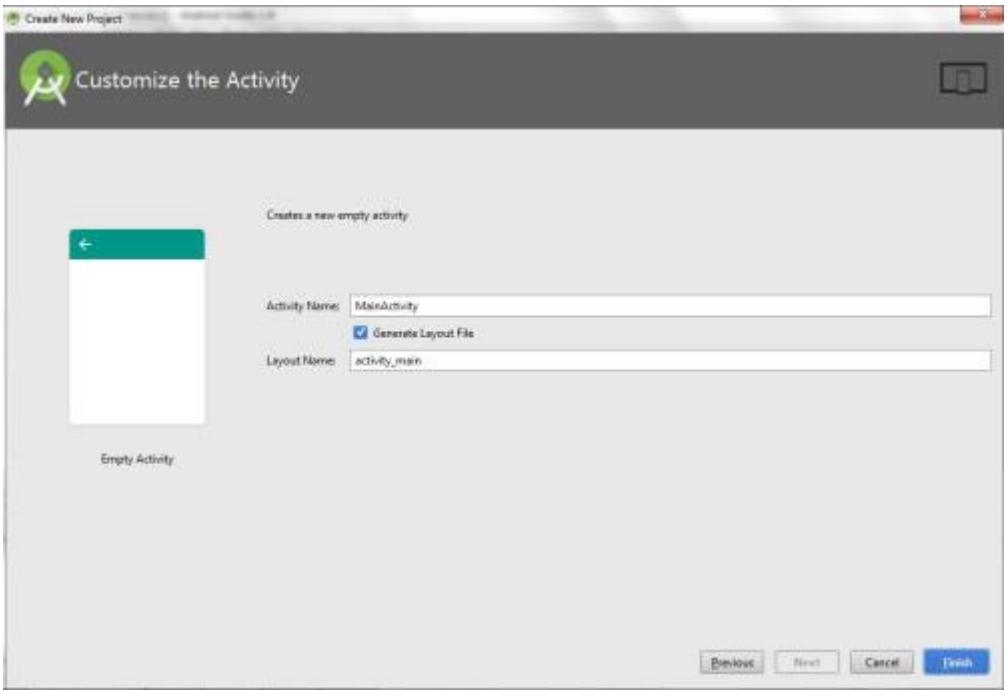
- Then select the **Minimum SDK** as shown below and click **Next**.



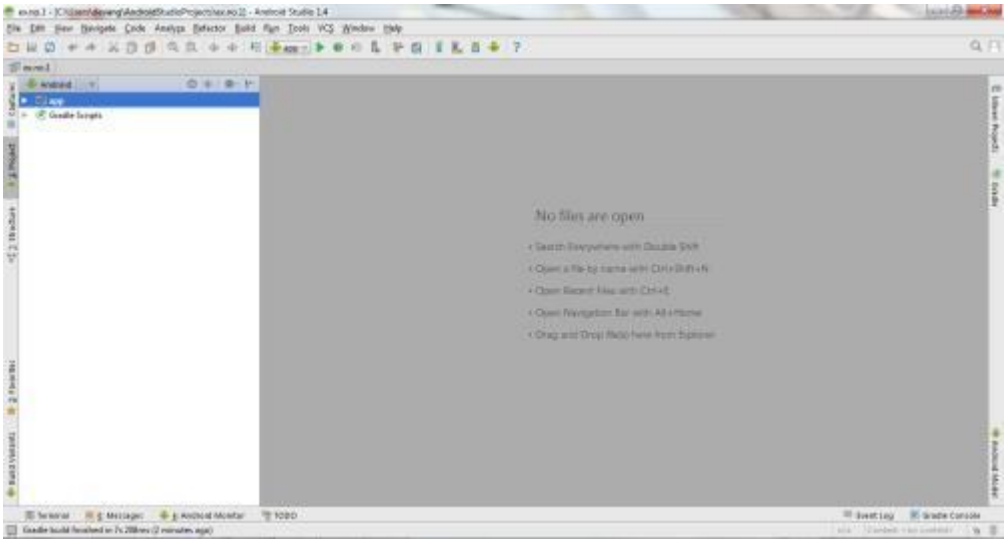
- Then select the **Empty Activity** and click **Next.**



- Finally click F**inish**.

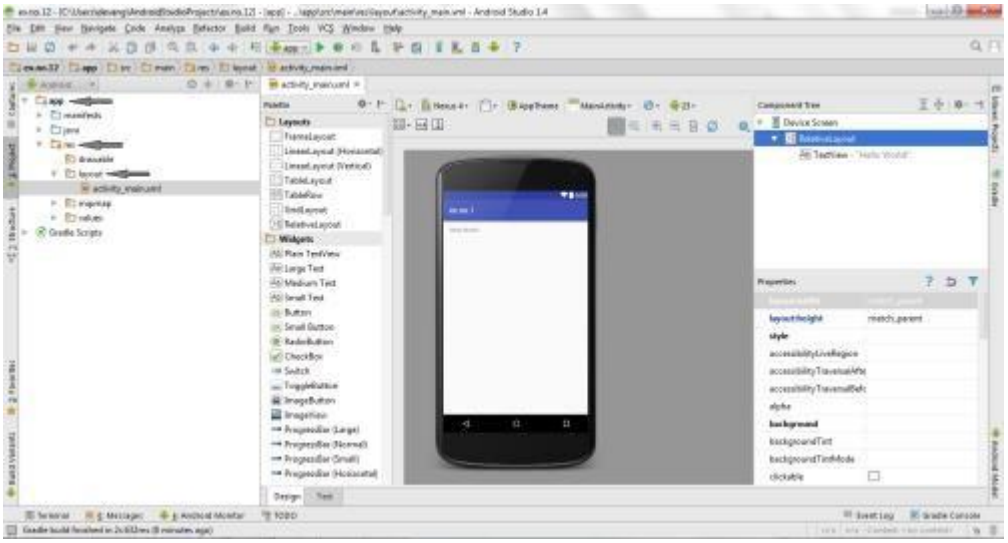


- It will take sometime to build and load the project.
- After completion it will look as given below.

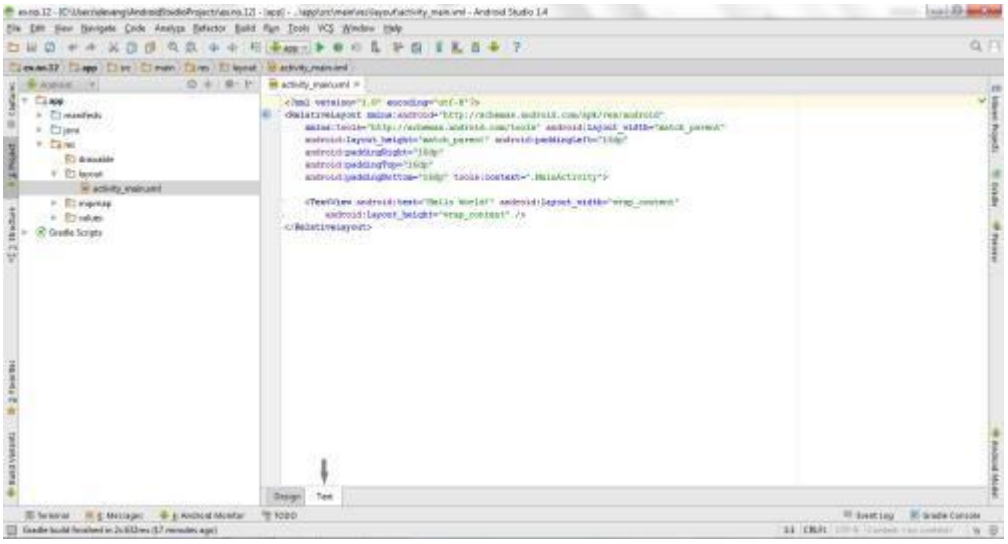


Designing layout for the Android Application:

- Click on **app->res-> layout->activity_main.xml.**



- Now click on **Text** as shown below.



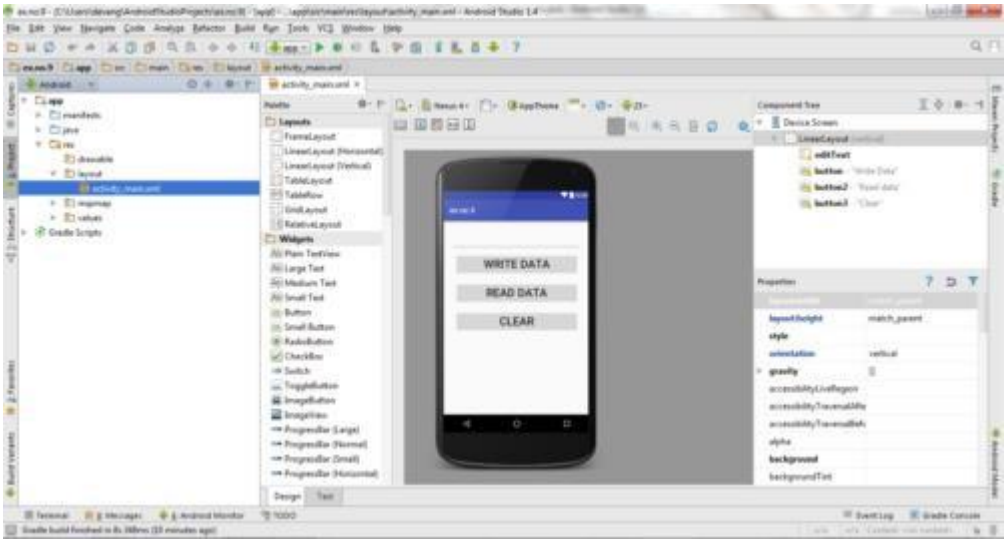
- Then delete the code which is there and type the code as given below.

Code forActivity_main.xml:

```
<?xmlversion="1.0"encoding="utf-8"?>
<LinearLayoutxmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="20dp"
    android:orientation="vertical">

    <EditText
        android:id="@+id/editText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:singleLine="true"android:textSize="3
        0dp"/>

    <Button
```

```
        android:id="@+id/button"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"android:text="Wr
        ite Data" android:textSize="30dp"/>

    <Button
        android:id="@+id/button2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"android:text="Re
        ad data" android:textSize="30dp"/>

    <Button
        android:id="@+id/button3"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:text="Clear"android:textSize
        ="30dp"/>

</LinearLayout>
```

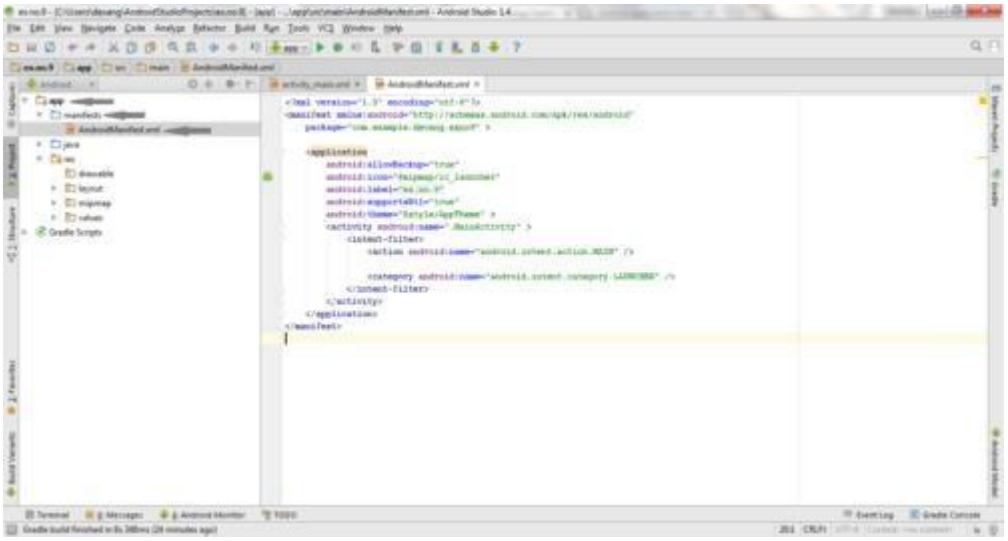
- Now click on **Design** and your application will look as given below.



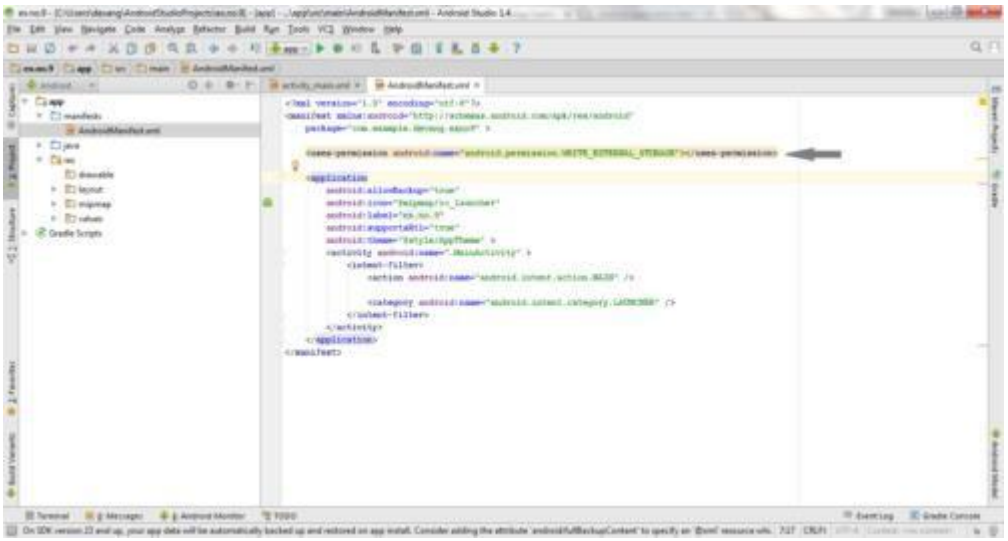
- So now the designing part is completed.

Adding permissions in Manifest for the Android Application:

- Click on **app ->manifests->AndroidManifest.xml**



- Now include the **WRITE_ EXTERNAL_ STORAGE** permissions in the AndroidManifest.xml file as shown below



Code for AndroidManifest.xml

```
<?xmlversion="1.0"encoding="utf-8"?>

<manifest...>
    <uses-permissionandroid:name="android.permission.ACCESS_FINE_LOCATION"
/>
   <uses-permissionandroid:name="android.permission.
ACCESS_COARSE_LOCATION"/>
   <uses-permissionandroid:name="android.permission.INTERNET"/>
</manifest>
```

ACCESS_COARSE_LOCATION is used when we use network location provider for our Android app.
But, ACCESS_ FINE_ LOCATION is providing permission for both providers.
INTERNET permission is must for the use of network provider.

Java file

```
packagecom.javapapers.android.geolocationfinder;

import android.os.Bundle;
import android.app.Activity;
importandroid.content.Context;
importandroid.location.Location;
importandroid.location.LocationListener;
import android.location.LocationManager;
import android.widget.TextView;

importandroid.util.Log;

publicclassMainActivityextendsActivityimplementsLocationListener{
protected LocationManager locationManager;
protectedLocationListenerlocationListener; protected
Context context;
TextViewtxtLat;
String lat;
Stringprovider;
protectedStringlatitude,longitude;
protectedbooleangps_enabled,network_enabled;

@Override
protectedvoidonCreate(BundlesavedInstanceState){
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
txtLat=(TextView)findViewById(R.id.textview1);

locationManager = (LocationManager)
getSystemService(Context.LOCATION_SERVICE);
locationManager.requestLocationUpdates(LocationManager.GPS_PROVIDER,0,0,
this);
}
@Override
publicvoidonLocationChanged(Locationlocation){
txtLat = (TextView) findViewById(R.id.textview1);
txtLat.setText("Latitude:"+location.getLatitude()+",Longitude:"+
location.getLongitude());
}

@Override
publicvoidonProviderDisabled(Stringprovider){
Log.d("Latitude","disable");
}

@Override
publicvoidonProviderEnabled(Stringprovider){
Log.d("Latitude","enable");
}

@Override
publicvoidonStatusChanged(Stringprovider,intstatus,Bundleextras){
Log.d("Latitude","status");
}
}
```

XML files for layout and android manifest are as shown below

```
<RelativeLayoutxmlns:android="http://schemas.android.com/apk/res/android"xmlns:too
    ls="http://schemas.android.com/tools"
```

```
    android:layout_width="match_parent"
    android:layout_height="match_parent"tools:contex
    t=".MainActivity">

    <TextView
        android:id="@+id/textview1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="true"
        android:layout_centerVertical="true"android:
        text="@string/hello_world" />

</RelativeLayout>
```

Manifest.xml

```
<?xmlversion="1.0"encoding="utf-8"?>
<manifestxmlns:android="http://schemas.android.com/apk/res/android"pa
    ckage="com.javapapers.android.geolocationfinder"
    android:versionCode="1"
    android:versionName="1.0">

    <uses-sdk
        android:minSdkVersion="8"
        android:targetSdkVersion="17"        />
    <uses-permissionandroid:name="android.permission.ACCESS_FINE_LOCATION"
/>
    <application
        android:allowBackup="true"
        android:icon="@drawable/ic_launcher"
        android:label="@string/app_name"android:them
        e="@style/AppBaseTheme">
        <activity

android:name="com.javapapers.android.geolocationfinder.MainActivity"android:label="
            @string/app_name">
            <intent-filter>
                <actionandroid:name="android.intent.action.MAIN"/>

                <categoryandroid:name="android.intent.category.LAUNCHER"
/>
            </intent-filter>
        </activity>
    </application>
</manifest>
```

OUTPUT:

RESULT:

Thus Android Application that implements GPS is developed and executed successfully.

EX.NO:09

DATE:

ANDROID APPLICATION THAT WRITES DATA TO THE SD CARD.

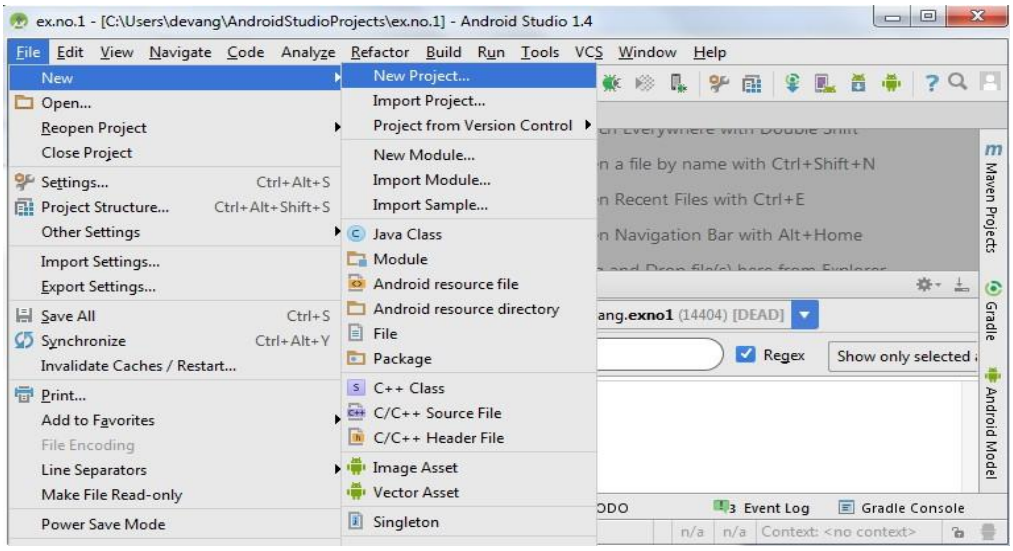
AIM:

To develop a Android Application that writes data to the SD Card.

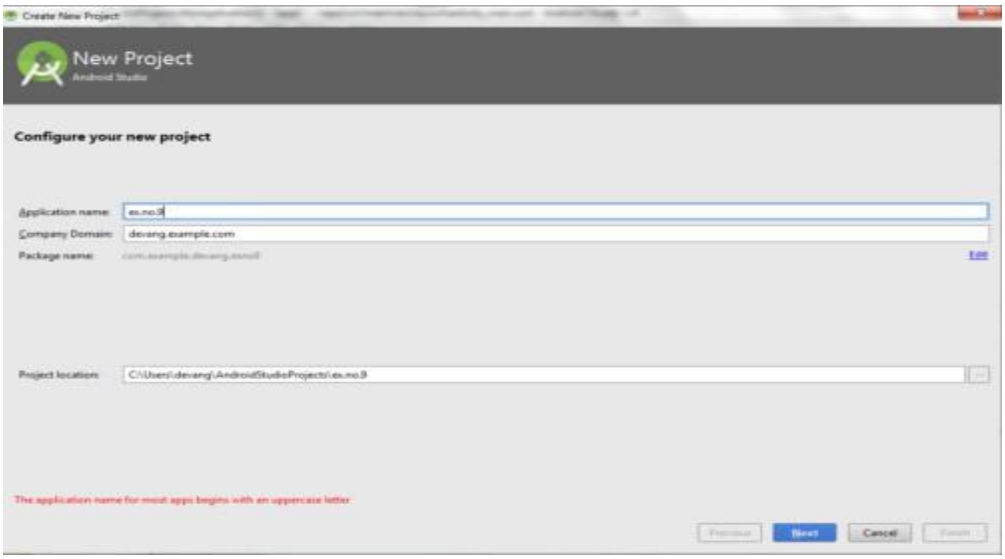
PROCEDURE:

Creating a New project:

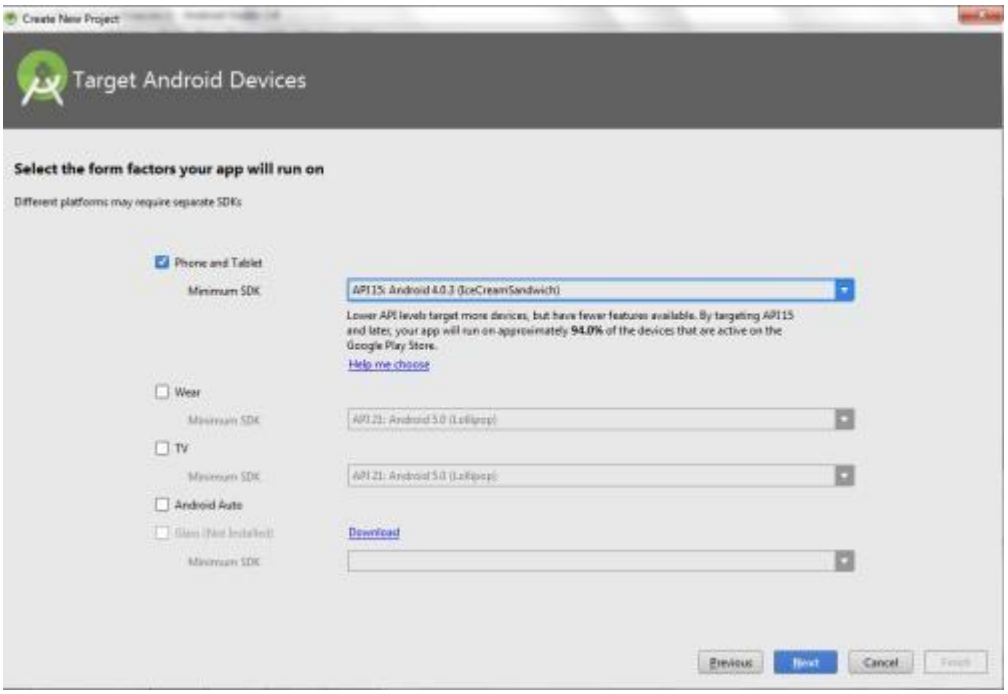
- Open Android Studio and then click on **File->New->New project.**



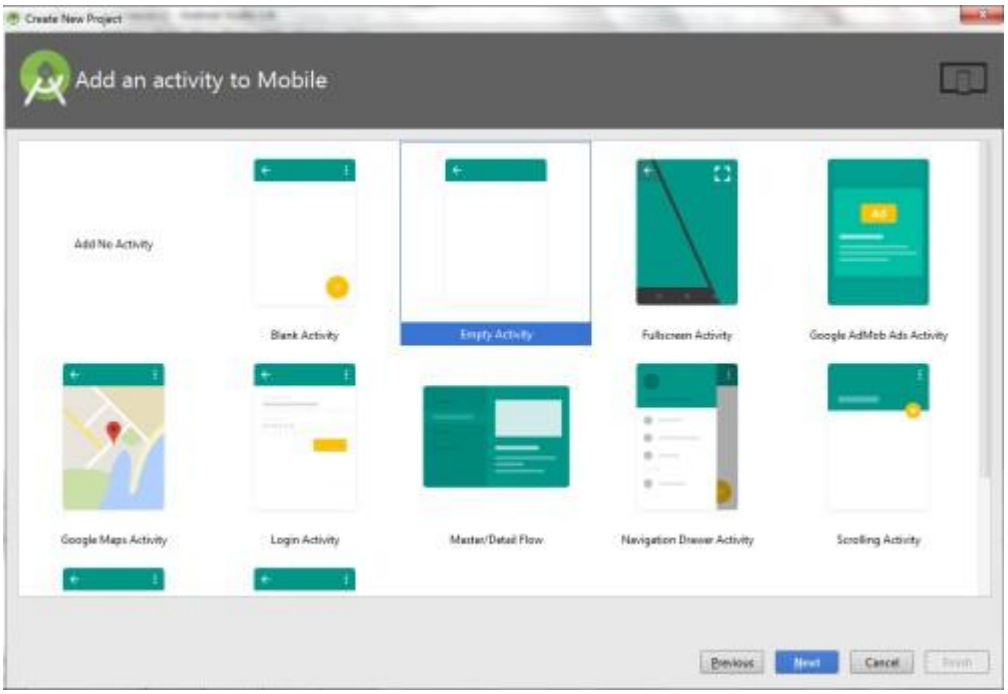
- Then type the Application name as“**ex.no.9″**and click**Next.**



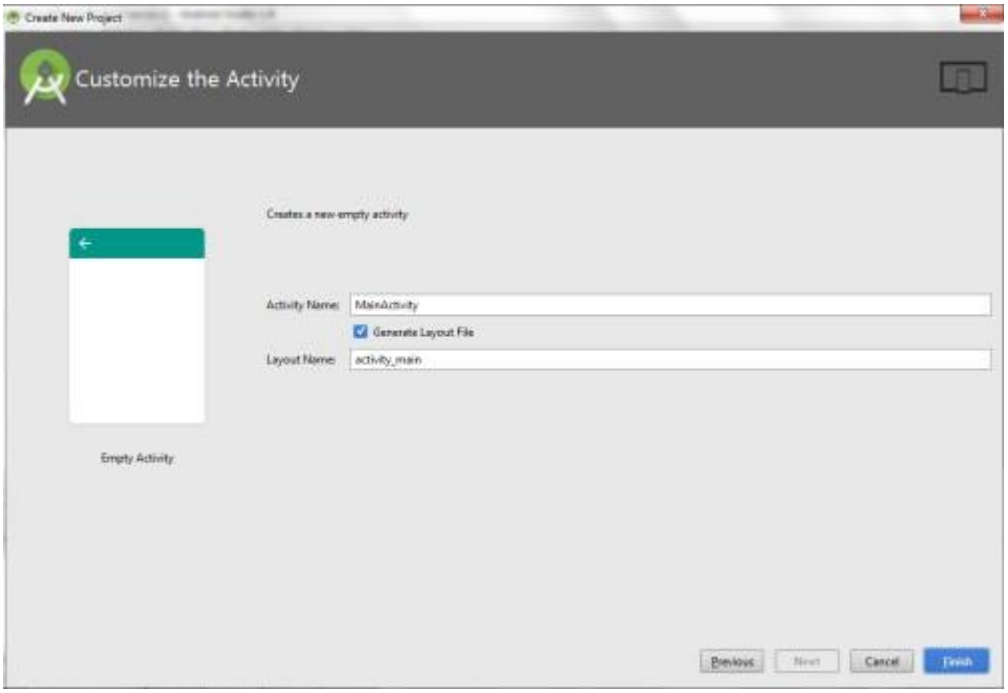
- Then select the **Minimum SDK** as shown below and click **Next**.



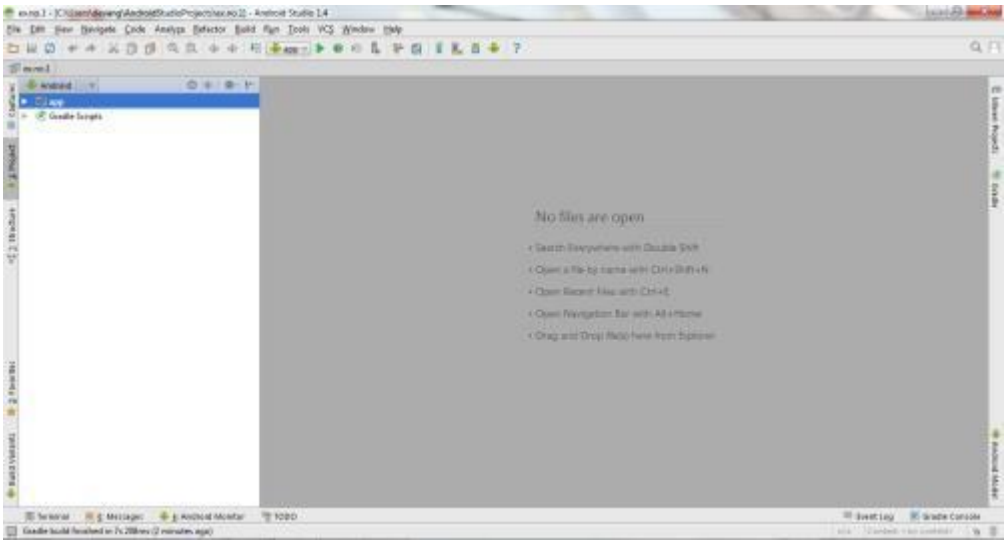
- Then select the **Empty Activity** and click **Next.**



- Finally click **Finish**.

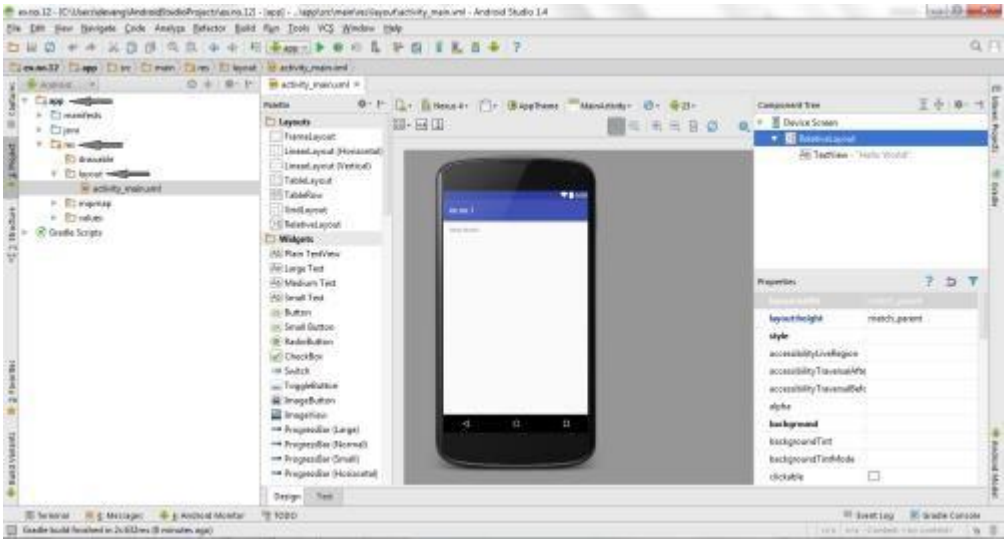


- It will take sometime to build and load the project.
- After completion it will look as given below.

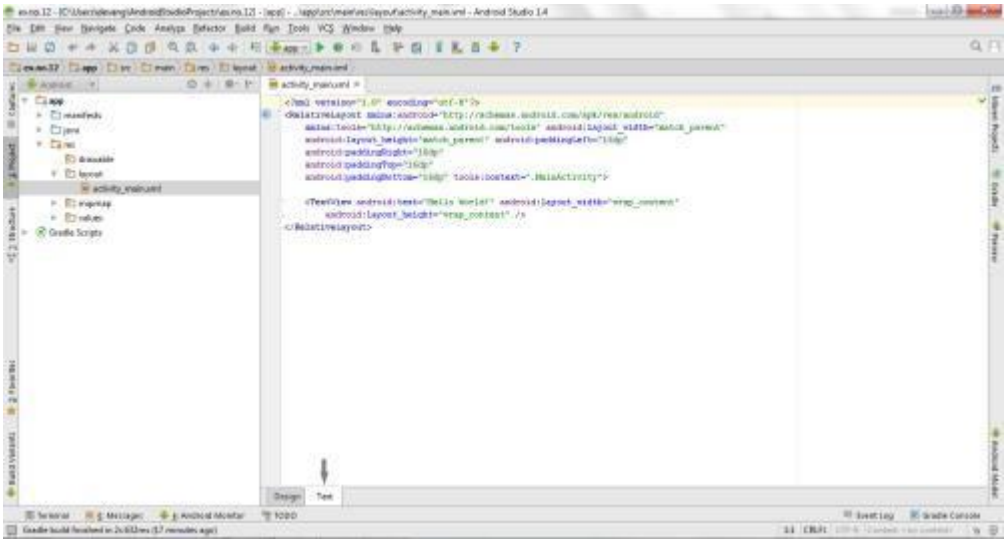


Designing layout for the Android Application:

- Click on **app->res-> layout->activity_main.xml.**



- Now click on **Text** as shown below.



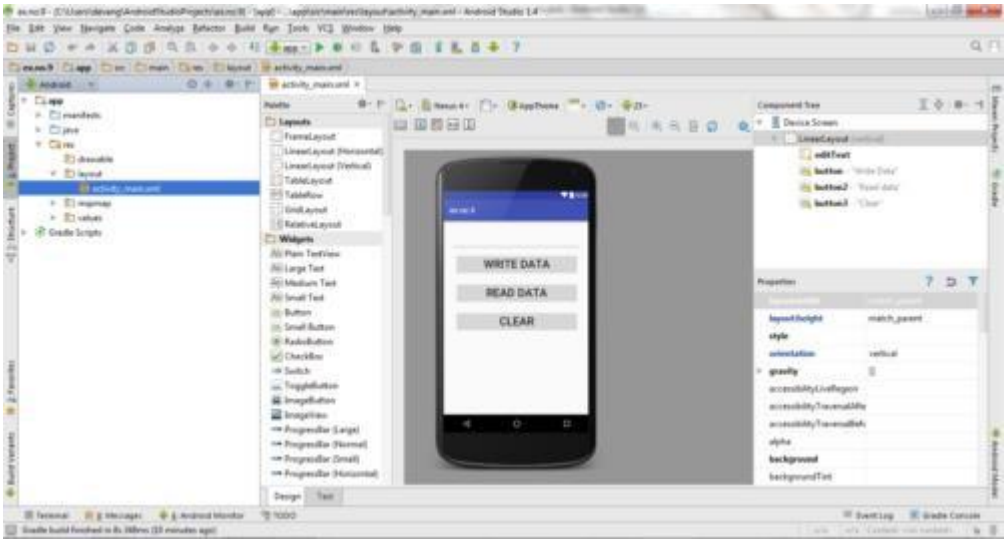
- Then delete the code which is there and type the code as given below.

Code forActivity_main.xml:

```
<?xmlversion="1.0"encoding="utf-8"?>
<LinearLayoutxmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="20dp"
    android:orientation="vertical">

    <EditText
        android:id="@+id/editText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:singleLine="true"android:textSize="3
        0dp"/>

    <Button
        android:id="@+id/button"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"android:text="Wr
        ite Data" android:textSize="30dp"/>

    <Button
        android:id="@+id/button2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"android:text="Re
        ad data" android:textSize="30dp"/>

    <Button
        android:id="@+id/button3"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:text="Clear"
```

```
        android:textSize="30dp"/>

</LinearLayout>
```

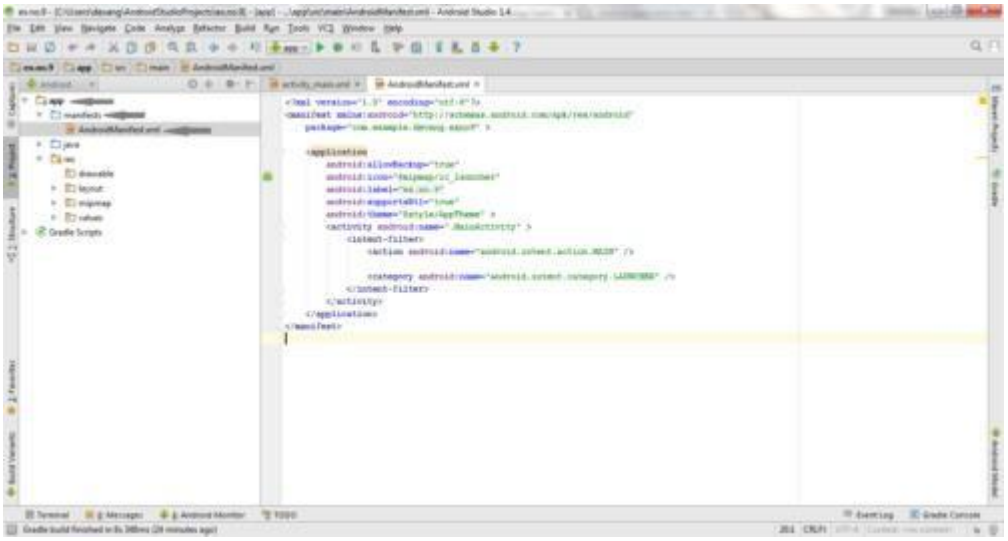
- Now click on **Design** and your application will look as given below.



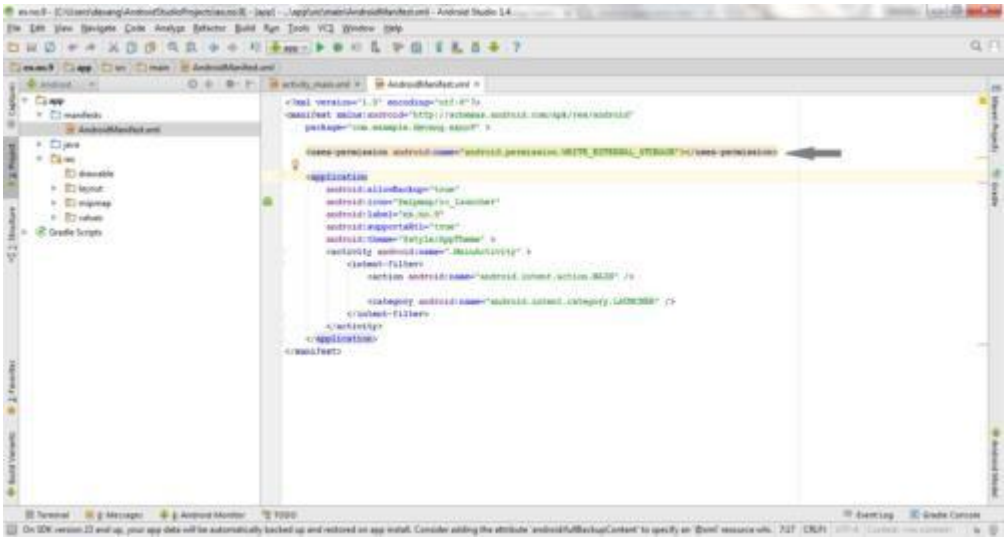
- So now the designing part is completed.

Adding permissions in Manifest for the Android Application:

- Click on **app ->manifests->AndroidManifest.xml**



- Now include the **WRITE_ EXTERNAL_ STORAGE** permissions in the AndroidManifest.xml file as shown below



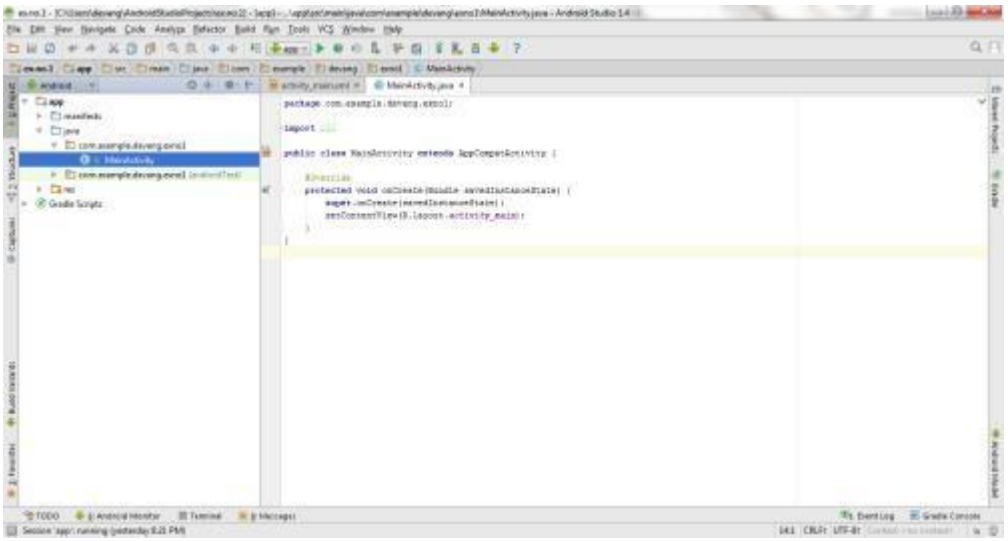
Code for Android Manifest.xml

```
<?xmlversion="1.0"encoding="utf-8"?>
<manifestxmlns:android="http://schemas.android.com/apk/res/android"package="com.exa
    mple.exno9">

    <uses-permission
android:name="android.permission.WRITE_EXTERNAL_STORAGE"></uses-permission>

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:supportsRtl="true"android:theme="@st
        yle/AppTheme">
        <activityandroid:name=".MainActivity">
            <intent-filter>
                <actionandroid:name="android.intent.action.MAIN"/>

                <categoryandroid:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>
    </application>
</manifest>
```

So now the Permissions are added in the Manifest.

Java Coding for the Android Application:

- Click on **app ->java->com.example.exno9->Main Activity.**



- Then delete the code which is there and type the code as given below.

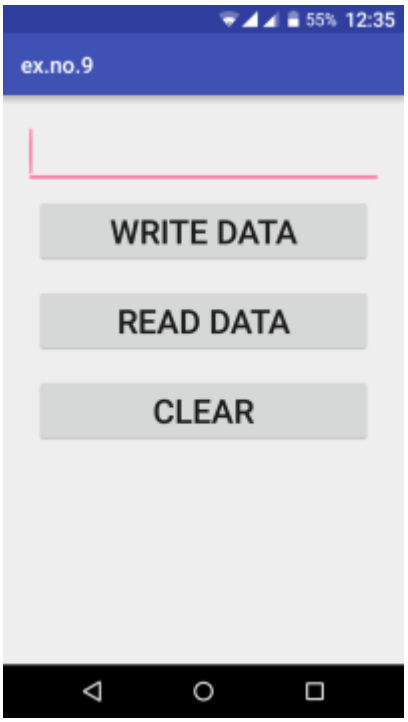
Code forMainActivity.java:

```
packagecom.example.exno9;

importandroid.os.Bundle;
importandroid.support.v7.app.AppCompatActivity;
importandroid.view.View;
importandroid.widget.Button;
importandroid.widget.EditText;
importandroid.widget.Toast;

importjava.io.BufferedReader;
importjava.io.File;
importjava.io.FileInputStream;
importjava.io.FileOutputStream;
importjava.io.InputStreamReader;

publicclassMainActivityextendsAppCompatActivity
{
    EditTexte1;
    Buttonwrite,read,clear;
    @Override
    protectedvoidonCreate(BundlesavedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        e1=(EditText)findViewById(R.id.editText);
        write= (Button) findViewById(R.id.button);
        read= (Button) findViewById(R.id.button2);
        clear=(Button)findViewById(R.id.button3);

        write.setOnClickListener(newView.OnClickListener()
        {
            @Override
            publicvoidonClick(Viewv)
            {
                Stringmessage=e1.getText().toString();
```

```
                try
                {
                    Filef=newFile("/sdcard/myfile.txt");
                    f.createNewFile();
                    FileOutputStreamfout=newFileOutputStream(f);
                    fout.write(message.getBytes());
                    fout.close();
                    Toast.makeText(getBaseContext(),"DataWrittenin
SDCARD",Toast.LENGTH_LONG).show();
                }
                catch(Exceptione)
                {
                    Toast.makeText(getBaseContext(),e.getMessage(),Toast.LE
NGTH_LONG).show();
                }
            }
        });

        read.setOnClickListener(newView.OnClickListener()
        {
            @Override
            publicvoidonClick(Viewv)
            {
                String message;
                Stringbuf=""; try
                {
                    File f = newFile("/sdcard/myfile.txt");
                    FileInputStreamfin=newFileInputStream(f);
                    BufferedReader br = newBufferedReader(new
InputStreamReader(fin));
                    while((message=br.readLine())!=null)
                    {
                        buf+=message;
                    }
                    e1.setText(buf);
                    br.close();
                    fin.close();
                    Toast.makeText(getBaseContext(),"DataRecivedfrom
SDCARD",Toast.LENGTH_LONG).show();
                }
                catch(Exceptione)
                {
                    Toast.makeText(getBaseContext(),e.getMessage(),
Toast.LENGTH_LONG).show();
                }
            }
        });

        clear.setOnClickListener(newView.OnClickListener()
        {
            @Override
            publicvoidonClick(Viewv)
            {
                e1.setText("");
            }
    });
     }
}
```

- So now the Coding part is also completed.
- Now run the application to see the output.

OUTPUT:



RESULT:

Thus Android Application that writes data to the SD Card is developed and executed successfully.

EX.NO:10
DATE:

ANDROID APPLICATION THAT CREATES AN ALERTUP ON RECEIVING A MESSAGE

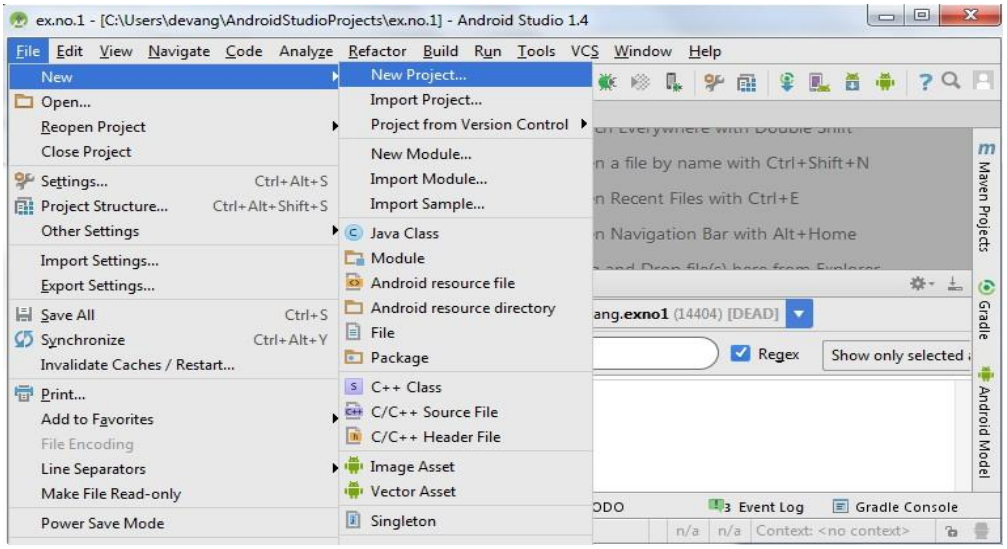
AIM:

To develop a Android Application that creates an alert upon receiving a message.

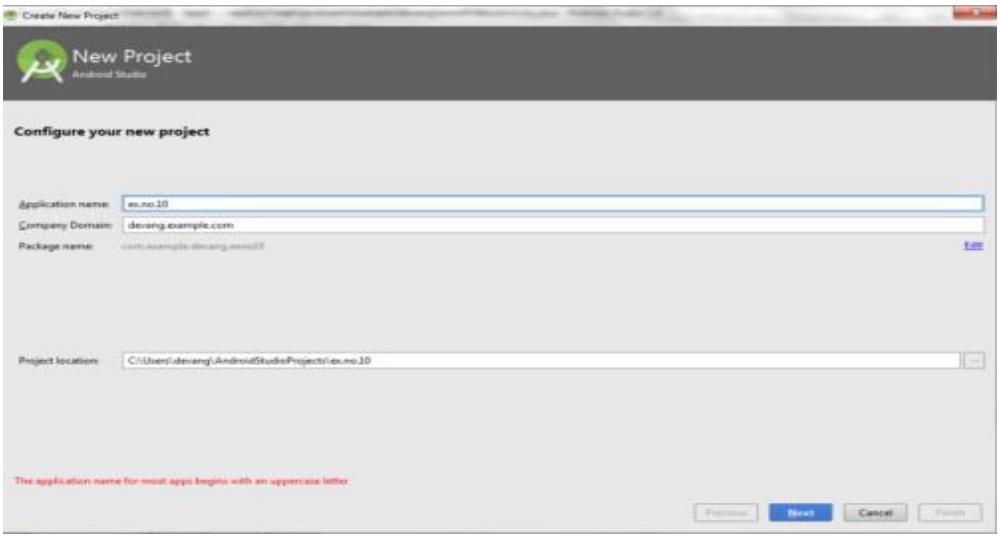
PROCEDURE:

Creating a New project:

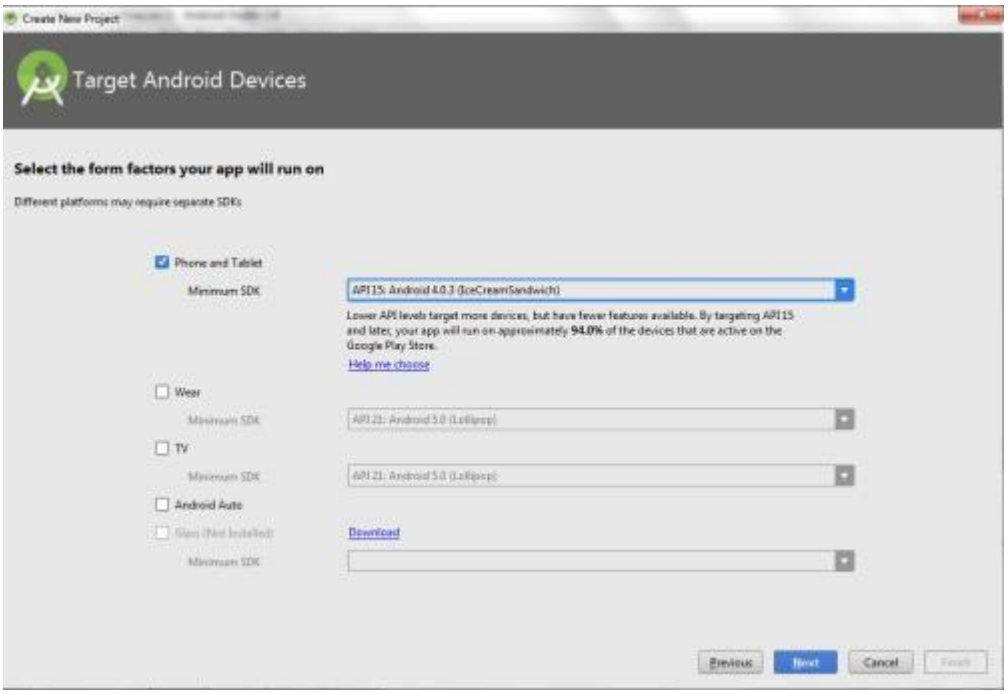
- Open Android Studio and then click on **File->New->New project.**



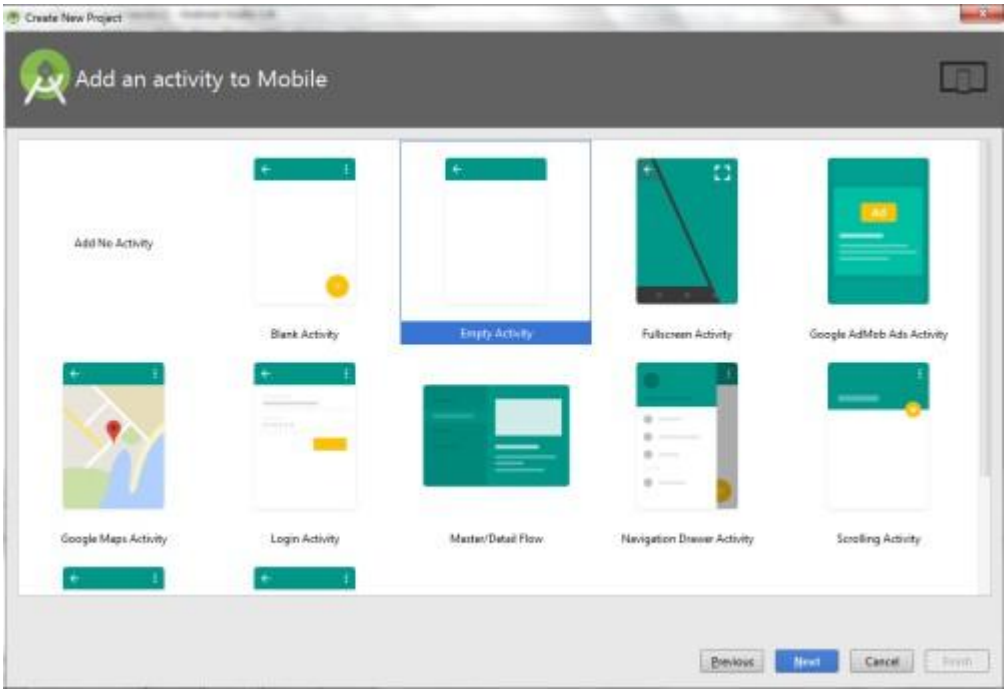
- Then type the Application name as"**ex.no.10"**and click**Next.**



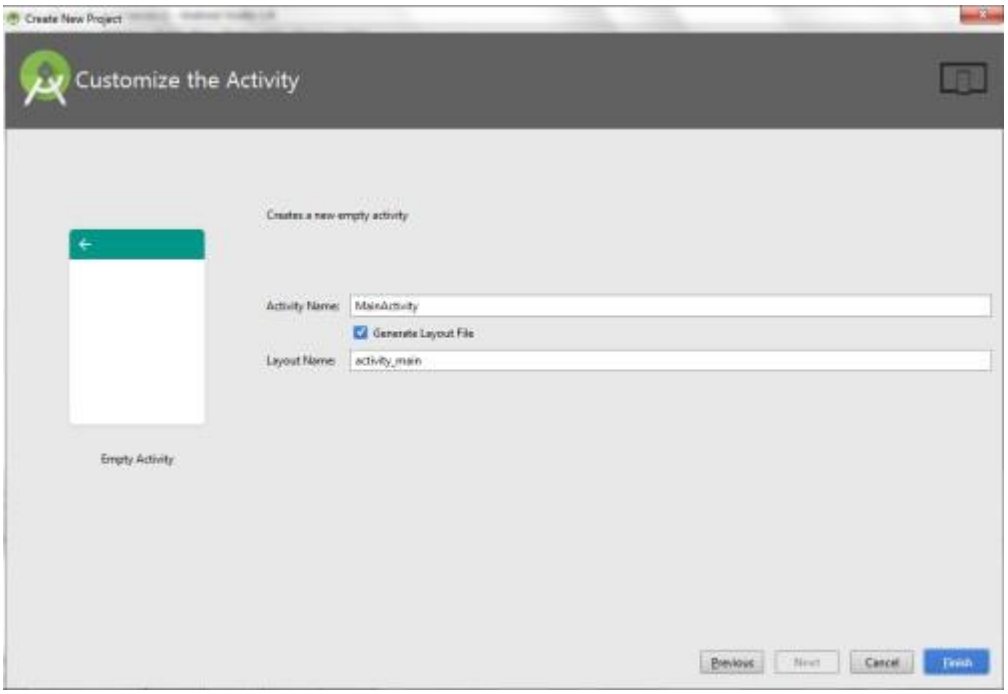
- Then select the **Minimum SDK** as shown below and click **Next**.



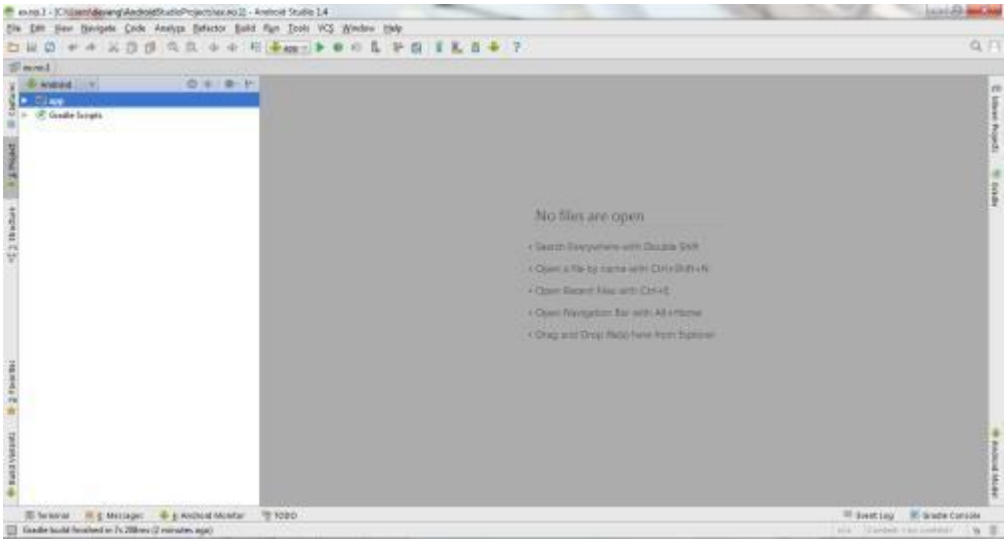
- Then select the **Empty Activity** and click **Next.**



- Finally click F**inish**.

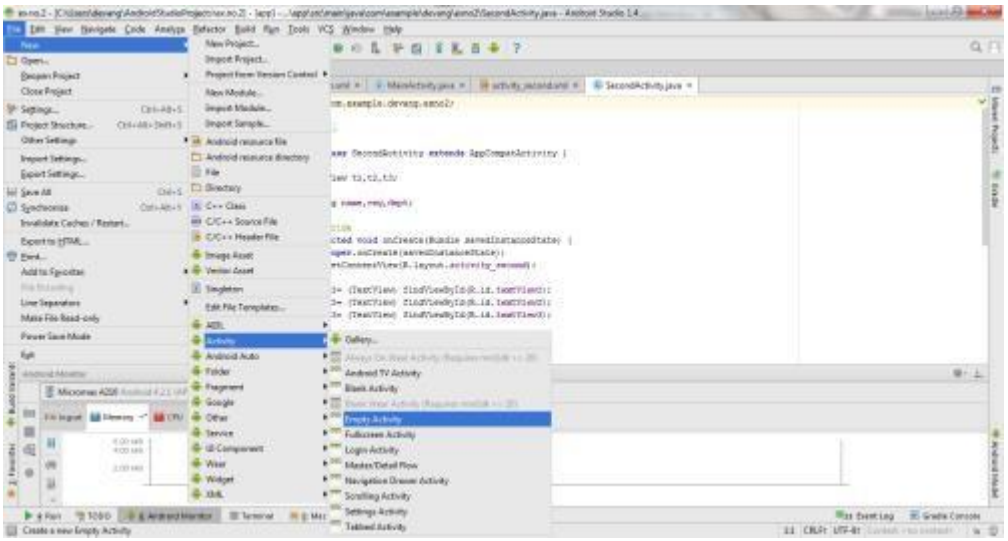


- It will take some time to build and load the project.
- After completion it will look as given below.

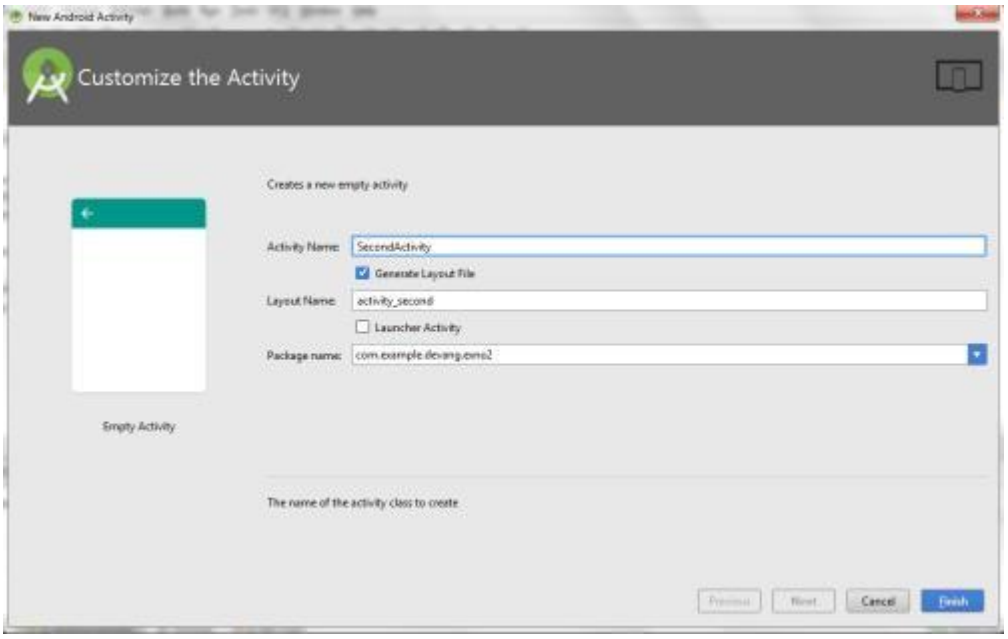


Creating Second Activity for the Android Application:

- Click on **File->New ->Activity->Empty Activity.**



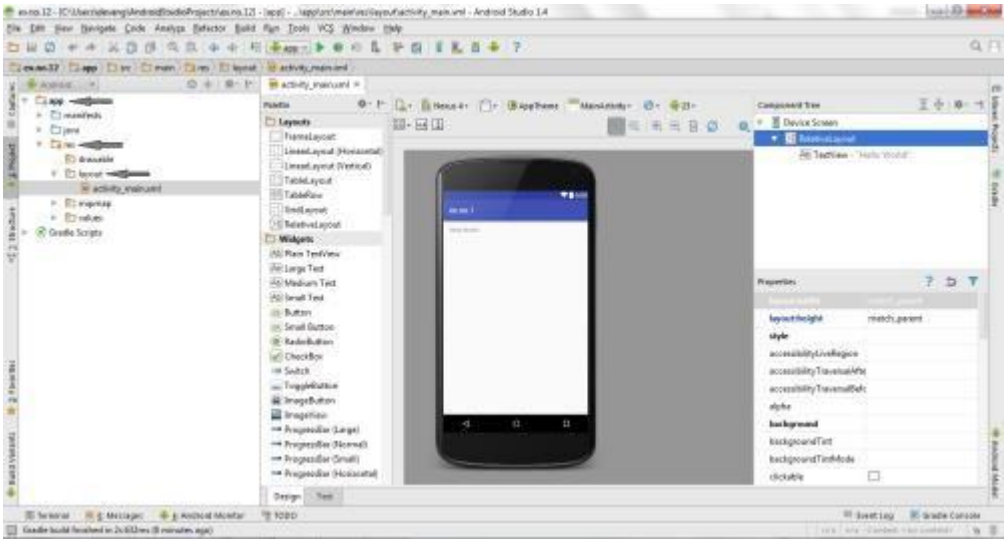
- Type the Activity Name as **Second Activity** and click **Finish** button.



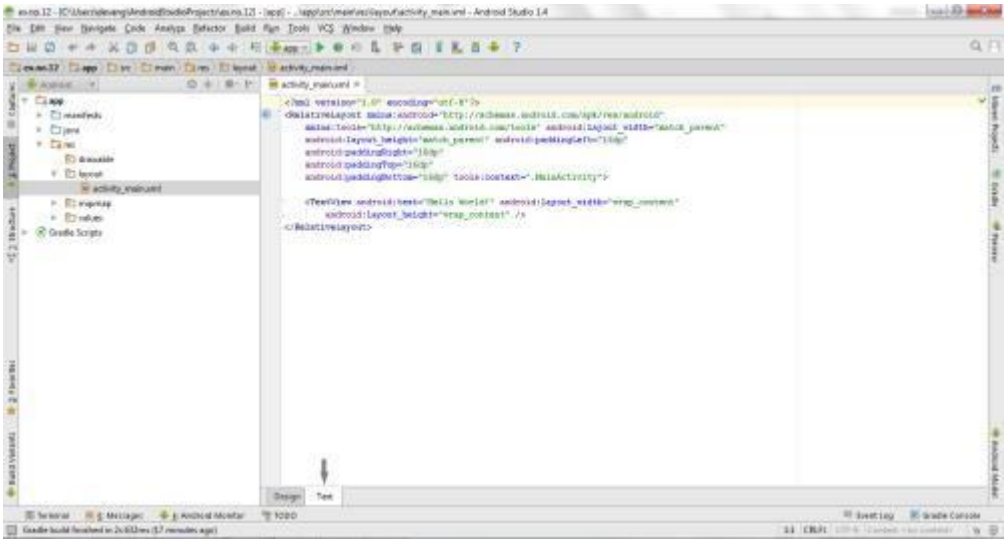
- Thus Second Activity For the application is created.

Designing layout for the Android Application:

- Clickon **app->res-> layout->activity_main.xml.**



- Now click on **Text** as shown below.



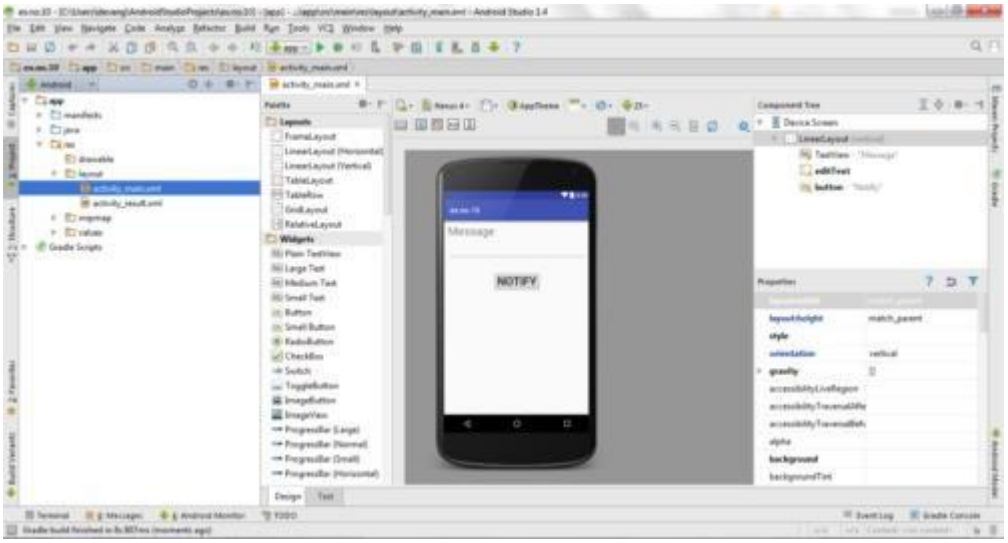
- Then delete the code which is there and type the code as given below.

Code for Activity_main.xml:

```
<?xmlversion="1.0"encoding="utf-8"?>
<LinearLayoutxmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="10dp"
    android:orientation="vertical">
     <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Message"android:textSize="30sp
        "/>

    <EditText
        android:id="@+id/editText"
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:singleLine="true"android:textSize="3
        0sp"/>

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="30dp"
        android:layout_gravity="center"
        android:text="Notify"
        android:textSize="30sp"/>

</LinearLayout>
```

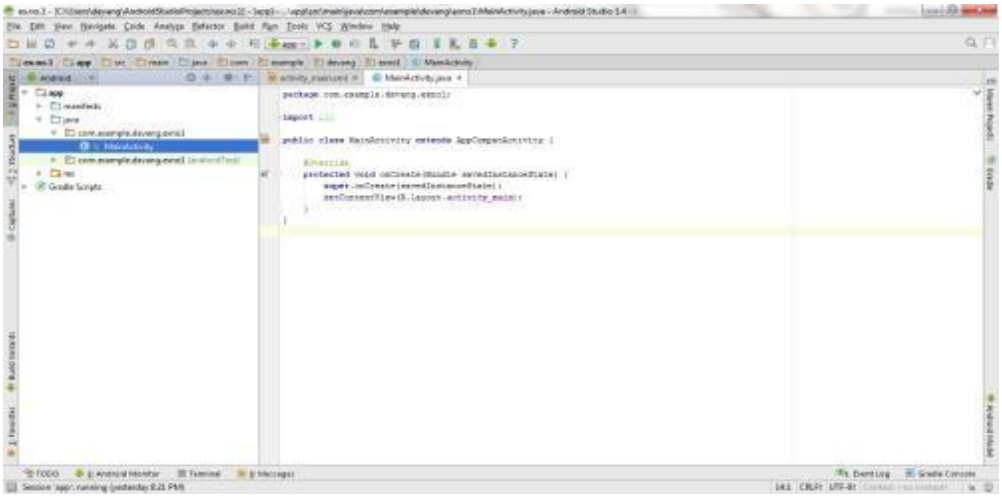
- Now click on **Design** and your application will look as given below.



- So now the designing part is completed.

Java Coding for the Android Application:

- Click on **app->java->com.example.exno10 ->Main Activity.**



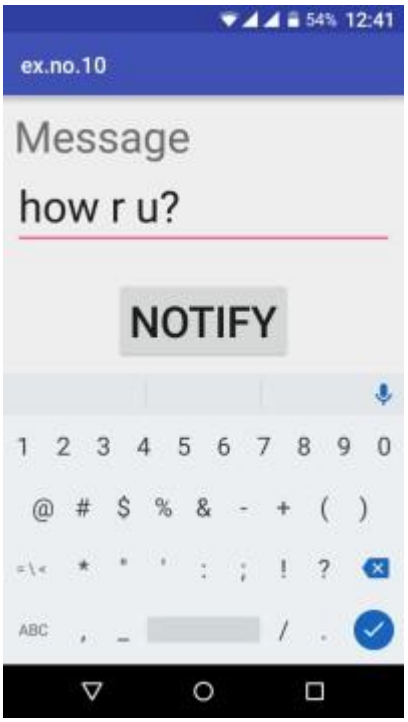
- Then delete the code which is there and type the code as given below.

Code forMainActivity.java:

```
packagecom.example.exno10;

importandroid.app.Notification;
importandroid.app.NotificationManager;
importandroid.app.PendingIntent;
importandroid.content.Intent;
importandroid.os.Bundle;
importandroid.support.v7.app.AppCompatActivity;
importandroid.view.View;
importandroid.widget.Button;
importandroid.widget.EditText;

publicclassMainActivityextendsAppCompatActivity
{
    Buttonnotify;
    EditText e;
    @Override
    protectedvoidonCreate(BundlesavedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        notify=(Button)findViewById(R.id.button); e=
        (EditText) findViewById(R.id.editText);

        notify.setOnClickListener(newView.OnClickListener()
        {
            @Override
            publicvoidonClick(Viewv)
            {
                Intentintent=newIntent(MainActivity.this,
SecondActivity.class);
                PendingIntent pending =
PendingIntent.getActivity(MainActivity.this,0,intent,0);
                Notification noti = new
Notification.Builder(MainActivity.this).setContentTitle("New
Message").setContentText(e.getText().toString()).setSmallIcon(R.mipmap.ic_l
auncher).setContentIntent(pending).build();
                NotificationManagermanager=(NotificationManager)
getSystemService(NOTIFICATION_SERVICE);
                noti.flags|=Notification.FLAG_AUTO_CANCEL;
                manager.notify(0, noti);
            }
        });
    }
}
```

- So now the Coding part is also completed.
- Now run the application to seethe output.

OUTPUT:



RESULT:

Thus Android Application that creates an alert up on receiving message is developed and executed successfully.

Ex.No:11

DATE:

ANDROID APPLICATION THAT CREATES ALARM CLOCK

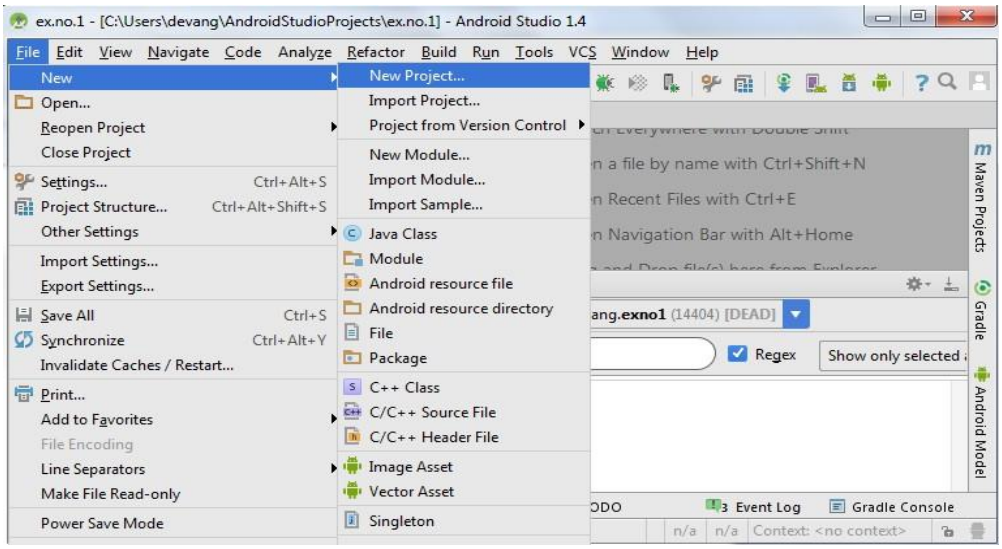
AIM:

To develop a Android Application that creates Alarm Clock.

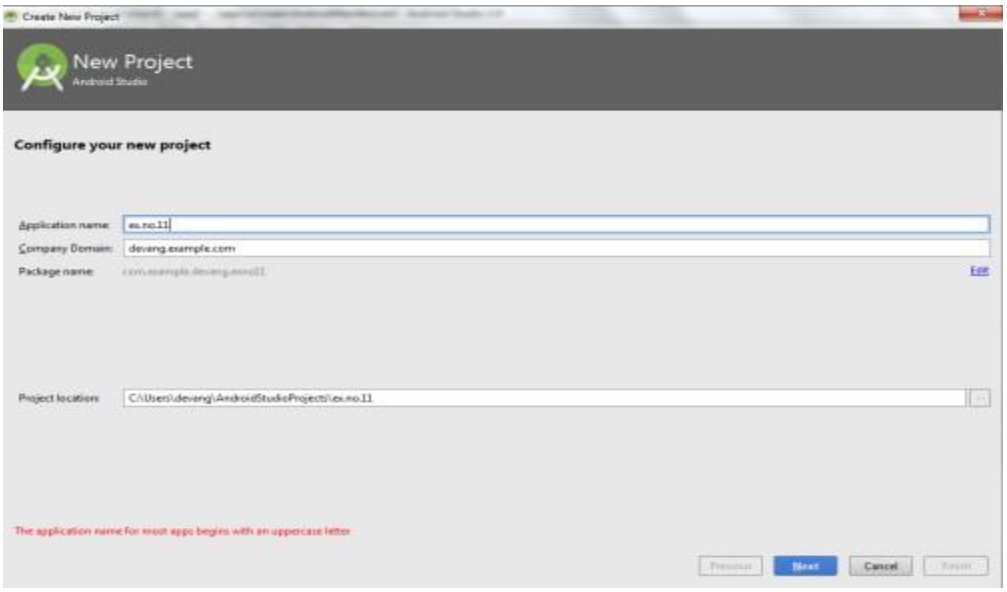
PROCEDURE:

Creating a New project:

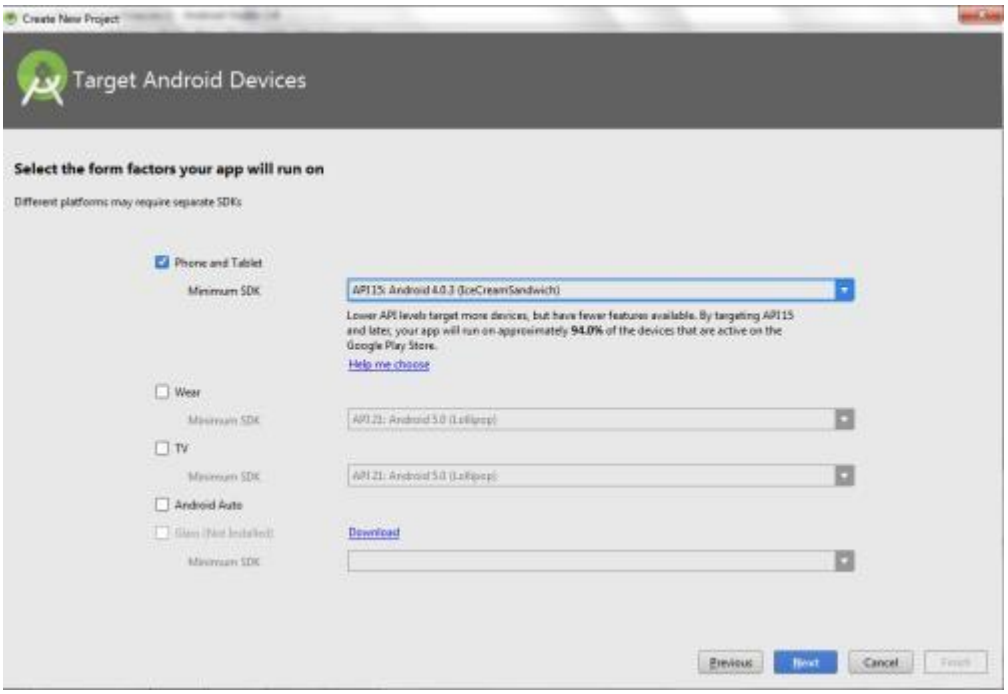
- Open Android Studio and then click on **File->New->New project.**



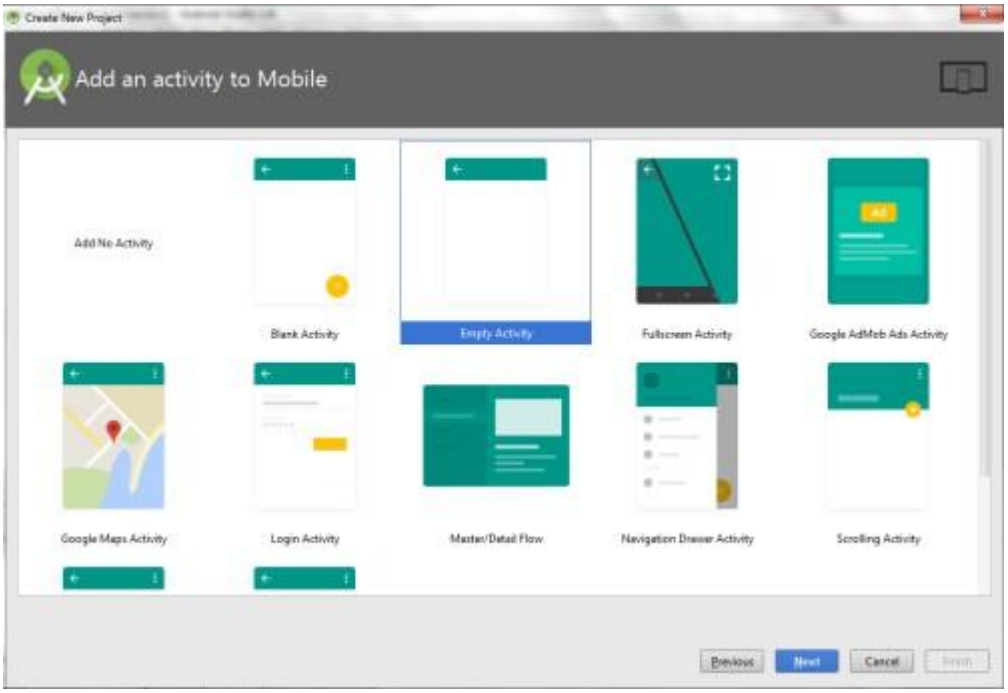
- Then type the Application name as“ **ex.no.11″**and click**Next.**



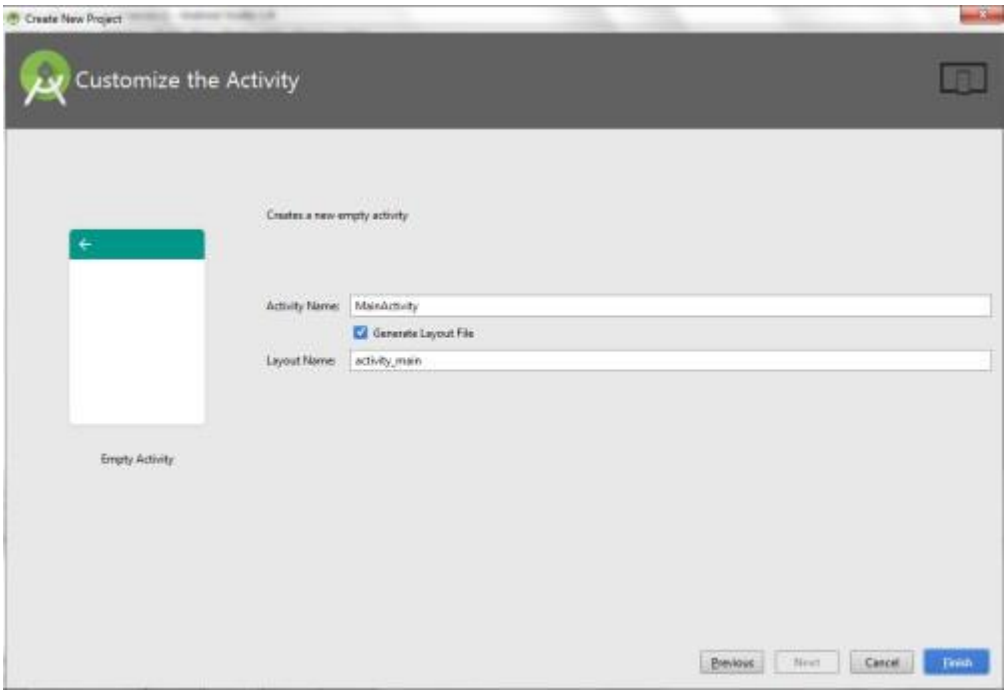
- Then select the **Minimum SDK** as shown below and click **Next**.



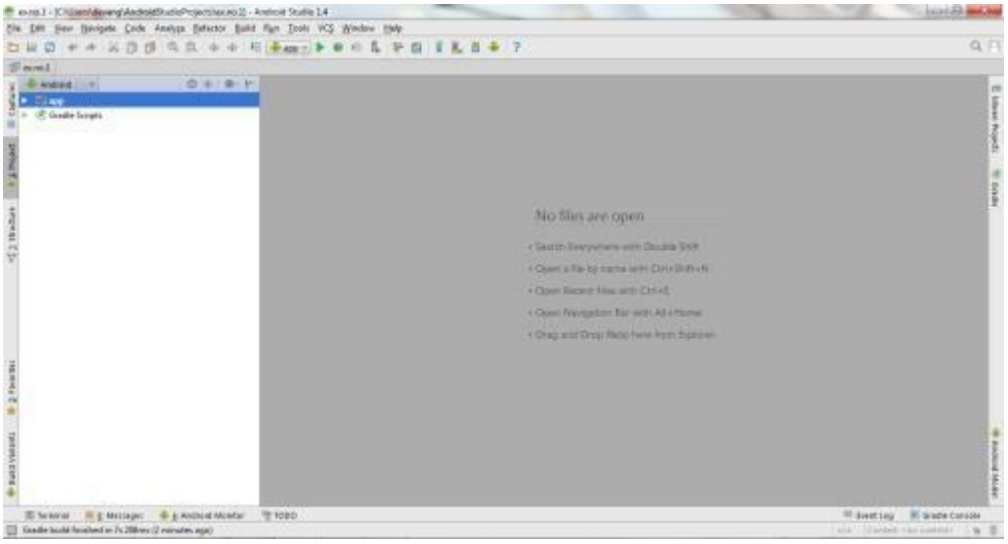
- Then select the **Empty Activity** and click **Next.**



- Finally click F**inish**.

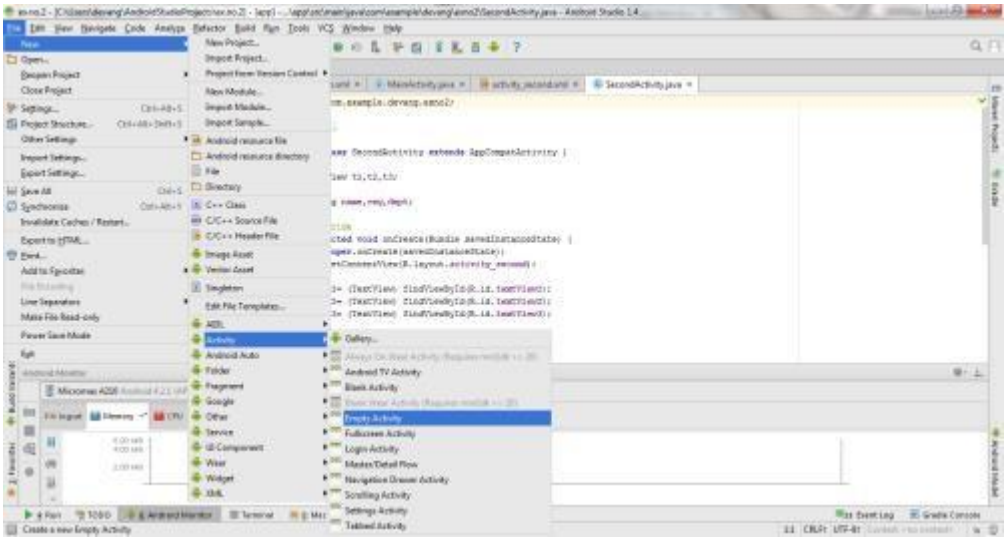


- It will take sometime to build and load the project.
- After completion it will look as given below.

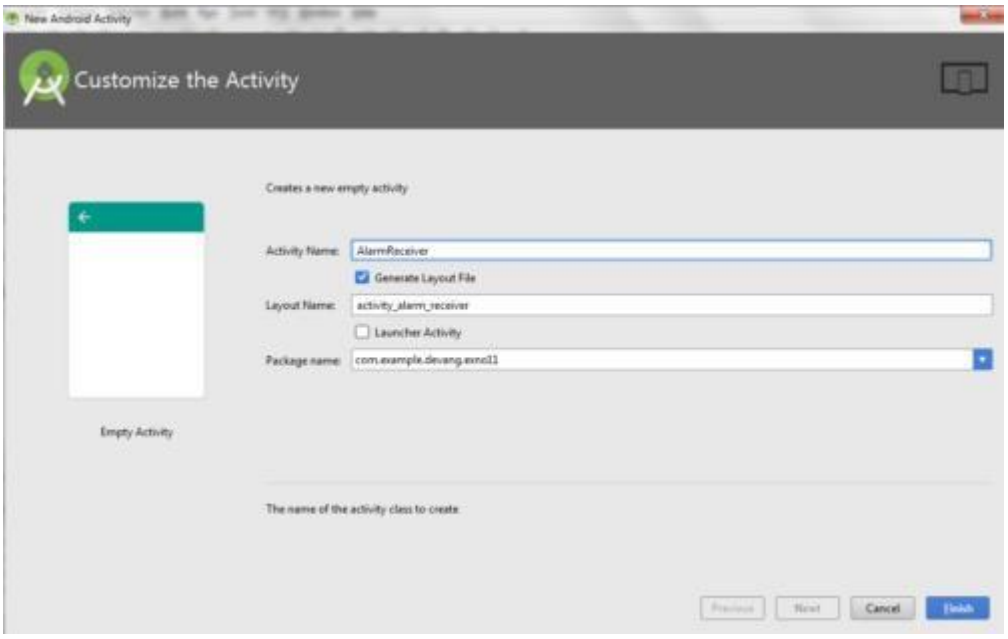


Creating Second Activity for the Android Application:

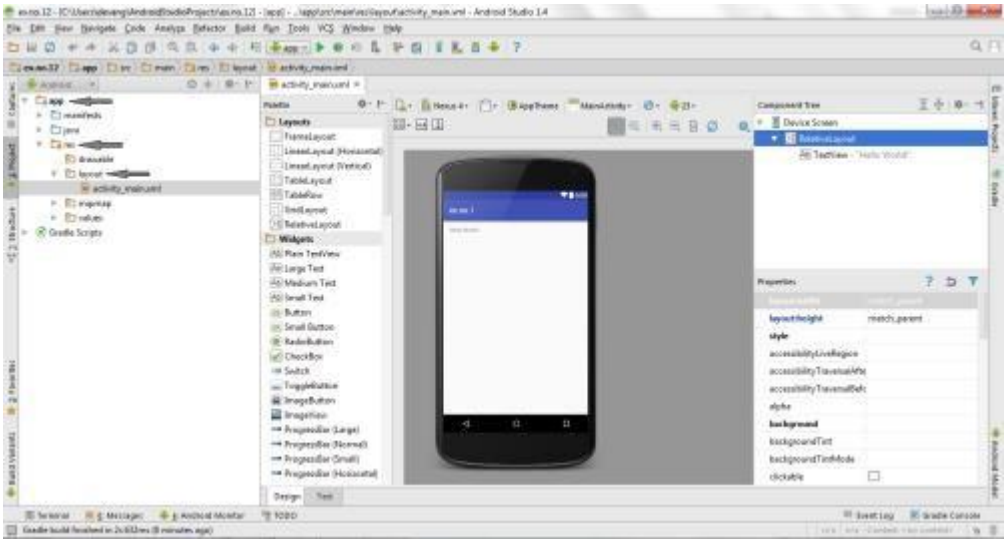
- Clickon **File->New ->Activity->Empty Activity.**



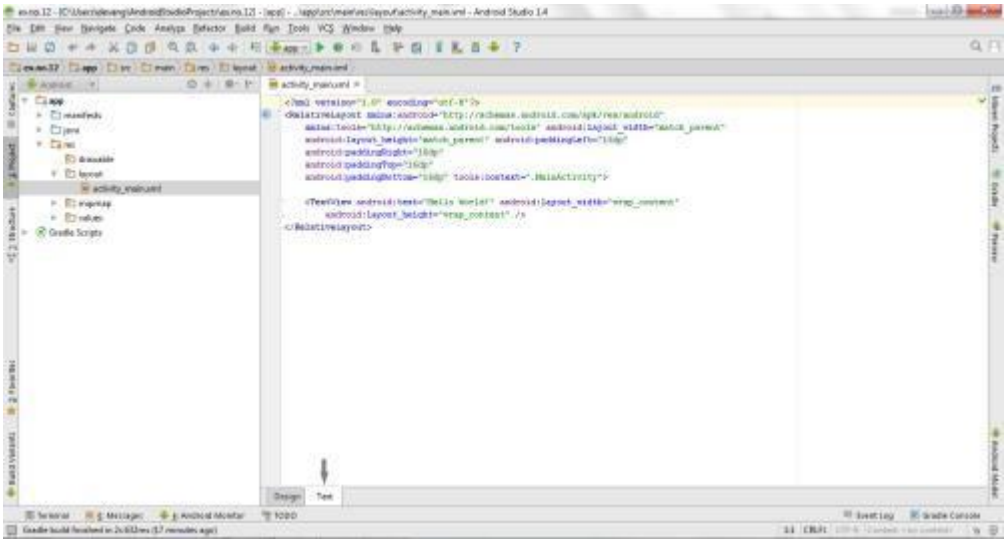
- Type the Activity Name as **Alarm Receiver** and click **Finish** button.



- Thus Second Activity For the application is created.
- Click on **app->res-> layout->activity_main.xml.**



- Now click on **Text** as shown below.



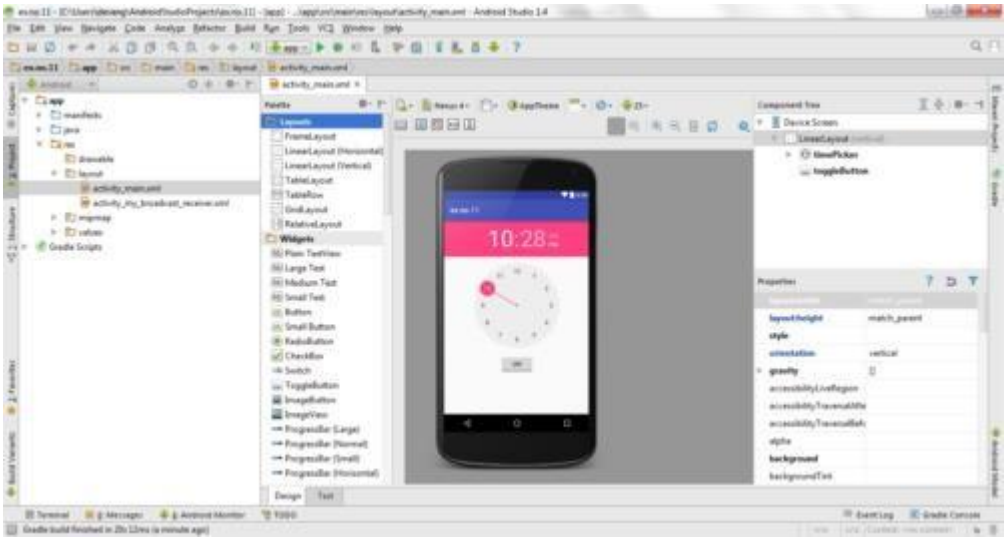
- Then delete the code which is there and type the code as given below.

Code forActivity_main.xml:

```
<?xmlversion="1.0"encoding="utf-8"?>
<LinearLayoutxmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"android:orientation="vertical">

    <TimePicker
        android:id="@+id/timePicker"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"android:layout_gravity="ce
        nter"/>

    <ToggleButton
        android:id="@+id/toggleButton"
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:layout_margin="20dp"
        android:checked="false"android:onClick="OnTo
        ggleClicked"/>
</LinearLayout>
```

- Now click on **Design** and your application will look as givenbelow.

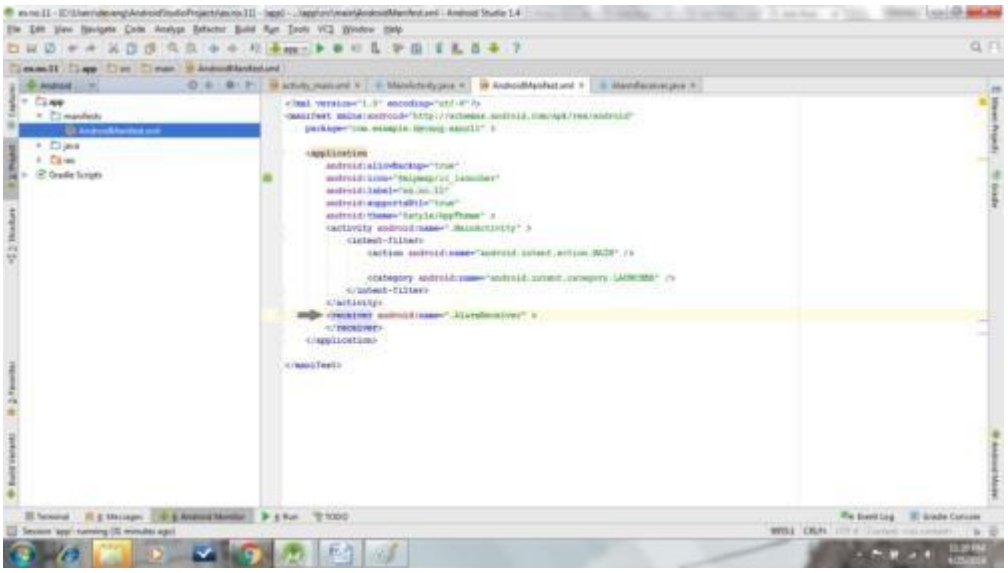


- So now the designing part is completed.

Changes in Manifest for the Android Application:

- Click on **app ->manifests->AndroidManifest.xml**

- Now change the **activity tag** to **receiver tag** in the Android Manifest.xml file as shown below



Code for Android Manifest.xml:

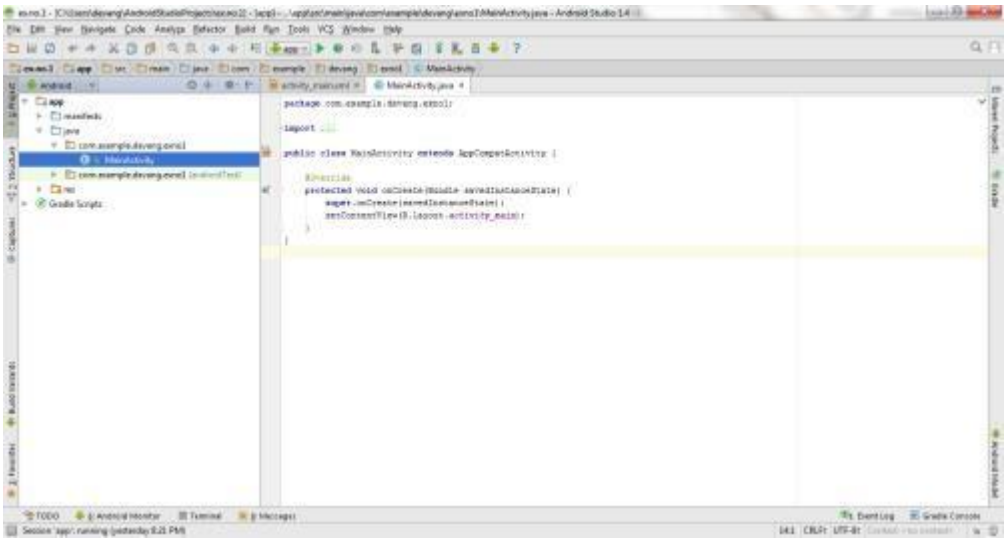
```
<?xmlversion="1.0"encoding="utf-8"?>
<manifestxmlns:android="http://schemas.android.com/apk/res/android"package="com.exa
    mple.exno11">

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:supportsRtl="true"android:theme="@st
        yle/AppTheme">
        <activityandroid:name=".MainActivity">
            <intent-filter>
                <actionandroid:name="android.intent.action.MAIN"/>
                 <categoryandroid:name="android.intent.category.LAUNCHER"
/>
            </intent-filter>
        </activity>
        <receiverandroid:name=".AlarmReceiver">
        </receiver>
    </application>

</manifest>
```

So now the changes are done in the Manifest.

Java Coding for the Android Application:

Java Coding for Main Activity:

- Clickon**app->java->com.example.exno11 ->Main Activity.**



- Then delete the code which is there and type the code as given below.

Code forMainActivity.java:

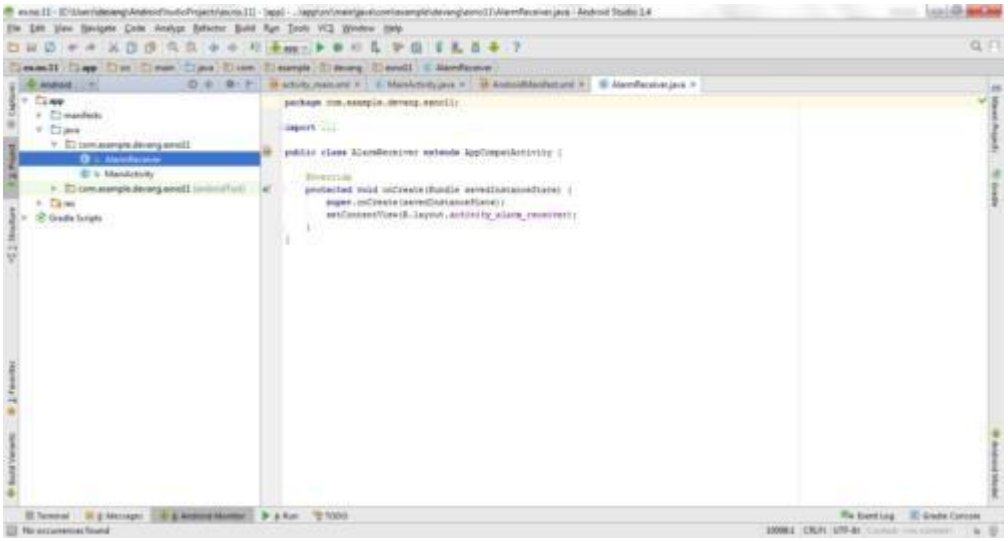
```
packagecom.example.exno11;

importandroid.app.AlarmManager;
importandroid.app.PendingIntent;
importandroid.content.Intent;
importandroid.os.Bundle;
importandroid.support.v7.app.AppCompatActivity;
importandroid.view.View;
importandroid.widget.TimePicker;
importandroid.widget.Toast;
importandroid.widget.ToggleButton;

importjava.util.Calendar;

publicclassMainActivityextendsAppCompatActivity
{
    TimePicker alarmTimePicker;
    PendingIntentpendingIntent;
    AlarmManager alarmManager;

    @Override
    protectedvoidonCreate(BundlesavedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        alarmTimePicker = (TimePicker) findViewById(R.id.timePicker);
        alarmManager=(AlarmManager)getSystemService(ALARM_SERVICE);
    }
    publicvoidOnToggleClicked(Viewview)
    {
        longtime;
        if(((ToggleButton)view).isChecked())
        {
            Toast.makeText(MainActivity.this,"ALARMON", Toast.LENGTH_SHORT).show();
            Calendarcalendar=Calendar.getInstance();
```

```
            calendar.set(Calendar.HOUR_OF_DAY,
alarmTimePicker.getCurrentHour());
            calendar.set(Calendar.MINUTE,
alarmTimePicker.getCurrentMinute());
            Intent intent = newIntent(this, AlarmReceiver.class);
            pendingIntent=PendingIntent.getBroadcast(this,0,intent,0);

            time=(calendar.getTimeInMillis()-
(calendar.getTimeInMillis()%60000));
            if(System.currentTimeMillis()>time)
            {
                if(calendar.AM_PM==0)
                    time=time+(1000*60*60*12); else
                    time=time+(1000*60*60*24);
            }
            alarmManager.setRepeating(AlarmManager.RTC_WAKEUP,time,10000,
pendingIntent);
        }
        else
        {
            alarmManager.cancel(pendingIntent);
            Toast.makeText(MainActivity.this,"ALARMOFF",
Toast.LENGTH_SHORT).show();
        }
    }
}
```

So now the Coding part of Main Activity is completed.

Java Coding for Alarm Receiver:

- Click on **app->java->com.example.exno11 ->Alarm Receiver.**



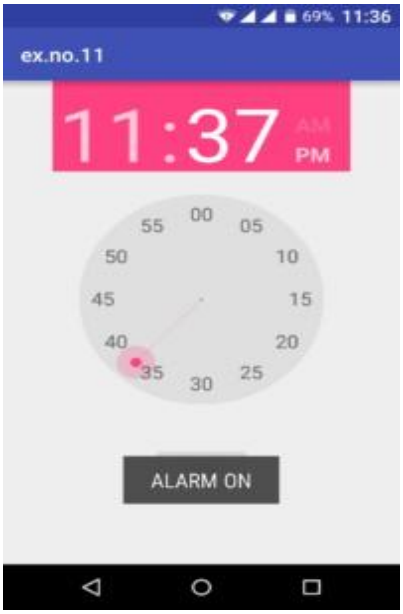
- Then delete the code which is there and type the code as given below.

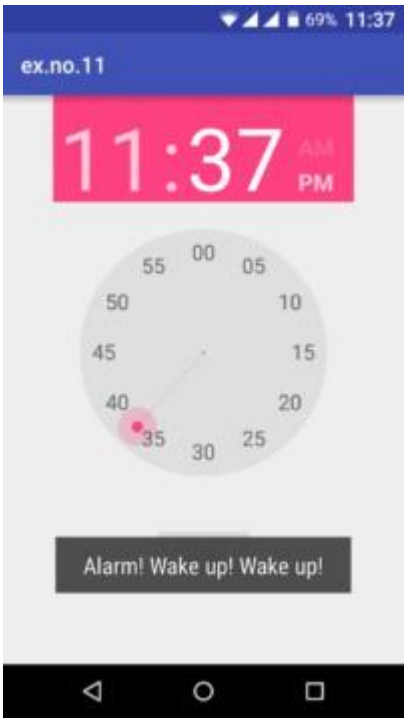
Code forAlarmReceiver.java:

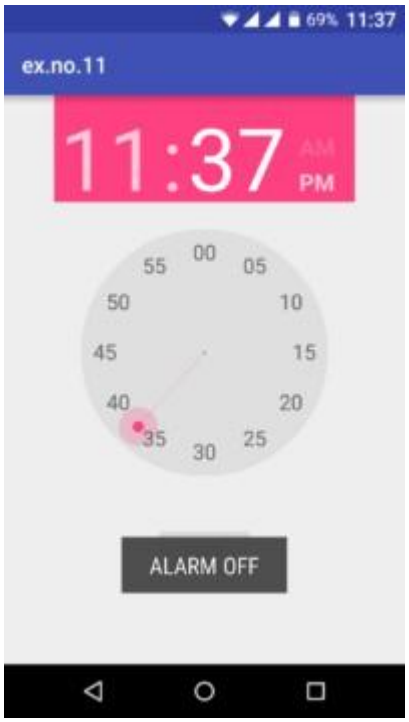
```
packagecom.example.exno11;
```

```
importandroid.content.BroadcastReceiver;
importandroid.content.Context;
importandroid.content.Intent;
importandroid.media.Ringtone;
importandroid.media.RingtoneManager;
importandroid.net.Uri;
importandroid.widget.Toast;

publicclassAlarmReceiverextendsBroadcastReceiver
{
    @Override
    publicvoidonReceive(Contextcontext,Intentintent)
    {
        Toast.makeText(context,"Alarm!Wakeup!Wakeup!", Toast.LENGTH_LONG).show();
        Uri alarmUri =
RingtoneManager.getDefaultUri(RingtoneManager.TYPE_ALARM);
        if(alarmUri==null)
        {
            alarmUri =
RingtoneManager.getDefaultUri(RingtoneManager.TYPE_NOTIFICATION);
        }
        Ringtoneringtone=RingtoneManager.getRingtone(context,alarmUri);
        ringtone.play();
    }
}
```

OUTPUT:







RESULT:

Thus Android Application that creates Alarm Clock is developed and executed successfully.